

Business Process Automation with VBA and Python

Mr. Eddie Chow / 23 May 2026



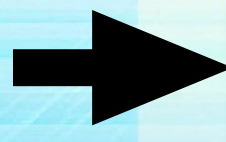
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Business Process automation with VBA

Business process automation with Python

Introduction to project management for business process automation



Development and implementation of business process automation

Final Group Presentation





Intended Learning Outcomes

1. explain the basic concepts and principles of web scraping, including the ethical considerations and legal implications involved in data extraction from websites.
2. demonstrate the ability to utilize Python libraries such as BeautifulSoup and Scrapy to efficiently extract, parse, and clean data from various web sources.
3. learn to visualize the scraped data using libraries like Matplotlib or Seaborn, enabling them to create informative graphs and charts that effectively communicate insights derived from the data.
4. apply data analysis techniques to interpret the visualized data, drawing conclusions that can inform decision-making processes in real-world scenarios, such as market analysis or competitive intelligence.

Agenda

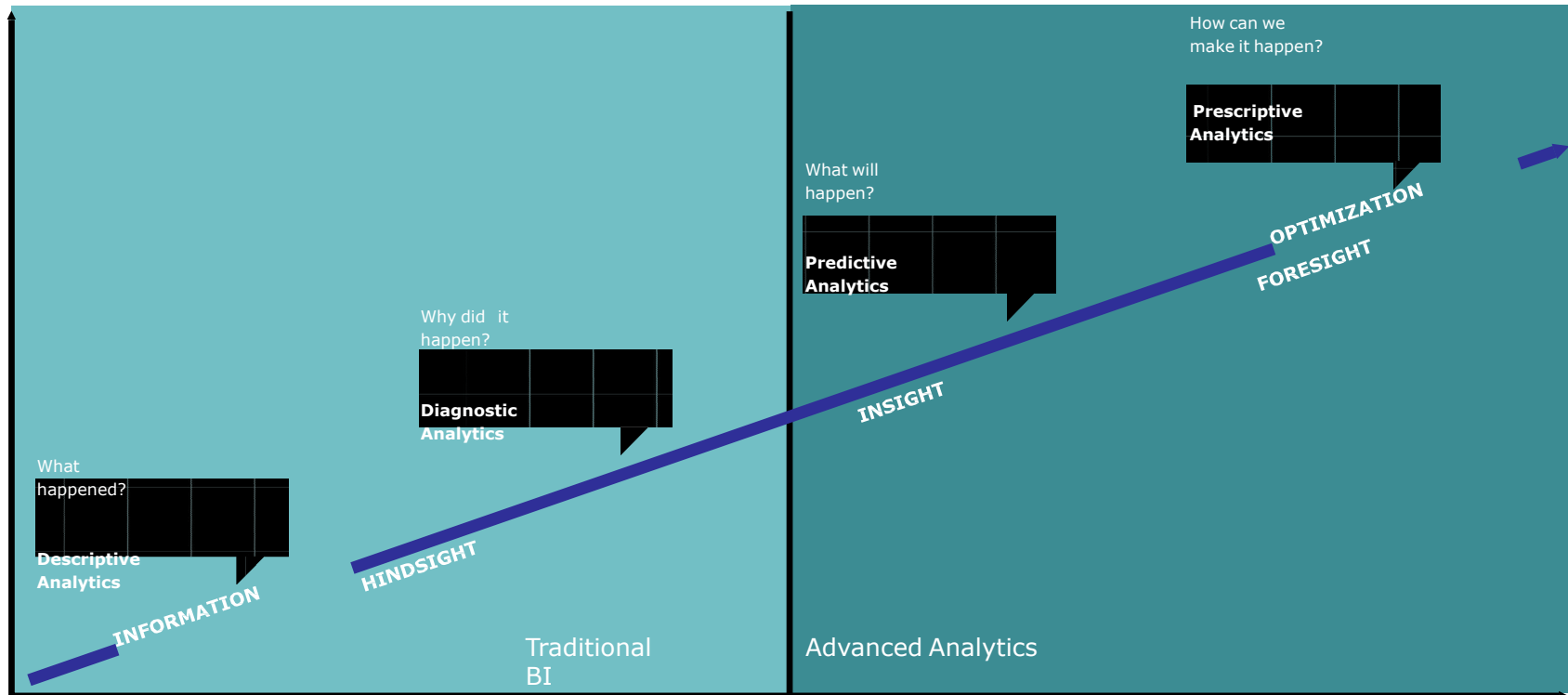
1. Introduction to Data Science
2. Concept of Structured, Unstructured, Discrete, Continuous Data
3. Applied Machine Learning Pipeline Overview
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5. Structured Data Collection - Web Crawler
6. Convert Unstructured Data to Structured Data - PDF, Paper Form, Image
7. Data Collection – Fetching Data using Pandas
8. Data Cleaning, Master Data List
9. Exploratory Data Analysis(EDA) and Data Visualization
10. Machine Learning Framework
11. How To Do Data Mining: Machine Learning Techniques
12. Introduction to Data Visualization

What Is Data Science?

From Wikipedia:

Data science, also known as data-driven science, is an interdisciplinary field about scientific processes and systems to extract knowledge or insights from data in various forms, either structured or unstructured, which is a continuation of some of the data analysis fields such as statistics, machine learning, data mining ,and predictive analytics similar to Knowledge Discovery in Databases (KDD).

From Digitalization To Data-Driven Decision Making



<https://arxiv.org/pdf/1904.13251>

Data-Driven vs Scenario-Driven Decision Making

Choosing the right approach for planning, execution, and risk management.

Aspect	Data-driven	Scenario-driven
Starting point	Observed data (logs, metrics, events)	Hypothetical or expected situations / “stories”
Main driver	Statistical patterns and KPIs	Business or operational logic of a situation
Best for	Optimization, performance tracking, and evidence-based management.	Strategic planning, experimentation, and uncertain environments.
Typical questions	“What is happening now?” What does the data predict?”	“What could happen?” “What should we do if it does?”

Use Case - Investment portfolio monitoring

Portfolio monitoring is used by VC, PE, wealth, and asset-management teams to follow performance, health, and risk after investment, while AI/ML helps analyze large financial datasets, detect patterns, and improve risk prediction.

Typical use case

A portfolio monitoring system ingests holdings, prices, cash flows, sector exposure, and company KPIs, then checks them against policy targets and benchmarks. It can automatically flag issues like concentration risk, style drift, liquidity problems, covenant breaches, or underperforming positions.

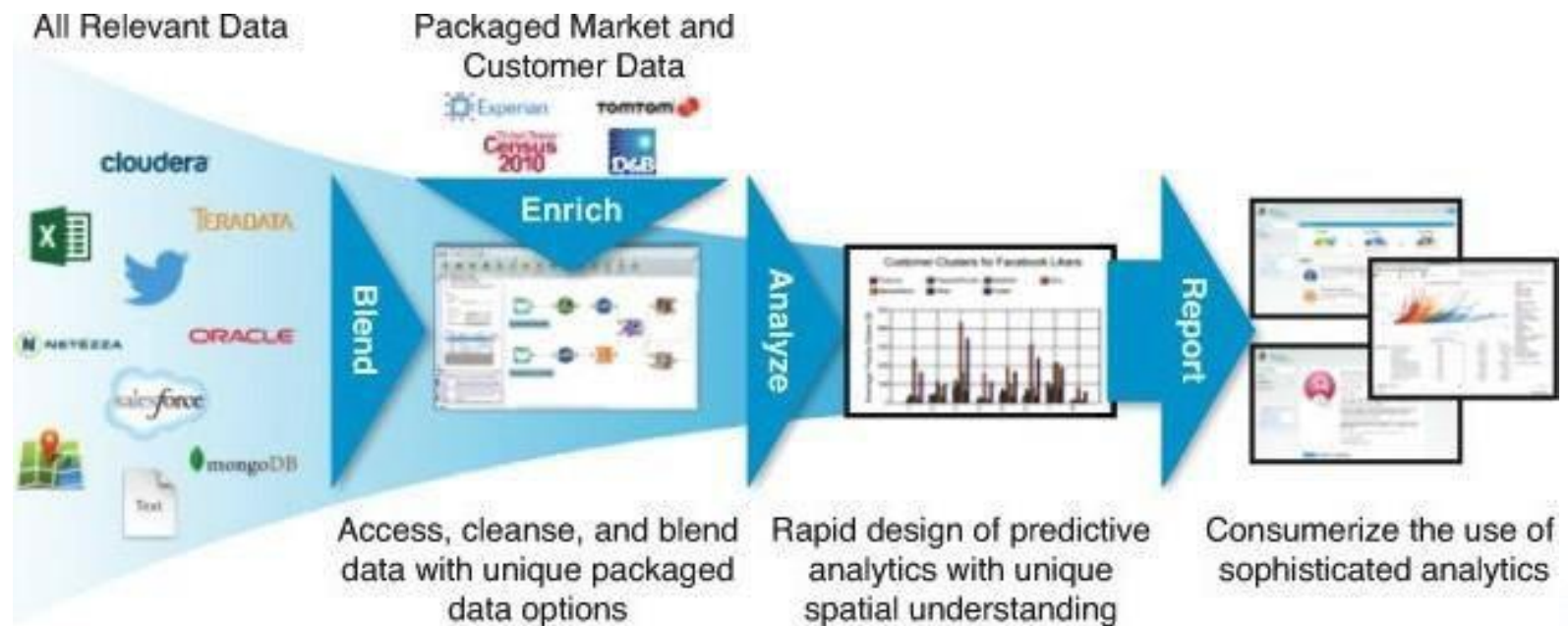
How ML and algorithms work together

- Algorithms handle deterministic logic: threshold alerts, rebalancing rules, benchmark comparisons, and risk-limit checks.
- Machine learning handles pattern detection: anomaly detection, predictive risk signals, and scoring which holdings are likely to deteriorate or outperform.
- The output is usually an alert, dashboard, or action recommendation for an investor, analyst, or portfolio manager.

Modern BI Process - Blend, Enrich, Analyze, and Report

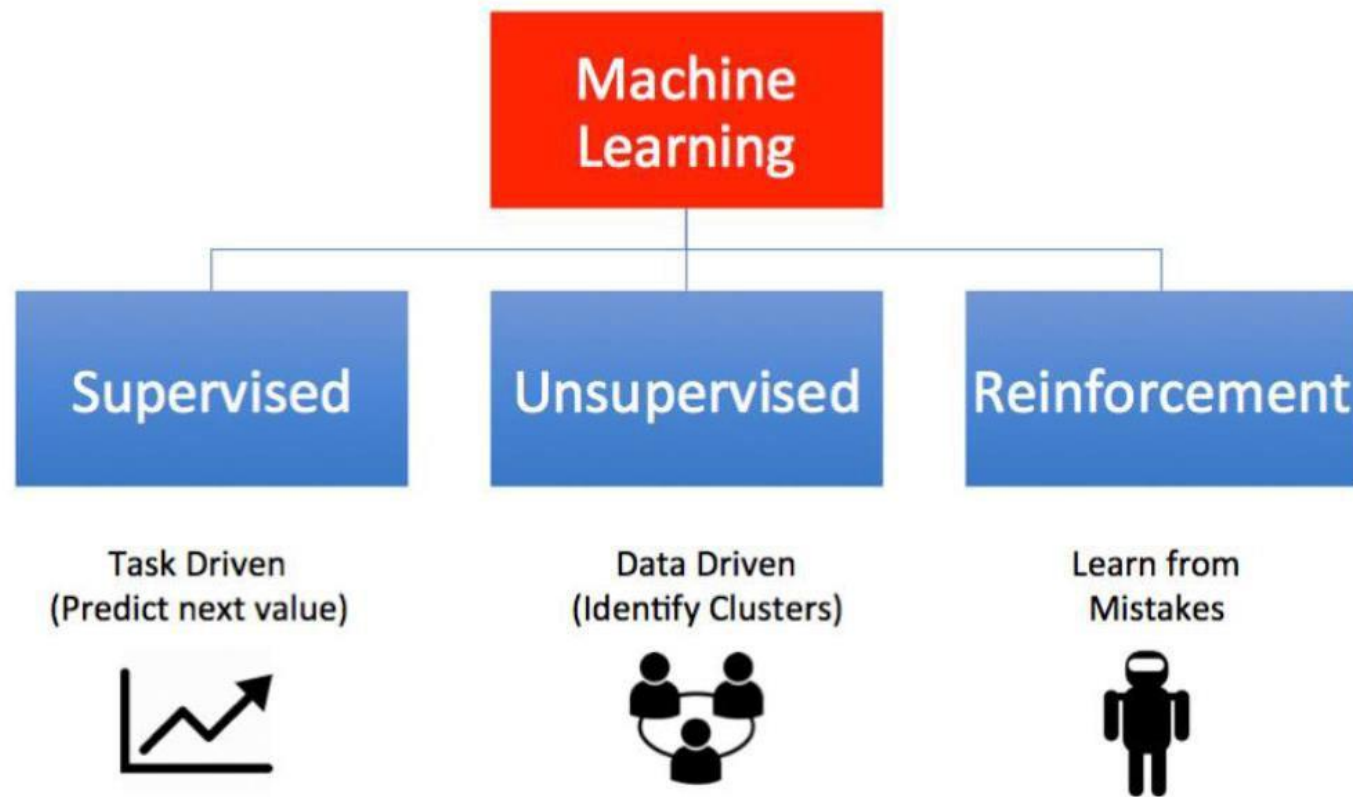
Predictive Analytics: By applying machine learning algorithms to historical project data, organizations can forecast outcomes more accurately and mitigate risks in project delivery

Business Intelligence: Automating the preparation of data for analytics tools allows organizations to gain actionable insights more rapidly, enhancing their decision-making capabilities.

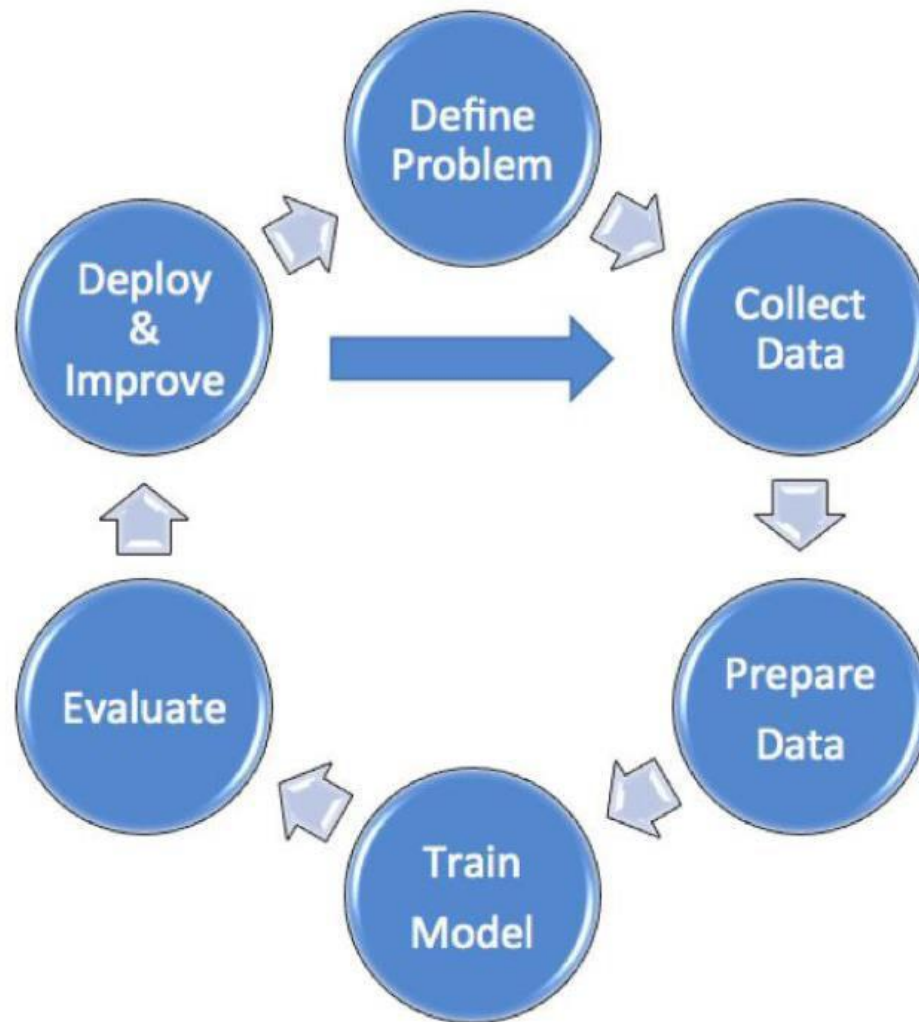


What is Business Analytics & Machine Learning?

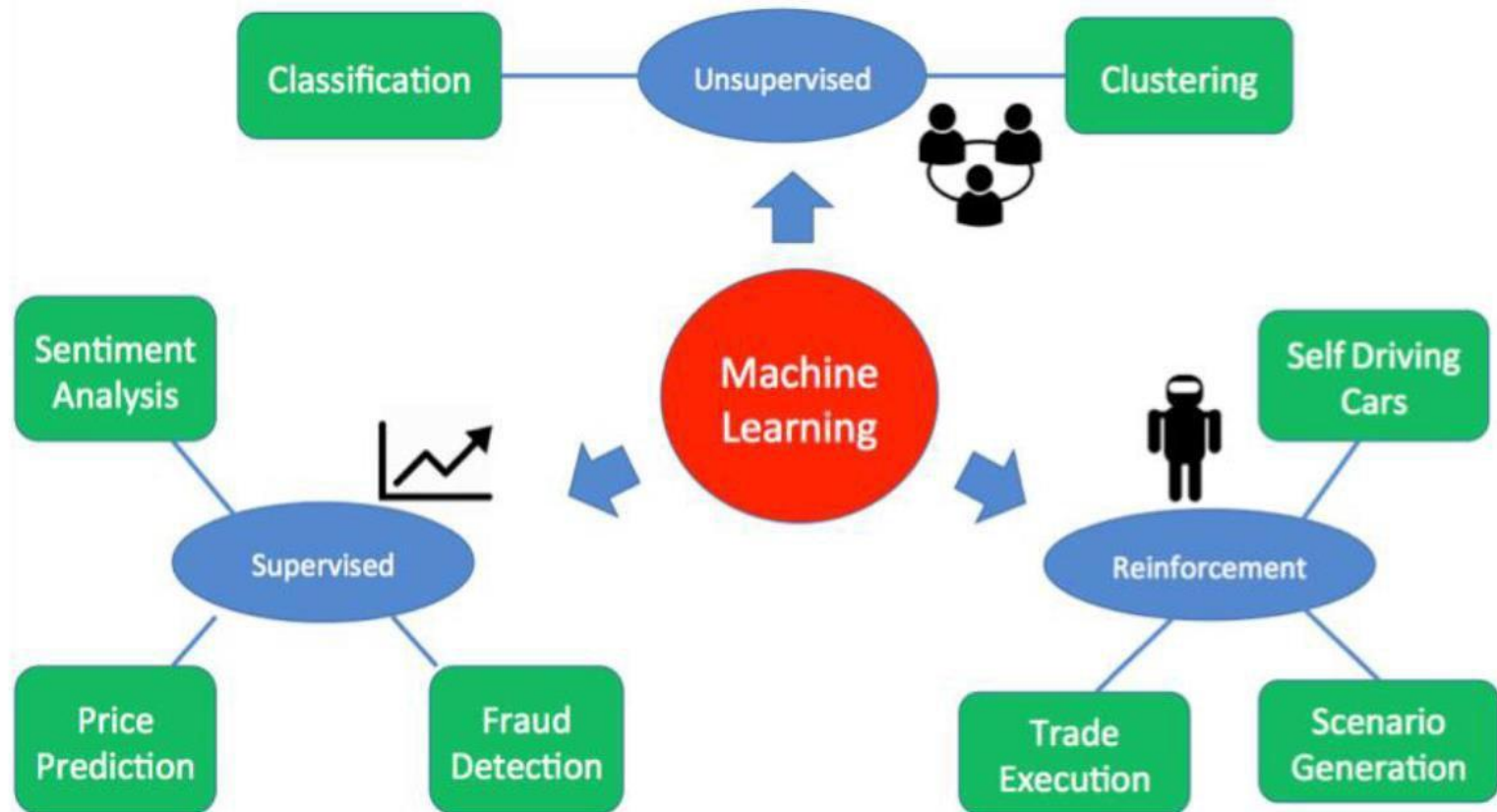
Types of Machine Learning



What is Business Analytics & Machine Learning?

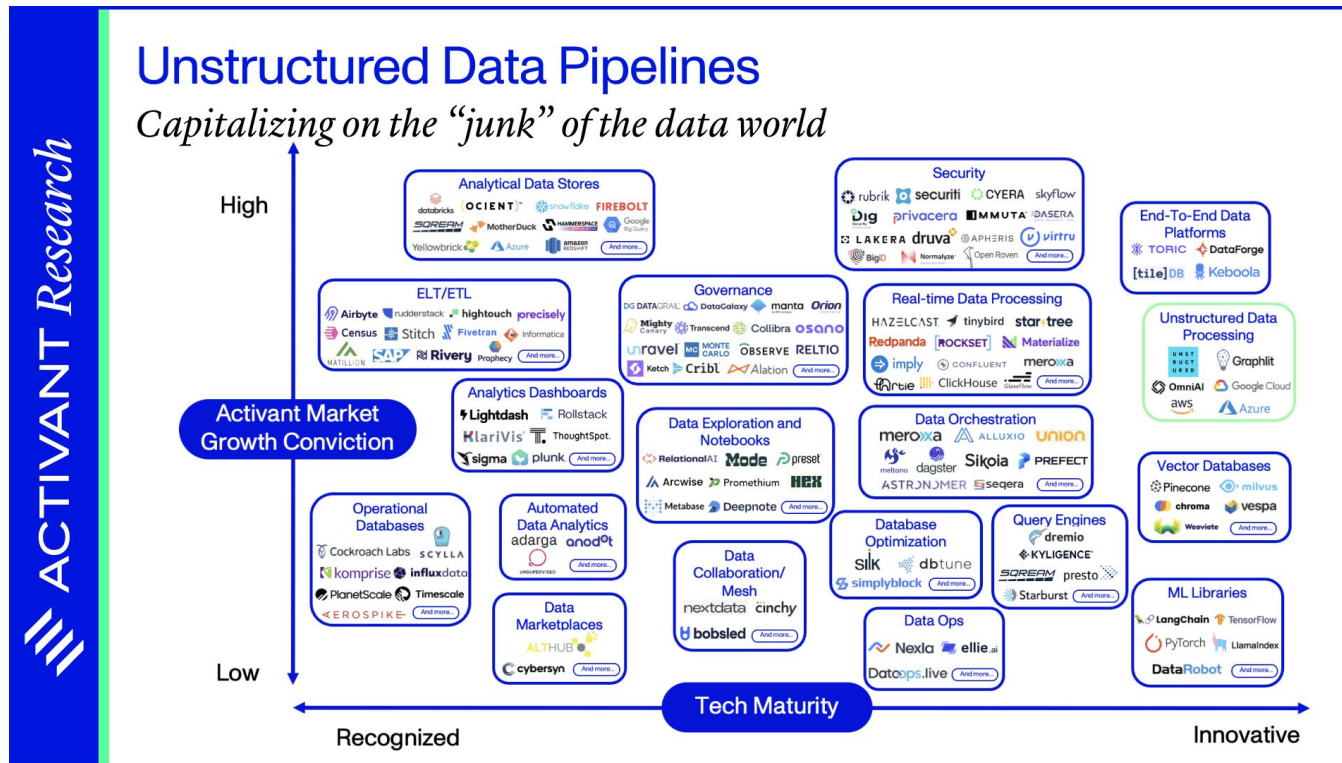


What is Business Analytics & Machine Learning?



IPA – Data Science

IPA's integration with data science is particularly impactful in several areas:
Unstructured Data Processing: IPA can convert unstructured data into structured formats suitable for analysis, enabling businesses to leverage insights from previously untapped data source.



Concept of Structured, Unstructured, Discrete, Continuous Data

Structured Data

- Fixed schema (rows/columns, e.g., SQL tables)
- Easy to query/search (names, prices)
- needs traditional ML models such as XGBoost, Random Forests techniques

Unstructured Data

- No fixed format (text, images, videos)
- needs deep learning vs NLP/ML techniques

Discrete Data

- Countable, distinct values (e.g., 0,1,2 items)
- needs traditional ML classification models like Logistic Regression or Decision Trees

Continuous Data

- Any value in range (e.g., 1.75m height)
- No gaps; line/histograms
- needs traditional ML regression models such as Linear Regression or Gradient Boosting

Data Processing & Analysis

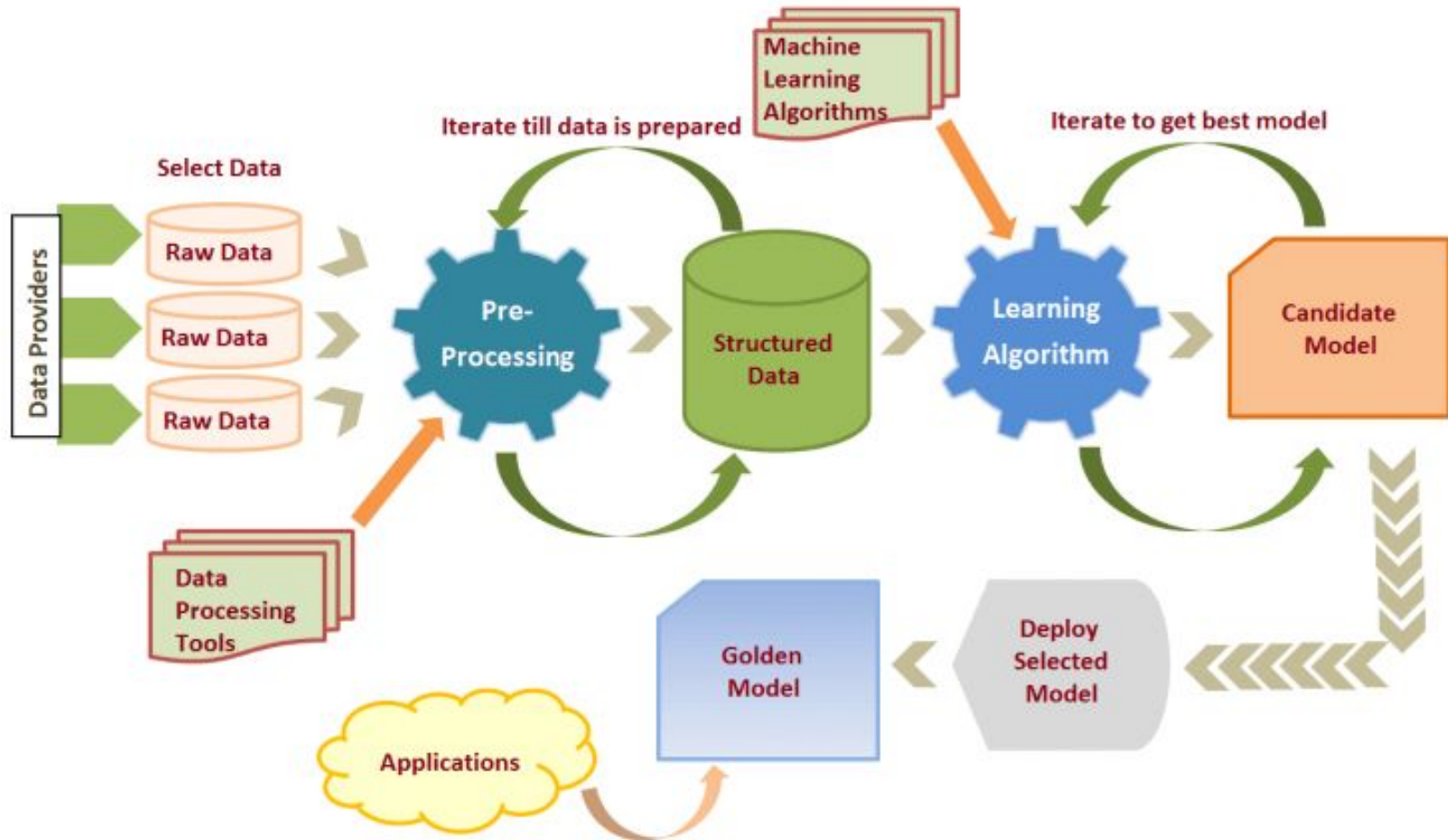
Practise 2

**Which Data Type is Structured/Unstructured,
Discret/Continuous?**

E-mails	
Digital Images	
Stock Market Logs	
Historical Gold Prices	
Credit Approval Records	
Social Media Friend Relationships	
Tweets & Trending Topics	
Sales Records	

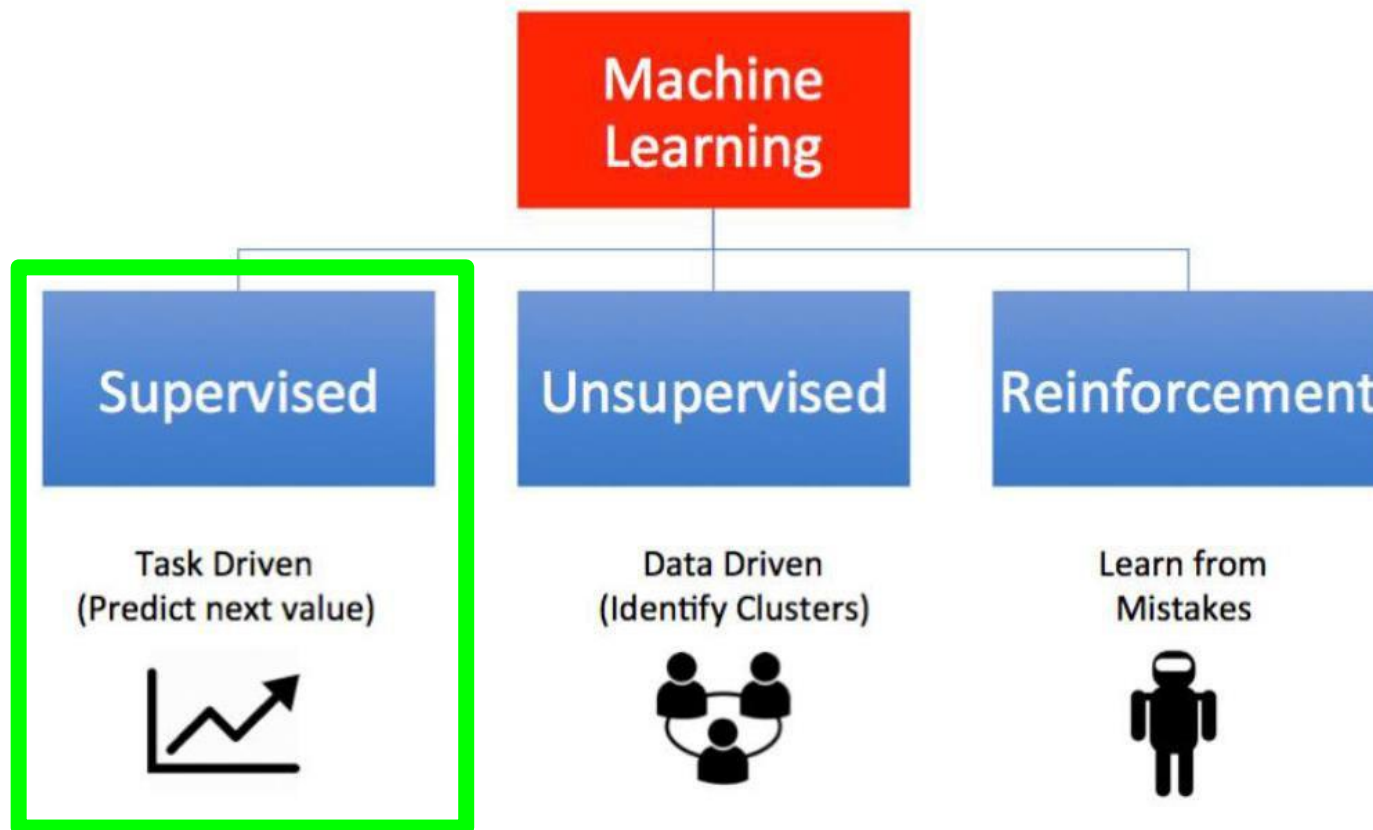
Source: Practical Data Analysis - Second Edition Hector Cuesta, Dr. Sampath Kumar OACKT Publishing BIRMINGHAM - MUMBAI

Applied Machine Learning Pipeline Overview



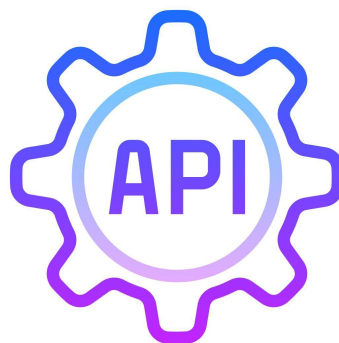
What is Business Analytics & Machine Learning?

Types of Machine Learning



Structured Data Collection Method

Main Data collection are get from Google Sheets, API and Web Scrapping



<https://www.kaggle.com/>



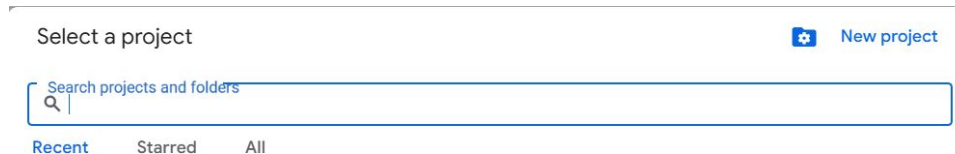
開放數據平台
DATA.GOV.HK

<https://data.gov.hk/en/>

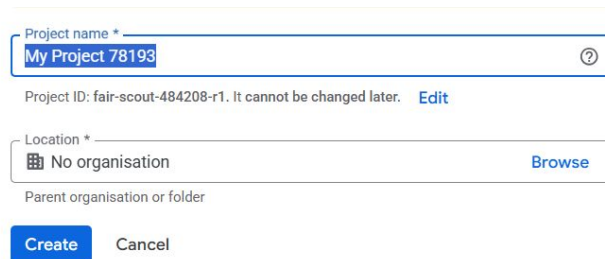
Data Collection - Reading Weather Data from Google Sheet

Enable the Google Sheets API with readpermission.

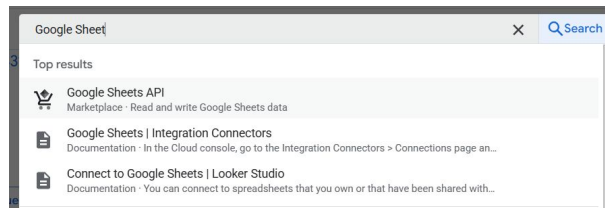
1. Open Browser and enter <https://console.cloud.google.com/>
2. Click “Select a project”
3. Click “New Project”



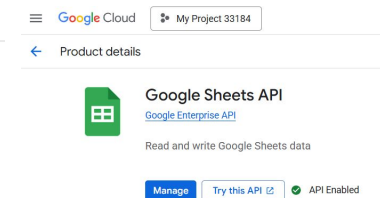
1. Enter information and click “Create”



1. Select your Project that create
2. in Search Bar, enter ‘Google Sheet’



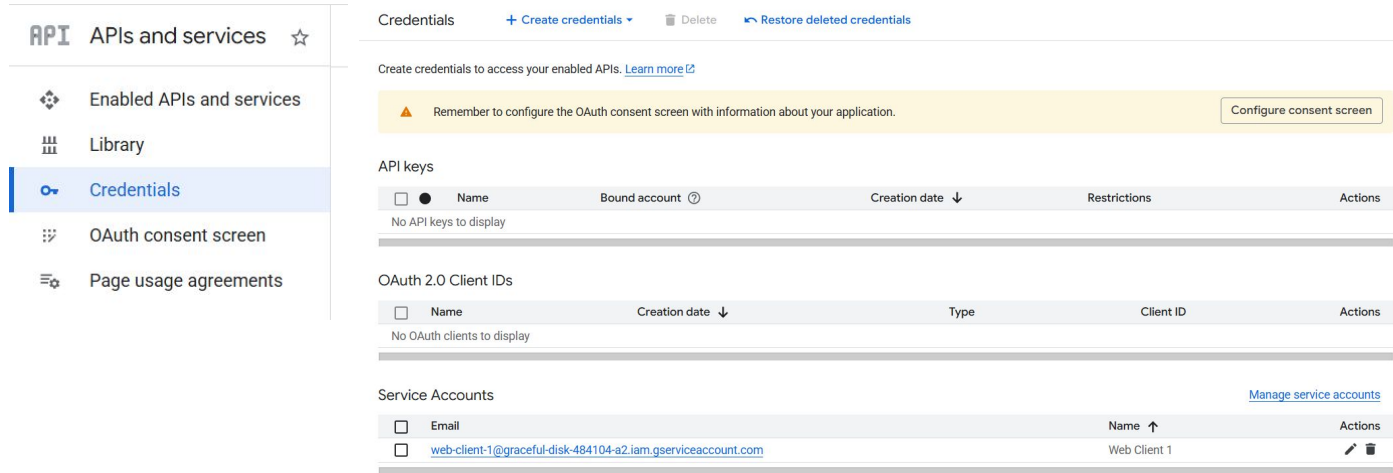
7. Select “Google Sheet API”
8. Click “Enable”



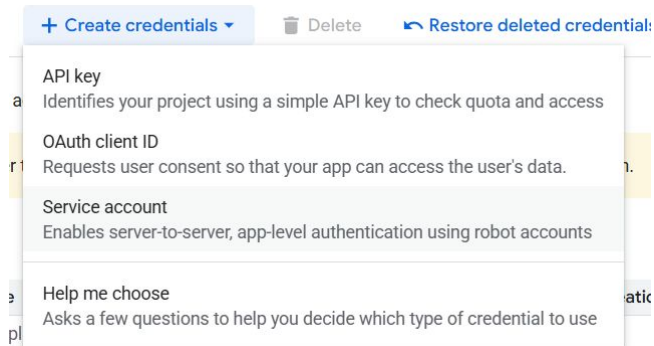
Data Collection - Reading Weather Data from Google Sheet

Credentials Setting

1. Click Credentials and whole screen is displayed



2. Click “Create credentials” and “Service Account”



Data Collection - Reading Weather Data from Google Sheet

Credentials Setting

3. In "Service account name", enter "Web Client 1" and Click "Create and Continue"

1 Create service account

Service account name
Web Client 1

Display name for this service account

Service account ID *
web-client-1-81

Email address: web-client-1-81@graceful-disk-484104-a2.iam.gserviceaccount.com

Service account description
Describe what this service account will do

Create and continue

4. In "Select a role", enter "Editor", click "Continue"

2 Create service account

Permissions (optional)

Grant this service account access to My Project 33184 so that it has permission to complete specific actions on the resources in your project.
[Learn more](#)

Role
Editor

IAM condition (optional) ⓘ
[+ Add IAM condition](#)

View, create, update and delete most Google Cloud resources. See the list of included permissions.

[+ Add another role](#)

[Help me choose roles](#)

Continue

5. Click "Done" to set credentials.

Service Accounts

[Manage service account](#)

☐	Email	Name ↑	Action
☐	web-client-1@graceful-disk-484104-a2.iam.gserviceaccount.com	Web Client 1	 

6. Copy the email like "web-client-1@graceful-disk-484104-a2.iam.gserviceaccount.com"

Data Collection - Reading Weather Data from Google Sheet

Credentials Setting

8. Click “Service Account Email”

nts / Service account: 112433876246315493925

← Web Client 1

Details Permissions Keys Metrics Logs Principals with access

Service account details

Name
Web Client 1 Save

Description Save

9. Click “Keys”

Details Permissions **Keys** Metrics Logs Principals with access

10. Click “Add Key” and “Create New Key”

Add key ▾

Create new key

Upload existing key

11. Enter “JSON”,

Create private key for 'Web Client 1'

Downloads a file that contains the private key. Store the file securely because this key cannot be recovered if lost.

Key type

☒ JSON
Recommended

☐ P12
For backward compatibility with code using the P12 format

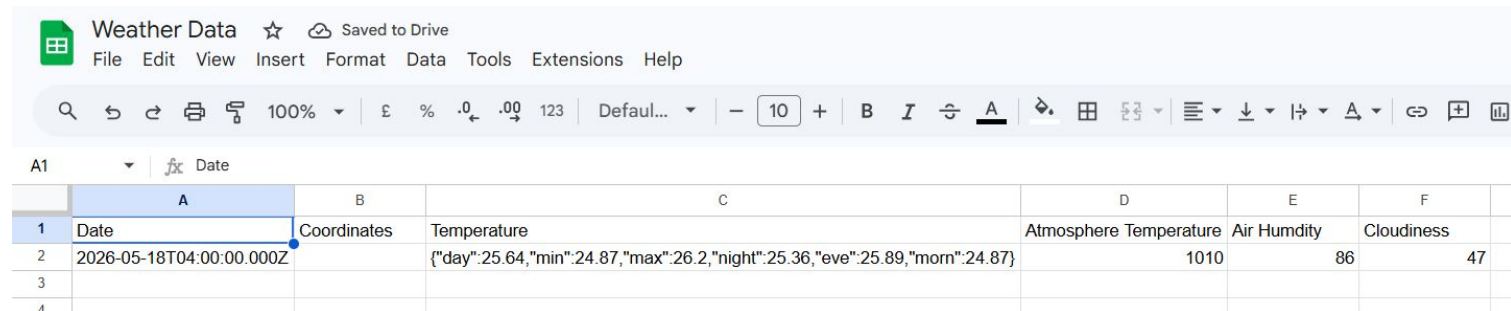
Cancel Create

12. A JSON Key is saved in your local PC

Data Collection - Reading Weather Data from Google Sheet

Google Sheet Settings

1. Open previous weather data google sheet



The screenshot shows a Google Sheet titled "Weather Data" with a menu bar (File, Edit, View, Insert, Format, Data, Tools, Extensions, Help) and a toolbar. The spreadsheet has columns A through F. Column A is labeled "Date", B is "Coordinates", C is "Temperature", D is "Atmosphere Temperature", E is "Air Humidity", and F is "Cloudiness". The data for the date "2026-05-18T04:00:00.000Z" is as follows:

A	B	C	D	E	F
Date	Coordinates	Temperature	Atmosphere Temperature	Air Humidity	Cloudiness
2026-05-18T04:00:00.000Z		{"day":25.64,"min":24.87,"max":26.2,"night":25.36,"eve":25.89,"morn":24.87}	1010	86	47
3					
4					

1. Click "File", Share, "Share with Others"
2. Enter "web-client-1@graceful-disk-484104-a2.iam.gserviceaccount.com"
3. Make sure it is in editor, click "Send"
4. From the google sheet website url like
<https://docs.google.com/spreadsheets/d/1BVdJkKIZQPTdGrzL-h7ymTYT6unhYKgJ68vqqUYsjP0/edit?gid=0#gid=0>
5. Copy sheet_id "**1BVdJkKIZQPTdGrzL-h7ymTYT6unhYKgJ68vqqUYsjP0**", range_name "**Sheet1!A1:F2**" in notepad

Data Collection - Reading Weather Data from Google Sheet

Python Coding

1. Open Anaconda Prompt, enter “pip install google-api-core google-auth google-api-python-client google-auth-http lib2 google-auth-oauthlib” to install google library

```
(base) C:\Users\Eddie Chow>pip install google-api-core google-auth google-api-python-client
Defaulting to user installation because normal site-packages is not writeable
WARNING: Ignoring invalid distribution -ransformers (c:\users\eddie chow\appdata\roaming\python\python39\site-packages)
WARNING: Ignoring invalid distribution -ransformers (c:\users\eddie chow\appdata\roaming\python\python39\site-packages)
Requirement already satisfied: google-api-core in c:\users\eddie chow\appdata\roaming\python\python39\site-packages (1.34.1)
Requirement already satisfied: google-auth in c:\users\eddie chow\appdata\roaming\python\python39\site-packages (1.35.0)
Requirement already satisfied: google-api-python-client in c:\users\eddie chow\appdata\roaming\python\python39\site-packages (2.184.0)
Requirement already satisfied: googleapis-common-protos<2.0dev,>=1.56.2 in c:\users\eddie chow\appdata\roaming\python\python39\site-packages (from google-api-core) (1.70.0)
Requirement already satisfied: protobuf!=3.20.0,!<3.20.1,!>4.21.0,!>4.21.1,!>4.21.2,!>4.21.3,!>4.21.4,!>4.21.5,<4.0.0dev,>=3.19.5 in c:\users\eddie chow\appdata\roaming\python\python39\site-packages (from google-api-core) (3.20.3)
Requirement already satisfied: requests<3.0.0dev,>=2.18.0 in c:\users\eddie chow\appdata\roaming\python\python39\site-packages (from google-api-core) (2.32.5)
Requirement already satisfied: cachetools<5.0,>=2.0.0 in c:\programdata\anaconda3\lib\site-packages (from google-auth) (4.2.2)
Requirement already satisfied: pyasn1-modules<=0.2.1 in c:\programdata\anaconda3\lib\site-packages (from google-auth) (0.2.8)
Requirement already satisfied: setuptools<=40.3.0 in c:\programdata\anaconda3\lib\site-packages (from google-auth) (61.2.0)
Requirement already satisfied: six<=1.9.0 in c:\programdata\anaconda3\lib\site-packages (from google-auth) (1.16.0)
Requirement already satisfied: rsa<5,>=3.1.4 in c:\programdata\anaconda3\lib\site-packages (from google-auth) (4.7.2)
Requirement already satisfied: http lib2<1.0.0,>=0.19.0 in c:\users\eddie chow\appdata\roaming\python\python39\site-packages (from google-api-python-client) (0.31.0)
Requirement already satisfied: google-auth-http lib2<1.0.0,>=0.2.0 in c:\users\eddie chow\appdata\roaming\python\python39\site-packages (from google-api-python-client) (0.2.0)
```

Data Collection - Reading Weather Data from Google Sheet

Python Coding

2. Open jupyter notebook
3. Type following code

Import Library

```
import os
import csv
from google.oauth2.service_account import Credentials
from googleapiclient.discovery import build
```

```
SCOPES = ['https://www.googleapis.com/auth/spreadsheets']
SERVICES_ACCOUNT_FILE = "graceful-disk-484104-a2-152ebd9c692e.json"
```

```
credentials = Credentials.from_service_account_file(SERVICES_ACCOUNT_FILE, scopes=SCOPES)
service = build('sheets', 'v4', credentials=credentials)
```

```
sheet_id = '1BVdJkKIZQPTdGrzL-h7ymTYT6unhYKgJ68vqqUYsjP0'
range_name = "Sheet1!A1:F2" # Specify the range with the sheet name
```

Data Collection - Reading Weather Data from Google Sheet

Retrieve the data

```
result = service.spreadsheets().values().get(sheetId=sheet_id, range=range_name).execute()
```

```
values = result.get('values', [])
```

```
print(values)
```

```
print(values)
[['Date', 'Coordinates', 'Temperature', 'Atmosphere Temperature', 'Air Humidity', 'Cloudiness'], ['2026-05-18T04:00:00.000Z', '', '{"day":25.64,"min":24.87,"max":26.2,"night":25.36,"eve":25.89,"morn":24.87}','1010', '86', '47']]
```

if not values:

```
    print('No data found.')
```

else:

Save to CSV file

```
with open('weatherforecastdata.csv', 'w', newline="", encoding='utf-8') as csv_file:
```

```
    writer = csv.writer(csv_file)
```

```
    writer.writerows(values) # Write all the rows
```

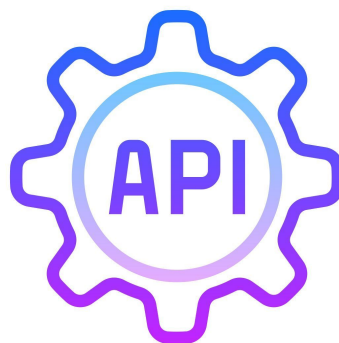
```
print('Data has been saved to weatherforecastdata.csv')
```

Structured Data Collection Method

Main Data collection are get from Google Sheets, API and Web Scrapping



Google Sheets



kaggle

<https://www.kaggle.com/>



開放數據平台
DATA.GOV.HK

<https://data.gov.hk/en/>

Definition

- a web crawler is a computer program that browses the World Wide Web in a methodical, automated manner. (Wikipedia)

Utilities:

- Gather pages from the Web.
- Support a search engine, perform data mining and so on.

Object:

- Text, video, image and so on.
- Link structure

Web Features of a Crawler

Should provide:

- Distributed
- Scalable
- Performance and efficiency
- Quality
- Freshness
- Extensible

Some scraping knowledge

1. HTTP: the communication protocol
2. HTML: the language in which web pages are defined (<https://www.w3schools.com/html/>)
3. JS: javascript (code executing in the browser) (<https://www.w3schools.com/js/default.asp>)
4. CSS: style sheets, how web pages are styled. Important, but does not contain data. (<https://www.w3schools.com/css/default.asp>)
5. JPG, PNG, BMP: images, usually not interesting
6. CSV / TXT / JSON / XML: data, interesting !!!

Obtaining Data – Web Scraping

Why scraping the web?

- Vast source of information; can combine with multiple datasets
- Companies have not provided APIs
- Automate tasks
- Keep up with sites / real-time data
- Fun!

Obtaining Data – Web Scraping

HTML

- Tags are denoted by angled brackets
- Almost all tags are in pairs e.g.,
`<p>Hello</p>`
- Some tags do not have a closing tag e.g.,
`<br/>`

Example

```
<!DOCTYPE html>
<html>
  <head>
    <title>Ttle</title>
  </head>
  <body>
    <h1>Body Title</h1>
    <p>Body Content</p>
  </body>
</html>
```

Obtaining Data – Web Scrapping

HTML

- `<html>`, indicates the start of an html page
- `<body>`, contains the items on the actual webpage (text, links, images, etc)
- `<p>`, the paragraph tag. Can contain text and links
- `<a>`, the link tag. Contains a link url, and possibly a description of the link
- `<input>`, a form input tag. Used for text boxes, and other user input
- `<form>`, a form start tag, to indicate the start of a form
- `<img>`, an image tag containing the link to an image

Obtaining Data – Web Scraping

How to Web scrape:

1. Get the webpage content

- Requests (Python library) gets a webpage for you

2. Parse the webpage content

- (e.g., find all the text or all the links on a page)
- BeautifulSoup (Python library) helps you parse the webpage.
- Documentation:

<http://crummy.com/software/BeautifulSoup>

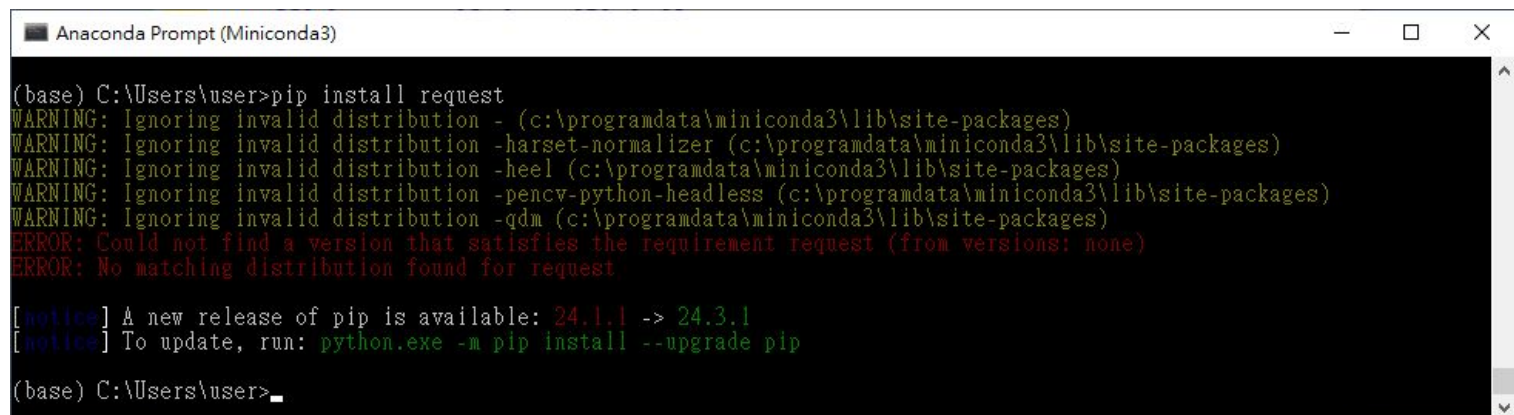
Obtaining Data – Web Scraping

1. Open tools “Anaconda Prompt” which will direct you to the command prompt.



```
Anaconda Prompt (Miniconda3)
(base) C:\Users\user>
```

2. Install python package requests by entering “pip install requests”



```
Anaconda Prompt (Miniconda3)
(base) C:\Users\user>pip install request
WARNING: Ignoring invalid distribution - (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -harset-normalizer (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -heel (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -pencv-python-headless (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -qdm (c:\programdata\miniconda3\lib\site-packages)
ERROR: Could not find a version that satisfies the requirement request (from versions: none)
ERROR: No matching distribution found for request

[notice] A new release of pip is available: 24.1.1 -> 24.3.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(base) C:\Users\user>_
```

Obtaining Data – Web Scraping

3. Install python package beautifulsoup4 by entering “pip install beautifulsoup4”

```
Anaconda Prompt (Miniconda3)

(base) C:\Users\user>pip install beautifulsoup4
WARNING: Ignoring invalid distribution - (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -harset-normalizer (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -heel (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -pencv-python-headless (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -qdm (c:\programdata\miniconda3\lib\site-packages)
Requirement already satisfied: beautifulsoup4 in c:\programdata\miniconda3\lib\site-packages (4.12.3)
Requirement already satisfied: soupsieve>1.2 in c:\programdata\miniconda3\lib\site-packages (from beautifulsoup4) (2.3.2.post1)
WARNING: Ignoring invalid distribution - (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -harset-normalizer (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -heel (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -pencv-python-headless (c:\programdata\miniconda3\lib\site-packages)
```

4. Enter “python” in the command

```
Anaconda Prompt (Miniconda3) - python

(base) C:\Users\user>python
Python 3.9.12 (main, Apr  4 2022, 05:22:27) [MSC v.1916 64 bit (AMD64)] :: Anaconda, Inc. on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> _
```

Obtaining Data – Web Scraping

1. Get the webpage content

Requests (Python library) gets a webpage for you

```
import requests
```

```
url = "https://www.nytimes.com/international/"
```

```
page = requests.get(url)
```

```
print(page.status_code)
```

```
print(page.content)
```

Returns the content of the response, in bytes.

Obtaining Data – Web Scrapping

2. Parse the webpage content

BeautifulSoup (Python library) helps you parse a webpage

```
from bs4 import BeautifulSoup
```

```
soup = BeautifulSoup(page.content,  
"html.parser")
```

```
soup.title
```

```
soup.title.text
```

Returns the text part of the title tag.

e.g.,

The New York Times - Breaking News

Web Scraping Practise 1

Open Browser and enter
<https://realpython.github.io/fake-jobs/>

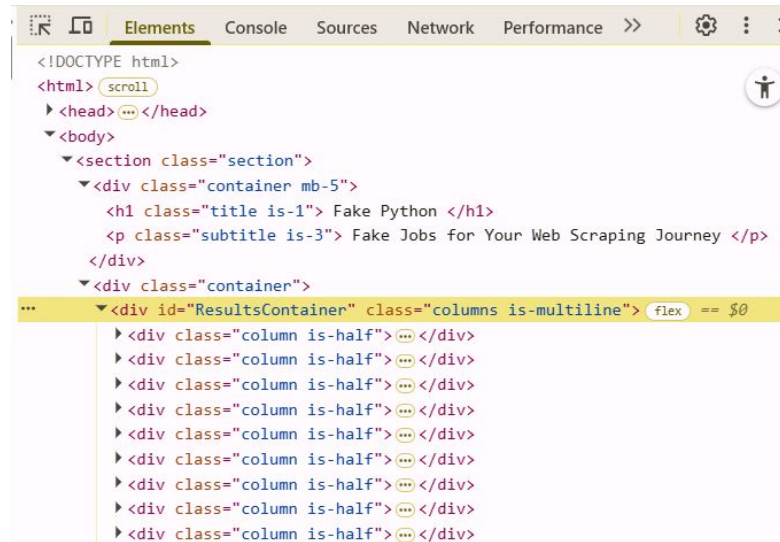
Fake Python

Fake Jobs for Your Web Scraping Journey

 Senior Python Developer Payne, Roberts and Davis Stewartbury, AA 2021-04-08 Learn Apply	 Energy engineer Vasquez-Davidson Christopherville, AA 2021-04-08 Learn Apply
 Legal executive Jackson, Chambers and Levy Port Ericaburgh, AA 2021-04-08 Learn Apply	 Fitness centre manager Savage-Bradley East Seanview, AP 2021-04-08 Learn Apply

Web Scraping Practise 1

Right Click and Select “Inspect”



Open jupyter notebook and enter following code

```
import requests
from bs4 import BeautifulSoup
```

```
URL = "https://realpython.github.io/fake-jobs/"
page = requests.get(URL)
```

Web Scraping Practise 1

```
soup = BeautifulSoup(page.content, "html.parser")
```

```
results = soup.find(id="ResultsContainer")  
print(results.prettify())
```

```
>>> print(results.prettify())  
<div class="columns is-multiline" id="ResultsContainer">  
  <div class="column is-half">  
    <div class="card">  
      <div class="card-content">  
        <div class="media">  
          <div class="media-left">  
            <figure class="image is-48x48">  
                
            </figure>  
          </div>  
          <div class="media-content">  
            <h2 class="title is-5">  
              Senior Python Developer  
            </h2>  
            <h3 class="subtitle is-6 company">  
              Payne, Roberts and Davis  
            </h3>  
          </div>  
        </div>  
      </div>  
    </div>  
  </div>
```

Web Scraping Practise 1

Find Elements by HTML Class Name

You've seen that every job posting is wrapped in a `<div>` element with the class `card-content`. Now you can work with your new object called `results` and select only the job postings in it. These are, after all, the parts of the HTML that you're interested in! You can pick out all job cards in a single line of code:

Python

```
>>> job_cards = results.find_all("div", class_="card-content")
```

Here, you call `.find_all()` on `results`, which is a BeautifulSoup object. It returns an `iterable` containing all the HTML for all the job listings displayed on that page.

Take a look at all of them:

Python

```
>>> for job_card in job_cards:
...     print(job_card, end="\n" * 2)
...
```

```
<div class="card-content">
<div class="media">
<div class="media-left">
<figure class="image is-48x48">

<h2 class="title is-5">Senior Python Developer</h2>
<h3 class="subtitle is-6 company">Payne, Roberts and Davis</h3>
</div>
</div>
<div class="content">
<p class="location">
    Stewartbury, AA
</p>
<p class="is-small has-text-grey">
<time datetime="2021-04-08">2021-04-08</time>
```

Extract title Content

```
>>> for job_card in job_cards:
...     title_element = job_card.find("h2", class_="title")
...     print(title_element)
...
<h2 class="title is-5">Senior Python Developer</h2>
<h2 class="title is-5">Energy engineer</h2>
<h2 class="title is-5">Legal executive</h2>
<h2 class="title is-5">Fitness centre manager</h2>
<h2 class="title is-5">Product manager</h2>
<h2 class="title is-5">Medical technical officer</h2>
<h2 class="title is-5">Physiological scientist</h2>
<h2 class="title is-5">Textile designer</h2>
<h2 class="title is-5">Television floor manager</h2>
<h2 class="title is-5">Waste management officer</h2>
<h2 class="title is-5">Software Engineer (Python)</h2>
<h2 class="title is-5">Interpreter</h2>
<h2 class="title is-5">Architect</h2>
<h2 class="title is-5">Meteorologist</h2>
```

Extract Content of title, company, location

Python

```
>>> for job_card in job_cards:
...     title_element = job_card.find("h2", class_="title")
...     company_element = job_card.find("h3", class_="company")
...     location_element = job_card.find("p", class_="location")
...     print(title_element)
...     print(company_element)
...     print(location_element)
...     print()
...
<h2 class="title is-5">Senior Python Developer</h2>
<h3 class="subtitle is-6 company">Payne, Roberts and Davis</h3>
<p class="location">
    Stewartbury, AA
</p>

<!-- ... -->
```

Full Article : <https://realpython.com/beautiful-soup-web-scraper-python/>

Web Scraping Practise 2

Open Browser and enter
<https://www.hk01.com/>

Right Click and Select "Inspect"



Select the Content that you want to extract

Web Scraping Practise 2

Right Click and click "Edit As HTML"
Copy part of the code until `</div>`

The screenshot shows a web browser window with a news article from HKU SPACE. The article is titled "李嘉誠治肝癌新儀器料歸三私院 陸志聰：公院引入須顧人手及預算" (Li Ka-shing's liver cancer treatment device may be used by private hospitals, Dr. Louis Loo: public hospitals need to consider manpower and budget). The article is dated 2 hours ago. Below the article, there are several promotional banners for PBX systems, including one that says "Reducing TCO Total Cost of Ownership by 44%" and another that says "Plug & Play Installation in under 4 Hours." A right-click context menu is open over the article, showing options like "Ask AI", "Add attribute", "Edit attribute", "Edit as HTML", "Duplicate element", "Delete element", "Cut", "Copy", "Paste", "Hide element", "Force state", "Break on", "Expand recursively", "Collapse children", "Capture node screenshot", "Scroll into view", "Focus", "Badge settings", and "Store as global variable". The "Edit as HTML" option is highlighted.

Web Scraping Practise 2

Open Perplexity at <https://www.perplexity.ai/>

Enter prompt "write python beautifulsoup to extract data from website url <https://www.hk01.com/> with html code {htmlcode} "

Where {htmlcode} = "<div class='page__layout-navbar' data-testid='page-layout-navbar'><div class='flex flex-none items-center md:hidden'><button data-testid='hamburger-menu-button' class='box-border w-6 px-0 py-3 m-0 transition-all duration-[0.6s] ease-[cubic-bezier(0.19,1,0.22,1)] [transform:translateZ(0)] border-none outline-none rounded-none cursor-pointer shadow-none bg-transparent focus:shadow-none focus:outline-none' aria-label='mobile-menu'></button></div>"

Web Scraping Practise 2

In original perplexity prompt, type the following prompt
"Add code that save it into csv file"

A	
1	text,url,class
2	./,
3	港聞,/zone/1/%E6%B8%AF%E8%81%9E,
4	兩會焦點,/issue/10355/%E5%85%A9%E6%9C%832025-%E5%9C%8B%E9%9A%9B%E5%B1%80%
5	娛樂,/zone/2/%E5%A8%9B%E6%A8%82,
6	最平酒店,https://clk.omgt3.com/?AID=2200313&PID=55029&uid=content&uid2=channel_旅遊_editor
7	國際,/zone/4/%E5%9C%8B%E9%9A%9B,
8	即時,/latest,
9	熱榜,/hot,
10	生活,/zone/8/%E7%94%9F%E6%B4%BB,
11	科技,/zone/11/%E7%A7%91%E6%8A%80%E7%8E%A9%E7%89%A9,
12	中國,/zone/5/%E4%B8%AD%E5%9C%8B,
13	體育,/zone/3/%E9%AB%94%E8%82%B2,
14	01深圳,/zone/25/01%E6%B7%B1%E5%9C%B3,
15	經濟,/zone/14/%E7%B6%93%E6%BF%9F,
16	觀點,/zone/12/%E8%A7%80%E9%BB%9E,
17	健康,/zone/24/%E5%81%A5%E5%BA%B7,
18	好吃玩飛,/zone/19/%E5%A5%BD%E9%A3%9F%E7%8E%A9%E9%A3%9B,
19	女生,/zone/6/%E5%A5%B3%E7%94%9F,
20	熱話,/zone/7/%E7%86%B1%E8%A9%B1,
21	藝文格物,/zone/9/%E8%97%9D%E6%96%87%E6%A0%BC%E7%89%A9,
22	社區,/zone/10/%E7%A4%BE%E5%8D%80,
23	教育,/zone/23/%E6%95%99%E8%82%B2,
24	簡,/,"font-light, pointer-events-none, !text-light-n5"
25	
26	
27	Mobile Menu Details:
28	Exists: True

Web Scraping Practise 2

Type following code to extract data and save in csv file

```
import requests
from bs4 import BeautifulSoup
from urllib.parse import urljoin
import pandas as pd

url = "https://www.hk01.com/"

resp = requests.get(url, timeout=20)

soup = BeautifulSoup(resp.text, "html.parser")

navbar = soup.select_one('div.page__layout-navbar[data-testid="page-layout-navbar"]')
links = []

if navbar:
    for a in navbar.select('div[data-testid="top-nav-option"] a[href]'):
        text = a.get_text(" ", strip=True)
        href = urljoin(url, a["href"])
        links.append({"text": text, "href": href})

for item in links:
    print(item["text"], item["href"])

df = pd.DataFrame(links)
df.to_csv("hk01_navbar_links.csv", index=False, encoding="utf-8-sig")

print(df)
```

How To Do Data Mining: Machine Learning Techniques

Features / Attributes

1. District
2. Floor

Outcome

1. Price Per Square

<https://www.28hse.com/en/rent/apartment/property-3496859>

Floor zone

Middle Floor

Saleable Area

807 ft²

Unit Price: @46.8

<https://www.28hse.com/en/rent/apartment>



♥BEST CHOICE♥ \$37800 Front line sea view...

Tsuen Wan Ocean Pride | Unit C, Mid Floor, Tower 1, Ocean Supreme, Middle Floor 12 minutes ago posted

Saleable Area: 807 ft² @46.8

美聯物業

Apartment Good view Sea view Elegant

Air conditioner Washing Machine Water Heaters Refrigerator

Induction Hob Cooker Hood

Lease HKD\$37,800



The largest room with sea view•Rarely availa...

Hung Hom Upper East | High Floor, Tower 1A 2 hours ago posted

Saleable Area: 375 ft² @49.3

Huicheng Property Company Limited

1 Bedroom, 1 Bathroom Western Style Bldg

Stand-alone Building Good view Sea view Elegant Microwave Oven

Air conditioner Washing Machine Water Heaters

Lease HKD\$18,500



3-bedroom Western-style building with open ...

Kwun Tong Hamden Court | High Floor, Block A 3 hours ago posted

Saleable Area: 481 ft² @35.8

Kam Lung Property Agency Co.

3 Bedrooms, 1 Bathroom Western Style Bldg Good view

Elegant

Lease HKD\$17,200

How To Do Data Mining: Machine Learning Techniques

Artificial Intelligence

Artificial Intelligence (AI) is the science of making things smart. Can be defined as:

“Human intelligence exhibited by machines”

A broad term for getting computers to perform human tasks. The scope of AI is disputed and constantly changing over time. Let's go deeper...

How To Do Data Mining: Machine Learning Techniques

Machine Learning

An approach to achieve artificial intelligence through 'software' systems that can *learn* from *experience* to find *patterns* in a *set of data*.

Write a computer program
with **explicit rules** to follow

```
if email contains V!agrå  
    then mark is-spam;  
if email contains ...  
if email contains ...
```

Traditional Programming

Write a computer program
to **learn from examples**

```
try to classify some emails;  
change self to reduce errors;  
repeat;
```

Machine Learning Programs

Business Process Automation with Python

- **API endpoint used**

- URL:

<https://data.weather.gov.hk/weatherAPI/opendata/weather.php?dataType=flw&lang=tc>

- Purpose: retrieves **local weather forecast** (flw) in **Traditional Chinese** (lang=tc)

- **Sample response structure**

- Example `weather_data` object:

```
weather_data = {  
    "generalSituation": "Mainly cloudy with a few showers",  
    "temperature": {  
        "value": 28,  
        "unit": "C"  
    },  
    "humidity": 85,  
    "wind": "Moderate southerly winds",  
    "updateTime": "2026-05-20 12:00"  
}  
print(weather_data["temperature"])  
print(weather_data["temperature"]["value"])  
print(weather_data["humidity"])
```

```
>>> print(weather_data["temperature"])  
{'value': 28, 'unit': 'C'}  
>>> print(weather_data["temperature"]['value'])  
28  
>>> print(weather_data["humidity"])  
85
```


Sequential Data Types

- **List**

- “Mutable” object (something you can change)
- Representation

```
▶ school_list = ['hku', 'cuhk', 'hkust']  
print(school_list)  
print(type(school_list))  
  
□ ['hku', 'cuhk', 'hkust']  
  <class 'list'>
```

- Indexing

```
▶ print(school_list[0])      # First 1 item  
  print(school_list[0:2])   # First 2 items  
  print(school_list[-1])    # Last one  
  
□ hku  
  ['hku', 'cuhk']  
  polyu
```

Sequential Data Types

- List
 - Adding items

```
• school_list = ['hku', 'cuhk', 'hkust'] + ['cityu', 'polyu']  
  school_list
```

```
□ ['hku', 'cuhk', 'hkust', 'cityu', 'polyu']
```

```
• school_list = ['hku', 'cuhk', 'hkust', 'cityu']  
  school_list.append('polyu')  
  school_list
```

```
□ ['hku', 'cuhk', 'hkust', 'cityu', 'polyu']
```

Sequential Data Types

- **List**

- Removing an item
 - By index → `pop()`

```
▶ school_list.pop(1)  
school_list  
☞ ['hku', 'hkust', 'cityu', 'polyu']
```

- By element → `remove()`

```
▶ school_list.remove('cityu')  
school_list  
☞ ['hku', 'hkust', 'polyu']
```

Business Process Automation with Python

Sequential Data Types

- Common operations
 - Membership Check → “in”

```
# Membership check
school_list = ['hku', 'cuhk', 'hkust', 'cityu', 'polyu']
print('hku' in school_list)
print('hkbu' in school_list)
```

```
True
False
```

Business Process Automation with Python

Sequential Data Types

- Common operations
 - Ordering → `max()` / `min()` / `reversed()` / `sorted()`

```
▶ number_list = [1, 5, 6, 2, 3]
  print(max(number_list))      # Max
  print(min(number_list))      # Min
  print(list(reversed(number_list))) # Reverse order
  print(sorted(number_list))    # Sort by ascending order
```



```
↳ 6
   1
   [3, 2, 6, 5, 1]
   [1, 2, 3, 5, 6]
```


Business Process Automation with Python

Sequential Data Types

- String (again)
 - string = sequential!



```
# Index:      0123456789
school_name = 'HKU School of Professional and Continuing Education'

print(len(school_name))           # There is a total of 51 letters
print(school_name.index('c'))     # First c is the 6th letter (counting 1 space)
print(school_name.count('o'))     # There are 7 O's
print('HKU' in school_name)      # Substring matching
print(school_name[4:10])         # Substring extraction
```

```
51
5
7
True
School
```


Business Process Automation with Python

Sequential Data Types

- String (again)
 - Conversions

```
# Conversion: String -> list  
print(list('hello'))
```

```
['h', 'e', 'l', 'l', 'o']
```

```
# Conversion: List -> String  
print(''.join(['h', 'e', 'l', 'l', 'o']))  
print(', '.join(['Welcome', 'Jackie!']))
```

```
hello  
Welcome, Jackie!
```

```
# Conversion: String <-> int?  
print(ord('a'))  
print(chr(98))
```

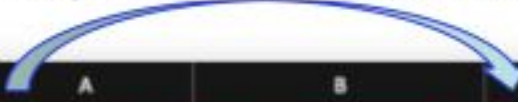
```
97  
b
```

Business Process Automation with Python

Data Type - Dictionary

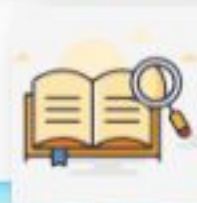
- **Motivation**

- Business problems often involve key-value pairs
- e.g., Employee ID → Full Name / Job Title



	A	B	C	D
1	Employee ID	Full Name	Job Title	Department
2	E02002	Kai Le	Controls Engineer	Engineering
3	E02003	Robert Patel	Analyst	Sales
4	E02004	Cameron Lo	Network Administrator	IT
5	E02005	Harper Castillo	IT Systems Architect	IT
6	E02006	Harper Dominguez	Director	Engineering
7	E02007	Ezra Vu	Network Administrator	IT
8	E02008	Jade Hu	Sr. Analyst	Accounting
9	E02009	Miles Chang	Analyst II	Finance
10	E02010	Gianna Holmes	System Administrator	IT
11	E02011	Jameson Thomas	Manager	Finance
12	E02012	Jameson Pena	Systems Analyst	IT
13	E02013	Bella Wu	Sr. Analyst	Finance

- Looking up a keyword
→ like finding words in **ictionaries**



Business Process Automation with Python

Data Type - Dictionary

- Representation

```
# Representation: option 1
school_rank_dict = {'hku': 21, 'cuhk': 38, 'hkust': 40}
print(type(school_rank_dict))

# Representation: option 2
school_rank_dict = dict(hku=21, cuhk=38, hkust=40)
print(school_rank_dict)

<class 'dict'>
{'hku': 21, 'cuhk': 38, 'hkust': 40}
```

Rank	University
21	 The University of Hong Kong Hong Kong, Hong Kong SAR
38	 The Chinese University of Hong Kong (CUHK) Hong Kong SAR, Hong Kong SAR
40	 The Hong Kong University of Science and Technology Hong Kong SAR, Hong Kong SAR

- Lookup → get()

```
# Lookup
print(school_rank_dict['cuhk'])          # Direct lookup
print(school_rank_dict.get('hkust'))     # get(): Success
print(school_rank_dict.get('test'))      # get(): Failed
print(school_rank_dict.get('test', 'N/A')) # get(): Error handling

38
40
None
N/A
```

Business Process Automation with Python

Data Type - Dictionary

- Adding an item

```
# Adding an item
school_rank_dict = {'hku': 21, 'cuhk': 38, 'hkust': 40}
school_rank_dict['cityu'] = 54
print(school_rank_dict)
```

```
{'hku': 21, 'cuhk': 38, 'hkust': 40, 'cityu': 54}
```

- Removing an item

```
# Removing an item
school_rank_dict = {'hku': 21, 'cuhk': 38, 'hkust': 40, 'cityu': 54}
school_rank_dict.pop('cityu')
print(school_rank_dict)
```

```
{'hku': 21, 'cuhk': 38, 'hkust': 40}
```


Business Process Automation with Python

Data Type - Dictionary

- Listing out the details

```
# Show all the keys
print(f'keys: {list(school_rank_dict.keys())}')

# Show all the values
print(f'values: {list(school_rank_dict.values())}')

# Show all the items
print(f'items: {list(school_rank_dict.items())}')

keys: ['hku', 'cuhk', 'hkust']
values: [21, 38, 40]
items: [('hku', 21), ('cuhk', 38), ('hkust', 40)]
```

- Membership Check

```
# Membership Check
print('hku' in school_rank_dict)

True
```

Business Process Automation with Python

Data Type - Dictionary

- Conversion
 - Cast from a list of tuples

	A	B	C
1	Employee ID ▼	Full Name ▼	Job Title ▼
2	E02002	Kai Le	Controls Engineer
3	E02003	Robert Patel	Analyst
4	E02004	Cameron Lo	Network Administrator

```
# Conversion: list => dict
employee_name_list = [('E02002', 'Kai Le'), ('E02003', 'Robert Patel')]
print(dict(employee_name_list))

{'E02002': 'Kai Le', 'E02003': 'Robert Patel'}
```


Business Process Automation with Python

Data Type - Dictionary

- **Conversion**

- zip() can be used to create tuples

	A	B	C
1	Employee ID	Full Name	Job Title
2	E02002	Kai Le	Controls Engineer
3	E02003	Robert Patel	Analyst
4	E02004	Cameron Lo	Network Administrator

```
employee_ids = ['E02002', 'E02003']  
full_names = ['Kai Le', 'Robert Patel']  
job_titles = ['Controls Engineer', 'Analyst']
```

```
print( list(zip(employee_ids, full_names)) )  
print( dict(zip(employee_ids, full_names)) )  
print( dict(zip(employee_ids, job_titles)) )
```

```
[ ('E02002', 'Kai Le'), ('E02003', 'Robert Patel') ]  
{ 'E02002': 'Kai Le', 'E02003': 'Robert Patel' }  
{ 'E02002': 'Controls Engineer', 'E02003': 'Analyst' }
```

Business Process Automation with Python

Conditional Statements

- If-else
 - Python tracks the scope using **indentation** (spaces)



VBA

```
Dim speed As Integer  
speed = 70  
  
If speed > 50 Then  
    Debug.Print ("Exceeded limit")  
Else  
    Debug.Print ("OK")  
End If
```

Python

```
speed = 70  
  
if speed > 50:  
    print('Exceeded limit')  
else:  
    print('OK')
```

Business Process Automation with Python

Conditional Statements

- If-else
 - Value assignment can be conditional too!

Multiline

```
speed = 70  
  
if speed > 50:  
    print('Exceeded limit')  
else:  
    print('OK')
```

1 line

```
speed = 70  
  
print('Exceeded limit' if speed > 50 else 'OK')
```

Business Process Automation with Python

Conditional Statements

- If-elif-else
 - elif → “Else If”

```
mark = 89
grade = None

if mark >= 90:
    grade = 'A'
elif mark >= 80:
    grade = 'B'
elif mark >= 70:
    grade = 'C'
else:
    grade = 'F'

print(grade)
```

Business Process Automation with Python

Conditional Statements

- If-elif-else
 - elif → “Else If”
 - May be broken down to if-else

```
mark = 89
grade = None

if mark >= 90:
    grade = 'A'
elif mark >= 80:
    grade = 'B'
elif mark >= 70:
    grade = 'C'
else:
    grade = 'F'

print(grade)
```

B

```
mark = 89
grade = None

grade = 'A' if mark >= 90 else ('B' if mark >= 80 else ('C' if mark >= 70 else 'F'))

print(grade)
```

B

Business Process Automation with Python

Conditional Statements

- If
 - “else” is optional

```
▶ salary = 10000
  salary_growth = 1.2
  years_of_service = 3

  # No increment for first year
  if years_of_service > 1:
      salary = salary * salary_growth
  print(salary)
```

12000.0

Business Process Automation with Python

Conditional Statements

- If-elif-else
 - Unlike VBA, scope is NOT defined with “If → End If”
 - Python tracks the scope using **indentation** (spaces)

VBA

```
Dim speed As Integer  
speed = 70  
  
If speed > 50 Then  
    Debug.Print ("Exceeded limit")  
Else  
    Debug.Print ("Speed is okay")  
End If
```

Python

```
speed = 70  
  
if speed > 50:  
    print('Exceeded limit')  
else:  
    print('Speed is okay')
```

Business Process Automation with Python

Loops – Using “for”

- Iterating through a range



	A	B	C
1	1		
2	2		
3	3		
4	4		
5	5		
6	6		
7	7		
8	8		
9	9		
10	10		

VBA

```
For i = 1 To 10  
    Debug.Print (i)  
Next i
```

Python

```
for i in range(1, 11):  
    print(i)
```

Business Process Automation with Python

Loops – Using “for”

- Different ways to specify ranges



	A	B	C
1	1		
2	2		
3	3		
4	4		
5	5		
6	6		
7	7		
8	8		
9	9		
10	10		

```
[11] for i in range(1, 11):  
    | print(i) # 1, 2, ..., 10
```

```
1  
2  
3  
4  
5  
6  
7  
8  
9  
10
```



```
for i in range(10):  
    | print(i + 1) # i = 0, 1, ..., 9
```

```
1  
2  
3  
4  
5  
6  
7  
8  
9  
10
```

Business Process Automation with Python

Loops – Using “for”

- Under the hood
 - Iterating through a sequence object

```
[21] print(range(10))  
range(0, 10)
```



	A	B	C
1	1		
2	2		
3	3		
4	4		
5	5		
6	6		
7	7		
8	8		
9	9		
10	10		

```
▶ list(range(10)) # In Python2, range() returns a list directly  
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
```



Business Process Automation with Python

Loops – Using “for”

- Be careful...

```
▶ item_prices = [200, 100]
total = 0
# Case 1: Apply $50 discount once
for price in item_prices:
    total = total + price
    if total >= 200:
        total = total - 50
print(total)
```

250

```
▶ item_prices = [200, 100]
total = 0
# Case 2: Apply $50 discount twice (accidentally..)
for price in item_prices:
    total = total + price
    if total >= 200:
        total = total - 50
print(total)
```

200

Business Process Automation with Python

Loops – Using “for”

- Looping through a list

```
▶ school_list = ['hku', 'cuhk', 'hkust']  
  
for school in school_list:  
    print(f'{school} is a good school in Hong Kong')  
  
☐ hku is a good school in Hong Kong  
   cuhk is a good school in Hong Kong  
   hkust is a good school in Hong Kong
```

Business Process Automation with Python

Loops – Using “for”

- Looping through a dictionary

```
▶ school_rank_dict = {'hku': 21, 'cuhk': 38, 'hkust': 40}

for school, rank in school_rank_dict.items():
    print(f'{school} is ranked {rank} in the world!')
```

```
↳ hku is ranked 21 in the world!
   cuhk is ranked 38 in the world!
   hkust is ranked 40 in the world!
```

Business Process Automation with Python

Loops – Using “for”

- Looping through a string
- Q: Why use “elif” instead of “else”?

```
# looping through a string
uppercase_letter_count = 0
lowercase_letter_count = 0

for letter in 'See you in Hong Kong':
    if letter.isupper():
        uppercase_letter_count = uppercase_letter_count + 1
    elif letter.islower():
        lowercase_letter_count = lowercase_letter_count + 1

print(f'Uppercase letters: {uppercase_letter_count}')
print(f'Lowercase letters: {lowercase_letter_count}')
```

Uppercase letters: 3
Lowercase letters: 13

Business Process Automation with Python

Loops – Using “for”

- List comprehension
→ compressing a loop in 1 line

```
▶ print(f'8 % 2 = {8%2}')
```

```
▶ print(f'9 % 2 = {9%2}')
```

```
⇒ 8 % 2 = 0
```

```
9 % 2 = 1
```

```
▶ # Traditional way
```

```
even_number_list = list()
```

```
for i in range(11):
```

```
    # Use the modulus operator to check for even number (Can be divided by 2)
```

```
    if i % 2 == 0:
```

```
        even_number_list.append(i)
```

```
print(even_number_list)
```

```
☐ [0, 2, 4, 6, 8, 10]
```

```
▶ # List comprehension
```

```
even_number_list = [i for i in range(11) if i % 2 == 0]
```

```
print(even_number_list)
```

```
☐ [0, 2, 4, 6, 8, 10]
```


Business Process Automation with Python

Loops – Using “for”

- List comprehension + Filtering

```
print(bool(''))  
print(bool(None))
```

```
False  
False
```

```
item_list = ['123', '456', '']
```

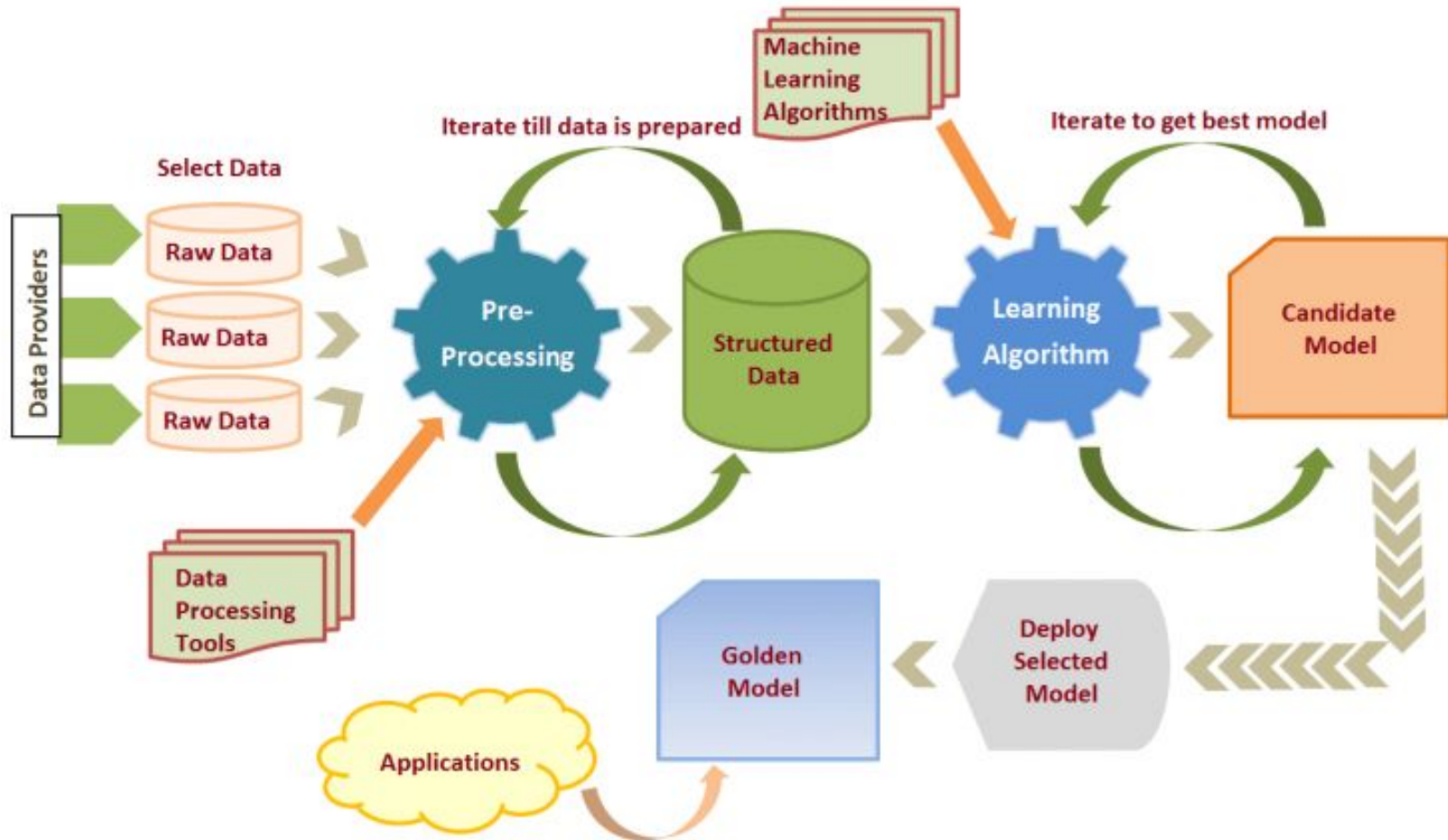
```
print([item for item in item_list])
```

```
print([item for item in item_list if bool(item)])
```

```
print([item for item in item_list if item])
```

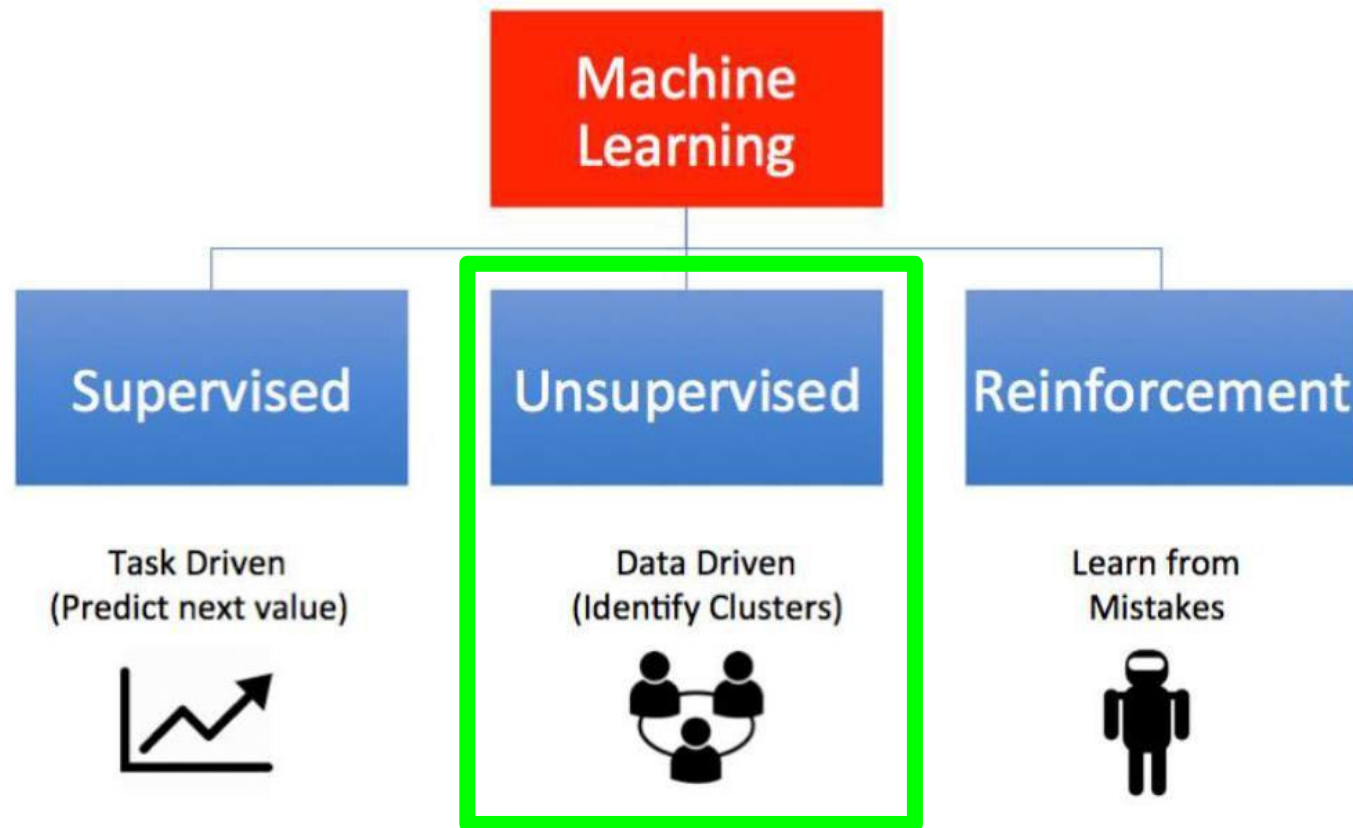
```
['123', '456', '']  
['123', '456']  
['123', '456']
```

Applied Machine Learning Pipeline Overview



What is Business Analytics & Machine Learning?

Types of Machine Learning



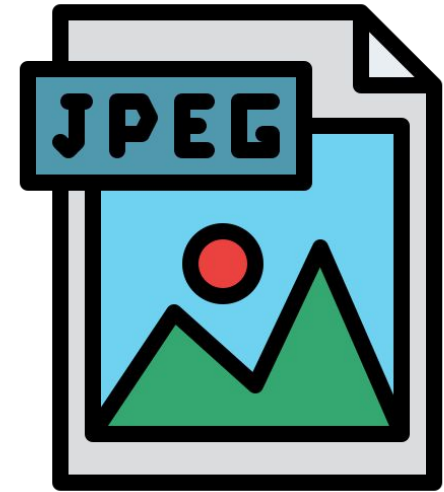
Convert Unstructured Data to Structured Data



PDF Document



Paper Form



Image, Photo

Processed Unstructured Data - PDF Document

Step 1 : Open Anaconda Prompt and enter “pip install pymupdf”

Step 2 : Open Jupyter notebook and enter following code

```
# RACAS Application Form Conversion (PDF to form)
def download_pdf(url, save_path):
    response = requests.get(url)
    if response.status_code == 200:
        with open(save_path, 'wb') as f:
            f.write(response.content)
        print(f"PDF downloaded and saved to {save_path}")
    else:
        print(f"Failed to download PDF: {response.status_code}")

import requests
pdf_url = ' '
# Usage example
pdf_file_path = "RACAS_applicationform_14012026.pdf"
# Step 1: Download the PDF
download_pdf(pdf_url, pdf_file_path)
```


Processed Unstructured Data - PDF Document

```
import fitz # PyMuPDF

pdf_file_path = "RACAS_applicationform_14012026.pdf"
pdf_document = fitz.open(pdf_file_path)

# Initialize a list to hold the table data
table_data = []

# Iterate through each page of the PDF
for page_num in range(len(pdf_document)):
    page = pdf_document[page_num]
    text = page.get_text("text") # Extract text from the page

    # Split the text into lines
    lines = text.splitlines()

    # Process lines to find tables (you may need to adjust this logic)
    for line in lines:
        # Example logic: Split lines by whitespace (this may need
        # adjustment)
        row = line.split()
        if row: # If the row is not empty
            table_data.append(row)

pdf_document.close()
# Print extracted table data
for row in table_data:
    print(row)
```

Raw PDF Document

HKAGAS001



香港安老院舍評審計劃申請表格

院舍名稱: _____ (中文)
_____ (英文)

院舍地址: _____

電話: _____ 傳真: _____

電郵: _____

院舍經營模式: 資助 / 自負盈虧 / 私營 / 合約院舍 (請刪去不適用者)

院舍類型: 安老院 / 護理安老院 / 護養院 / 安老院暨護理安老院 / 護養院暨護理安老院 / 其他類型請
註明 _____ (請刪去不適用者)

社會福利署/衛生署發出牌照院舍名稱: _____

實位後院舍床位數目: _____ (請附上社會福利署實位協議書副本)

院舍開始運作日期: _____ 年 _____ 月 _____ 日

申請費: 一千元以劃線支票形式 (支票抬頭: 香港老年學會) 寄回下列地址:

九龍尖沙咀金巴利道35號
金巴利中心一樓金閣
香港老年學會「香港安老院舍評審計劃」。

團體/院舍(蓋印): _____

負責人簽署: _____
姓名(正楷): _____ (先生/女士)
職銜: _____
電話: _____
傳真: _____
電郵: _____
日期: _____

初版日期: 04/2005 上次修定日期: 6/2015 最新修定日期: 1/2020

Structured Data

```
['HKAGAS001']
['初版日期:', '04/2005', '上次修定日期: 6/2015',
 '最新修定日期: 1/2020']
['香港老年學會']
['香港安老院舍評審計劃']
['九龍尖沙咀金巴利道35', '號金巴利中心1', '樓 /']
['新界葵涌石籬(一)邨石興樓地下18-20', '號']
['電話號碼:', '27273250 /', '37052383', '傳真號碼:',
 '37052507']
['電郵:', 'racas.adm@hkag.org', '網頁:', 'http://
www.hkag.org']
['香港安老院舍評審計劃申請表格']
['院舍名稱: ', '_____']
_____ (中文)']
```

Processed Unstructured Data - Image/Photo

**Step 1 : Open Anaconda Prompt and enter “pip install paddleocr
paddlepaddle”**

Step 2: Open Jupyter notebook and enter following code

```
#Paddle OCR Layout Analysis Module
```

```
#https://www.paddleocr.ai/main/en/version3.x/module_usage/layout_analysis.html#iii-quick-integration
```

```
from paddleocr import LayoutDetection
```

```
# Initialize the LayoutDetection model
```

```
model = LayoutDetection(model_name="PP-DocLayoutV2") #Better than PP-DocLayout_plus-L
```

```
# Predict layout from the image
```

```
output = model.predict("invoice.jpg", batch_size=1, layout_nms=True)
```

```
# Prepare to save results
```

```
results = []
```

```
# Process each result
```

```
for res in output:
```

```
    # Print the result details
```

```
    res.print()
```

```
    # Save the annotated image
```

```
    res.save_to_img(save_path="./output/")
```

```
    res.save_to_json(save_path="./output/invoice_res.json")
```

```
{'res': [{'input_path': 'invoice.jpg', 'page_index': None, 'boxes': [{'cls_id': 12, 'label': 'header', 'score': 0.5108081, 'coordinate': [503.62277, 267.39325, 638.60474, 348.99844]}, {'cls_id': 17, 'label': 'paragraph_title', 'score': 0.7680234909057617, 'coordinate': [784.45374, 255.98231, 1145.298, 355.98138]}, {'cls_id': 17, 'label': 'paragraph_title', 'score': 0.8592896461486816, 'coordinate': [502.58075, 445.82178, 701.0267, 487.18213]}, {'cls_id': 22, 'label': 'text', 'score': 0.8447363376617432, 'coordinate': [501.88065, 573.3299, 704.94745, 609.7668]}, {'cls_id': 22, 'label': 'text', 'score': 0.8478004932403564, 'coordinate': [501.92218, 493.01996, 666.6064, 528.4518]}, {'cls_id': 17, 'label': 'paragraph_title', 'score': 0.8744010329246521, 'coordinate': [503.01636, 1553.1962, 829.45074, 1590.8115]}, {'cls_id': 22, 'label': 'text', 'score': 0.8731784224510193, 'coordinate': [502.15244, 532.2916, 869.65027, 569.0419]}, {'cls_id': 22, 'label': 'text', 'score': 0.965853750705719, 'coordinate': [504.39917, 1605.239, 1503.1549, 1681.4655]}, {'cls_id': 22, 'label': 'text', 'score': 0.6948447822524475, 'coordinate': [1132.0269, 573.3034, 1333.7224, 609.73065]}, {'cls_id': 22, 'label': 'text', 'score': 0.6929407119750977, 'coordinate': [1132.066, 531.9936, 1498.3948, 569.1969]}, {'cls_id': 22, 'label': 'text', 'score': 0.6627132167916162, 'coordinate': [1132.5897, 492.5825, 1296.9484, 528.977]}, {'cls_id': 17, 'label': 'paragraph_title', 'score': 0.525238037109375, 'coordinate': [1134.1367, 445.6676, 1386.406, 487.97943]}, {'cls_id': 21, 'label': 'table', 'score': 0.9788541793823242, 'coordinate': [453.17435, 682.1853, 1532.3761, 1501.2639]}, {'cls_id': 9, 'label': 'footer_image', 'score': 0.8323854207992554, 'coordinate': [519.22656, 1738.9999, 662.2307, 178.0.7227]}, {'cls_id': 8, 'label': 'footer', 'score': 0.8681032061576843, 'coordinate': [688.5701, 1745.3066, 776.73505, 1.778.9103]}, {'cls_id': 8, 'label': 'footer', 'score': 0.9132580161094666, 'coordinate': [957.31256, 1747.4482, 1085.8271, 1.776.3594]}, {'cls_id': 8, 'label': 'footer', 'score': 0.9150736331939697, 'coordinate': [1294.7247, 1746.2638, 1464.4.988, 1.777.5459]}]}
```

Processed Unstructured Data - Image/Photo

```
#Display original image and labelled image  
from IPython.display import Image, display
```

```
# Specify the path to your image file
```

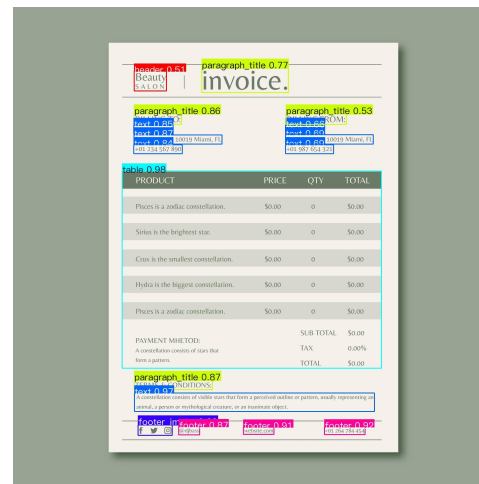
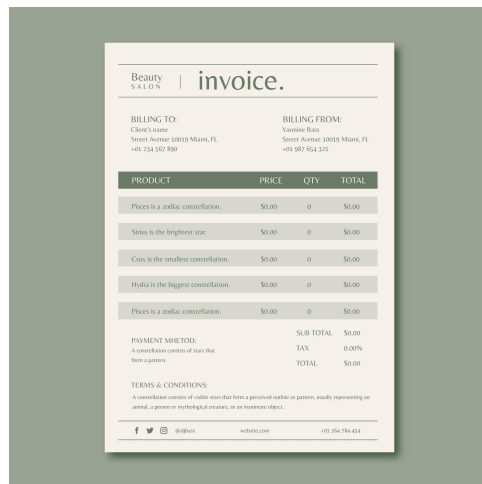
```
orig_image_path = "invoice.jpg"
```

```
label_image_path = "./output/invoice_res.jpg"
```

```
# Display the images with specified resolution
```

```
display(Image(filename=orig_image_path, width=640, height=480))
```

```
display(Image(filename=label_image_path, width=640, height=480))
```

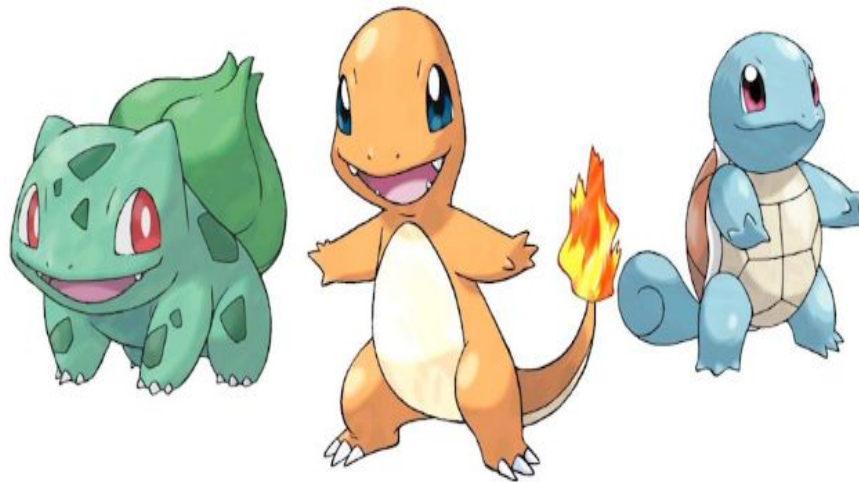


Download the data

Download the Pokemon dataset from:

https://github.com/innoviai/IPA_Courses-May2026/blob/main/pokemon_dataset.csv

Save the pokemon dataset file in a location you'll remember.



Data Collection – Fetching Data using Pandas

What / Why?

- Pandas is an *open-source* Python library (anyone can contribute)
- Allows for high-performance, easy-to-use data structures and data analysis
- Unlike NumPy library which provides multi-dimensional arrays, Pandas provides 2D table object called **DataFrame** (akin to a spreadsheet with column names and row labels).
- Used by *a lot* of people

Data Collection – Fetching Data using Pandas

How

- import **pandas** library (convenient to rename it)
- Use **read_csv()** function

```
import pandas as pd  
dataframe = pd.read_csv("yourfile.csv")
```

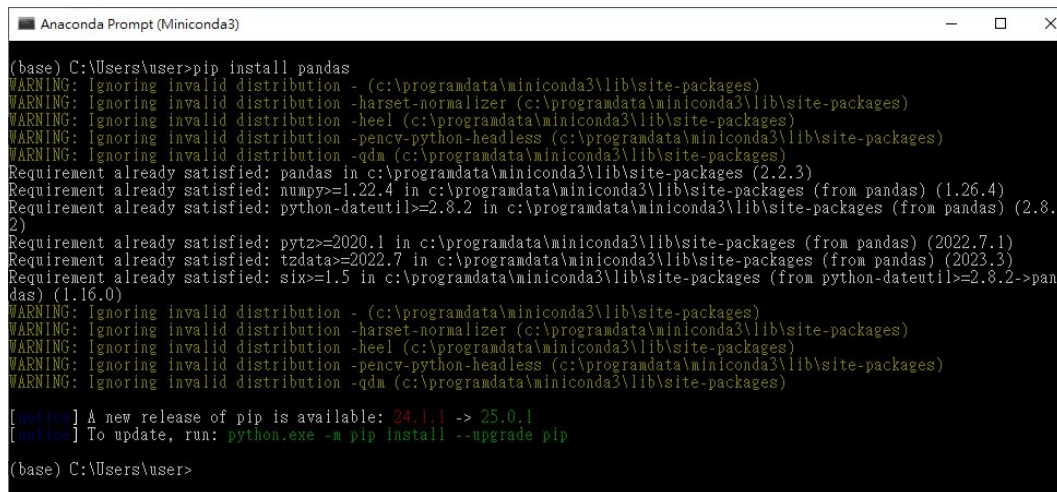
Library Installation in Python

1. Open tools “Anaconda Prompt” which will direct you to the command prompt.



```
Anaconda Prompt (Miniconda3)
(base) C:\Users\user>
```

2. Install python package requests by entering “pip install pandas”



```
Anaconda Prompt (Miniconda3)
(base) C:\Users\user>pip install pandas
WARNING: Ignoring invalid distribution - (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -harset-normalizer (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -heel (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -pencv-python-headless (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -qdm (c:\programdata\miniconda3\lib\site-packages)
Requirement already satisfied: pandas in c:\programdata\miniconda3\lib\site-packages (2.2.3)
Requirement already satisfied: numpy>=1.22.4 in c:\programdata\miniconda3\lib\site-packages (from pandas) (1.26.4)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\programdata\miniconda3\lib\site-packages (from pandas) (2.8.2)
Requirement already satisfied: pytz>=2020.1 in c:\programdata\miniconda3\lib\site-packages (from pandas) (2022.7.1)
Requirement already satisfied: tzdata>=2022.7 in c:\programdata\miniconda3\lib\site-packages (from pandas) (2023.3)
Requirement already satisfied: six>=1.5 in c:\programdata\miniconda3\lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.16.0)
WARNING: Ignoring invalid distribution - (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -harset-normalizer (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -heel (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -pencv-python-headless (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -qdm (c:\programdata\miniconda3\lib\site-packages)
[notice] A new release of pip is available: 24.1.1 -> 25.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(base) C:\Users\user>
```

Reading in the data

First we import the Python packages we are going to use.
Then we use Pandas to load in the dataset as a data frame.

```
import numpy as np
import pandas as pd
from matplotlib import pyplot as plt
import seaborn as sns
```

```
df1 = pd.read_csv("pokemon_dataset.csv", index_col=0, encoding="ISO-8859-1")
```

NOTE: The argument `index_col` argument states that we'll treat the first column of the dataset as the ID column.

NOTE: The encoding argument allows us to by pass an input error created by special characters in the data set.

Examine the data set

```
print(df1.head())
```

	Name	Type 1	Type 2	Total	...	Sp. Def	Speed	Stage	Legendary
#					...				
1	Bulbasaur	Grass	Poison	318	...	65	45	1	False
2	Ivysaur	Grass	Poison	405	...	80	60	2	False
3	Venusaur	Grass	Poison	525	...	100	80	3	False
4	Charmander	Fire	NaN	309	...	50	65	1	False
5	Charmeleon	Fire	NaN	405	...	65	80	2	False

```
print(df1.describe())
```

	Total	HP	Attack	...	Sp. Def	Speed	Stage
count	151.000000	151.000000	151.000000	...	151.000000	151.000000	151.000000
mean	407.07947	64.211921	72.549669	...	66.019868	68.933775	1.582781
std	99.74384	28.590117	26.596162	...	24.197926	26.746880	0.676832
min	195.000000	10.000000	5.000000	...	20.000000	15.000000	1.000000
25%	320.000000	45.000000	51.000000	...	49.000000	46.500000	1.000000
50%	405.000000	60.000000	70.000000	...	65.000000	70.000000	1.000000
75%	490.000000	80.000000	90.000000	...	80.000000	90.000000	2.000000
max	680.000000	250.000000	134.000000	...	125.000000	140.000000	3.000000

Data Collection – Fetching Data using Pandas

Common Panda functions

High-level viewing:

- `head()` – first N observations
- `tail()` – last N observations
- `columns()` – names of the columns
- `describe()` – statistics of the quantitative data
- `dtypes()` – the data types of the columns

Data Collection – Fetching Data using Pandas

Common Panda functions

Accessing/processing:

- `df["column_name"]`
- `Df.column_name`
- `.max()`, `.min()`, `.idxmax()`, `.idxmin()`
- `<dataframe> <conditional statement>`
- `.loc[]` – label-based accessing
- `.iloc[]` – index-based accessing
- `.sort_values()`
- `.isnull()`, `.notnull()`

How To Do Data Mining: Machine Learning Techniques

Features / Attributes

1. District – Get from HK Building Record
2. Floor – Get from HK Building Record
3. School Net Numbers - POA School Net Look-up Table

Outcome

Price Per Square - Get from HK Building Record

<https://www.28hse.com/en/rent/apartment/property-3496859>

Floor zone

Middle Floor

Saleable Area

807 ft²

Unit Price: @46.8

<https://www.28hse.com/en/rent/apartment>



♥BEST CHOICE♥ \$37800 Front line sea view...

Tsuen Wan Ocean Pride | Unit C, Mid Floor, Tower 1, Ocean Supreme, Middle Floor 12 minutes ago posted

Saleable Area: 807 ft² @46.8

美聯物業

Apartment Good view Sea view Elegant

Air conditioner Washing Machine Water Heaters Refrigerator

Induction Hob Cooker Hood

Lease HKD\$37,800



The largest room with sea view•Rarely availa...

Hung Hom Upper East | High Floor, Tower 1A 2 hours ago posted

Saleable Area: 375 ft² @49.3

Huicheng Property Company Limited

1 Bedroom, 1 Bathroom Western Style Bldg

Stand-alone Building Good view Sea view Elegant Microwave Oven

Air conditioner Washing Machine Water Heaters

Lease HKD\$18,500



3-bedroom Western-style building with open ...

Kwun Tong Hamden Court | High Floor, Block A 3 hours ago posted

Saleable Area: 481 ft² @35.8

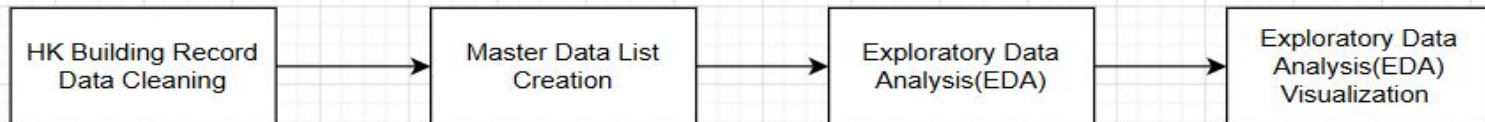
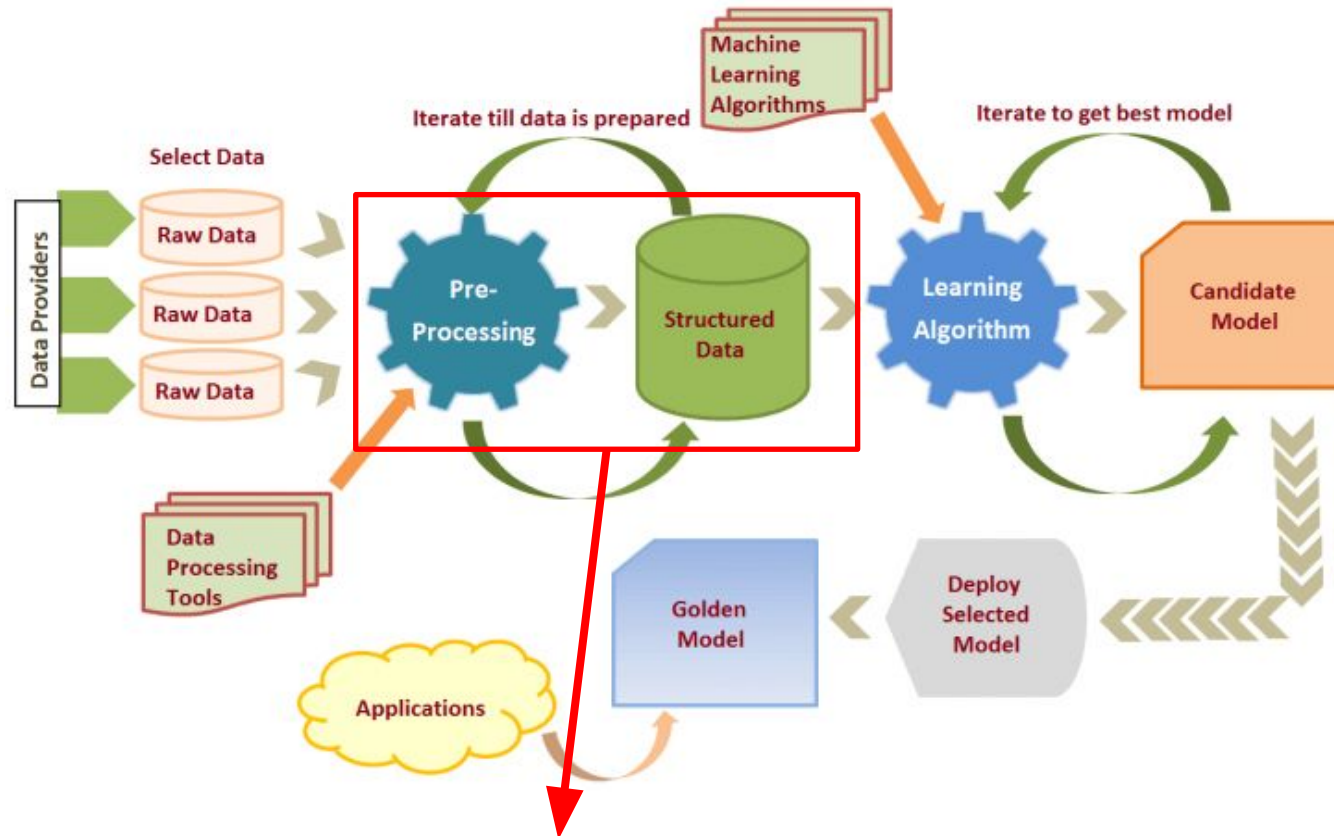
Kam Lung Property Agency Co.

3 Bedrooms, 1 Bathroom Western Style Bldg Good view

Elegant

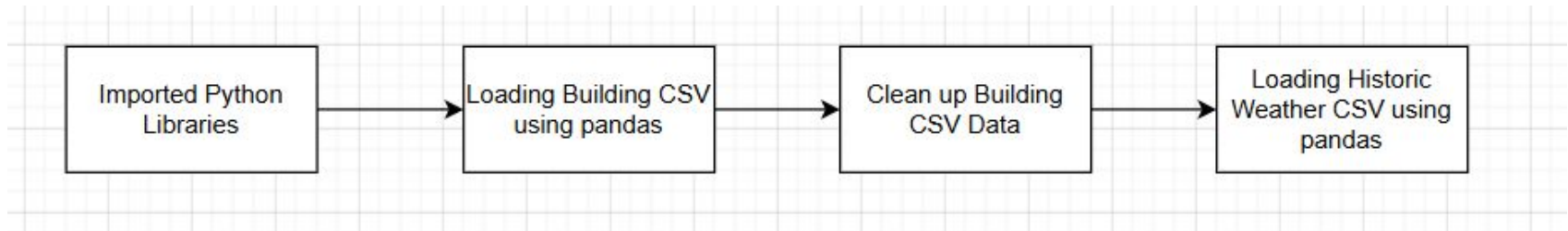
Lease HKD\$17,200

Applied Machine Learning Pipeline Overview



HK Building Record Data Cleaning

Below are the Flow Chart with the Process of Data Collection to Data Cleaning on Historic Building Data



This Tasks is to conduct Data Collection and Cleaning

Step 1: Import the required libraries:

```
import pandas as pd
```

Step 2: Loading Historic Weather CSV using pandas

#Loading original CSV file containing historic property data:

```
hkbuildingdata = pd.read_csv('hkbuildinginfo.csv', encoding='ISO-8859-1')
```

```
hkbuildingdata.head(5)
```

HK Building Record Data Cleaning

Step 3: Clean up the data

Check for missing values

```
print("\nMissing values in each column:")
```

```
print(hkbuildingdata.isnull().sum())
```

#Handle missing values by Dropping rows with missing values

```
hkbuildingdata.drop(columns=['withpre', 'rootid', 'fatherid', 'catid', 'source', 'block', 'state',  
                             'date_dm'],inplace=True, errors='ignore')
```

Assuming 'floor' is the column causing the error

```
hkbuildingdata['floor'] = hkbuildingdata['floor'].astype(str) # Convert to string type
```

Now you can safely use the .str accessor

```
hkbuildingdata['floor'] = hkbuildingdata['floor'].str.replace(r'[^\d]', '', regex=True) #Remove non-numeric  
character
```

Convert back to numeric if needed

```
hkbuildingdata['floor'] = pd.to_numeric(hkbuildingdata['floor'], errors='coerce') # Convert to numeric
```

Fill missing values for 'floor' and 'room'

```
median_value = hkbuildingdata['floor'].median()
```

```
hkbuildingdata['floor'] = hkbuildingdata['floor'].fillna(median_value ) # Fill with median for numeric data
```

```
mode_value = hkbuildingdata['room'].mode()[0]
```

```
hkbuildingdata['room'] = hkbuildingdata['room'].fillna(mode_value) # Fill with mode for categorical data
```

Missing values in each column:

date	0
withpre	23564
id	0
rootid	0
fatherid	0
catid	0
catname	0
catfathername	0
url_father	0
url_cat	0
source	11556
contract	0
memo	0
price	0
price_value	0
holddate	0
winloss_flag	0
winloss	0
act_area	0
area	0
arearaw	4
sq_price	0
sq_price_value	0
sq_actprice	0
sq_actprice_value	0
month	0
day	0
date_dm	0
date_y	0
block	32308
state	77003
floor	32
room	258
addr	0
Unnamed: 34	80000
dtype: int64	

HK Building Record Data Cleaning

#Continue from previous page

Rename columns to more meaningful names

```
hkbuildingdata.rename(columns={'floor': 'Floor', 'sq_price_value': 'Price Per Square', 'catname': 'Building Name', 'catfathername': 'District'}, inplace=True)
```

Display the cleaned DataFrame and check for remaining missing values

```
print("\nCleaned DataFrame:")
```

```
print(hkbuildingdata.head())
```

```
print("\nRemaining Missing Values:")
```

```
print(hkbuildingdata.isnull().sum())
```

Step 4: Loading Weather CSV using pandas

#Loading original CSV file containing historic weather data:

```
weatherdata = pd.read_csv("historic_weatherdata.csv")
```

```
print(weatherdata.head(5))
```

```
weatherdata['date'] = pd.to_datetime(weatherdata['date']).dt.strftime('%Y-%m-%d')
```

```
Cleaned DataFrame:
   date      id  Building Name  District \
0  2020-11-27  683333  Bel Air Heights  Diamond Hill
1  2020-11-27  683332  Fa Yuen Plaza      Mong Kok
2  2020-11-27  683331  Caldecott Hill  Yau Yat Tsuen
3  2020-11-27  683330  Pang Ching Court  Wong Tai Sin
4  2020-11-27  683329  Metro Harbour View  Tai Kok Tsui

   url_father \
0  https://data.28hse.com/en/kl/diamond-hill
1  https://data.28hse.com/en/kl/mong-kok
2  https://data.28hse.com/en/kl/yau-yat-tsuen
3  https://data.28hse.com/en/kl/wong-tai-sin
4  https://data.28hse.com/en/kl/tai-kok-tsui

   url_cat  contract \
0  https://data.28hse.com/en/kl/diamond-hill/2058...  Agreement
1  https://data.28hse.com/en/kl/mong-kok/2712_fa-...  Agreement
2  https://data.28hse.com/en/kl/yau-yat-tsuen/309...  Agreement
3  https://data.28hse.com/en/kl/wong-tai-sin/5035...  Agreement
4  https://data.28hse.com/en/kl/tai-kok-tsui/2564...  Agreement

   memo  price  price_value  ...  Price Per Square  sq_actprice \
0  20112702480015  $10M  10000000  ...  12106.54  $15221
1  20112702470044  $4.28M  4280000  ...  12334.29  $20000
2  20112702440018  $11M  11000000  ...  8580.34  $11100
3  20112702410017  $4.9M  4900000  ...  8404.80  --
4  20112702400032  $7.15M  7150000  ...  12016.81  $17354

   sq_actprice_value  month  day  date_y  Floor  room  addr \
0  15220.70  Nov  27  2020  2.0  E  BLOCK 2 2#/F Room E
1  20000.00  Nov  27  2020  1.0  C  1#/F Room C
2  11099.90  Nov  27  2020  7.0  E  TOWER 1 7/F Room E
3  0.00  Nov  27  2020  1.0  22  1#/F Room 22
4  17354.37  Nov  27  2020  1.0  B  BLOCK 8 1#/F Room B
```

HK School Net Number CSV File

Data from Manual Input

District and POA School Net Look-up Table

https://www.edb.gov.hk/attachment/en/edu-system/primary-secondary/spa-systems/primary-1-admission/school-lists/District_Net_LookUpTable.pdf

區域 District	小一學校網編號 POA School Net Number
中西區 Central & Western District	11
灣仔 Wan Chai	12
北角/筲箕灣/柴灣 North Point/ Shau Kei Wan/ Chai Wan	14, 16
薄扶林/香港仔/黃竹坑/赤柱 Pok Fu Lam/ Aberdeen/ Wong Chuk Hang/ Stanley	18
油麻地/尖沙咀/旺角 Yau Ma Tei/ Tsim Sha Tsui/ Mong Kok	31, 32
何文田/紅磡/九龍城 Ho Man Tin/ Hung Hom/ Kowloon City	34, 35, 41
深水埗 Sham Shui Po	40
黃大仙 Wong Tai Sin	43, 45
九龍灣/牛頭角/觀塘/油塘 Kowloon Bay/ Ngau Tau Kok/ Kwun Tong/ Yau Tong	46, 48
荃灣 Tsuen Wan	62
葵涌/荔景/大窩口/青衣 Kwai Chung/ Lai King/ Tai Wo Hau/ Tsing Yi	64, 65, 66
屯門 Tuen Mun	70, 71
天水圍/元朗 Tin Shui Wai/ Yuen Long	72, 73, 74
上水/粉嶺/北區 Sheung Shui/ Fanling/ North District	80, 81, 83
大埔 Tai Po	84
沙田 Shatin	88, 89, 91
西貢/將軍澳 Sai Kung/ Tseung Kwan O	95
離島/東涌 Islands/ Tung Chung	96, 97, 98, 99

Master Data List Combination System Architecture

HK Historic
Weather

Data from API

historic_weath
erdata.csv

Key: date

HK Building
Record

Data from Web
Wrapping

hkbuildinginfo.
csv

Key: District

HK School
Net Number

Data from
Manual Input

Primary Sch_Net
Numbers By
District.csv

Key: District

Master Data List Combination System Architecture

historic_weatherdata.csv

Key: date

date	wind	weather	maxtemp	mintemp	icon	psr
27/11/2020	North force 4 to 5.	Mainly cloudy with a few light rain patches at first. Becoming fine	20	15	51	Low
26/11/2020	North to northeast force 4, force 5 at first.	Fine. Rather cool in the morning. Very dry during the day.	19	13	92	Low
25/11/2020	Northeast force 4, east force 5 later.	Fine. Rather cool in the morning. Very dry during the day.	19	13	50	Low

Key: date

hkbuildinginfo.csv

date	withpre	id	rootid	fatherid	catid	catname	catfathername	url_father	url_cat	source	contract	memo	price
27/11/2020	1	683333	2	79	2058	Bel Air Hei Diamond I	https://data	https://data	Land Regis Agreement	2.01E+13	\$10M		
27/11/2020	1	683332	2	71	2712	Fa Yuen Pl Mong Kok	https://data	https://data	Land Regis Agreement	2.01E+13	\$4.28M		
27/11/2020	1	683331	2	78	3094	Caldecott F Yau Yat Ts	https://data	https://data	Land Regis Agreement	2.01E+13	\$11M		
27/11/2020		683330	2	57	5035	Pang Ching Wong Tai	https://data	https://data	Land Regis Agreement	2.01E+13	\$4.9M		
27/11/2020	1	683329	2	67	2564	Metro Hart Tai Kok Ts	https://data	https://data	Land Regis Agreement	2.01E+13	\$7.15M		
27/11/2020	1	683328	2	65	2521	Nob Hill Mei Foo	https://data	https://data	Land Regis Agreement	2.01E+13	\$2.46M		
27/11/2020	1	683327	2	62	5281	Chi Fung C Sham Shui	https://data	https://data	Land Regis Agreement	2.01E+13	\$3.2M		
27/11/2020	1	683326	2	56	2080	Telford Ga Kowloon E	https://data	https://data	Land Regis Agreement	2.01E+13	\$7.68M		
27/11/2020	1	683325	2	57	2104	Tsui Chuk Wong Tai	https://data	https://data	Land Regis Agreement	2.01E+13	\$3.67M		
27/11/2020	1	683324	2	77	2548	Grand wate To Kwa W	https://data	https://data	Land Regis Agreement	2.01E+13	\$9M		
27/11/2020	1	683323	2	77	3073	Sun Cheon To Kwa W	https://data	https://data	Land Regis Agreement	2.01E+13	\$4.2M		
27/11/2020	1	683322	2	77	5461	Siu Fung C To Kwa W	https://data	https://data	Land Regis Agreement	2.01E+13	\$4.9M		
27/11/2020	1	683321	2	71	8587	SKYPARK Mong Kok	https://data	https://data	Land Regis Agreement	2.01E+13	\$7.35M		
27/11/2020	1	683320	2	127	5263	Aria Ngau Chi V	https://data	https://data	Land Regis Agreement	2.01E+13	\$8.25M		
27/11/2020	1	683319	2	60	5948	One Victor Ho Man Ti	https://data	https://data	Land Regis Agreement	2.01E+13	\$8.83M		
27/11/2020	1	683318	2	77	22478	Wan Tung To Kwa W	https://data	https://data	Land Regis Agreement	2.01E+13	\$5.43M		
27/11/2020		683317	2	56	2082	Richland C Kowloon E	https://data	https://data	Land Regis Agreement	2.01E+13	\$5.67M		
27/11/2020		683316	2	62	2429	Cheung Hc Sham Shui	https://data	https://data	Land Regis Agreement	2.01E+13	\$3.73M		
27/11/2020	1	683315	2	60	2243	Dragon Vie Ho Man Ti	https://data	https://data	Land Regis Agreement	2.01E+13	\$9.8M		

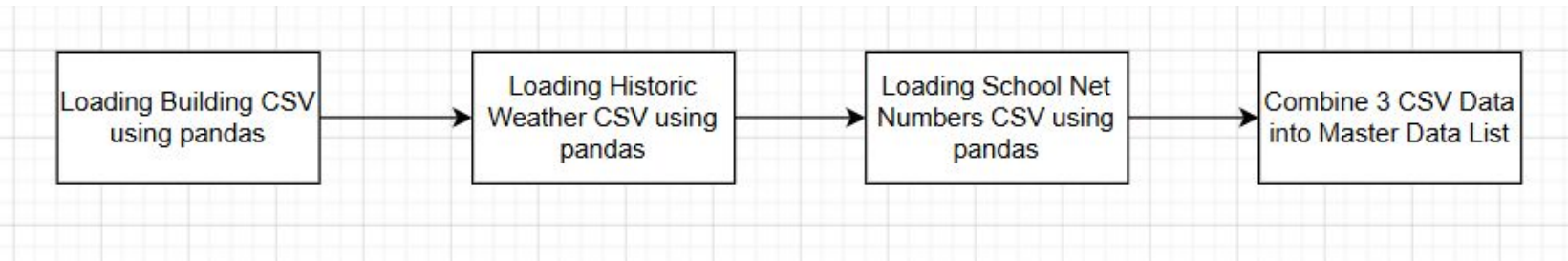
Primary Sch_Net
Numbers By District.csv

Key: District

Sch_Net_No	District Areas	Districts
1	CENTRAL	Hong Kong Island
1	THE PEAK	Hong Kong Island
1	SHEUNG WAN	Hong Kong Island
1	SAI YING PUN	Hong Kong Island
1	SHEK TONG TSUI	Hong Kong Island
1	KENNEDY TOWN	Hong Kong Island
1	WAN CHAI	Hong Kong Island
1	CAUSEWAY BAY	Hong Kong Island
1	SO KON PO	Hong Kong Island
1	HAPPY VALLEY	Hong Kong Island
1	TAI HANG	Hong Kong Island
1	JARDINE'S LOOKOUT	Hong Kong Island
1	NORTH POINT	Hong Kong Island
1	QUARRY BAY	Hong Kong Island
1	LEI KING WAN	Hong Kong Island
1	SHAU KEI WAN	Hong Kong Island
1	SAI WAN HO	Hong Kong Island
1	SHEK O	Hong Kong Island

Master Data List Combination System Architecture

Below are the Flow Chart with the Process of Combining 3 CSV Data to a master list



#Combine Weather Data with Building Data

```
hkbuildingdata['date'] = pd.to_datetime(hkbuildingdata['date']).dt.strftime('%Y-%m-%d')  
merged_df = hkbuildingdata.merge(weatherdata[['date', 'wind', 'maxtemp', 'mintemp']], on='date', how='left')
```

#Combine Combined Data with School Net Number Data

```
#Loading original CSV file from school net number data  
schoolnetnumber_df = pd.read_csv("Primary Sch_Net Numbers By District.csv")  
schoolnetnumber_df.head(5)
```


Combining Weather, Building Information and School Net Numbers

Normalize the 'District' column in both DataFrames to lowercase

```
merged_df['District'] = merged_df['District'].str.lower()
```

```
schoolnetnumber_df['District'] = schoolnetnumber_df['District Areas'].str.lower()
```

Merge schooldistrict_df with districtcode_df on the normalized 'District' column

```
merged_df = merged_df.merge(schoolnetnumber_df[['District', 'Sch_Net_No']], on='District', how='left')
```

Create the 'School_Netnumber' column and fill in 0 for not record found

```
merged_df['School_Netnumber'] = merged_df['Sch_Net_No'].fillna(0).astype(int)
```

Save the merged_df into csv with name “combined_building_withweather_school.csv”

```
merged_df.to_csv('combined_building_withweather_school.csv', index=False)
```

```
merged_df.head(5)
```

[27]:	date	id	Building Name	District	url_father	url_cat	contract	memo	price	price_va
0	2020-11-27	683333	Bel Air Heights	diamond hill	https://data.28hse.com/en/kd/diamond-hill	https://data.28hse.com/en/kd/diamond-hill/2058...	Agreement	20112702480015	\$10M	100000
1	2020-11-27	683332	Fa Yuen Plaza	mong kok	https://data.28hse.com/en/kd/mong-kok	https://data.28hse.com/en/kd/mong-kok/2712_fa-...	Agreement	20112702470044	\$4.28M	42800
2	2020-11-27	683331	Caldecott Hill	yau yat tsuen	https://data.28hse.com/en/kd/yau-yat-tsuen	https://data.28hse.com/en/kd/yau-yat-tsuen/309...	Agreement	20112702440018	\$11M	110000
3	2020-11-27	683330	Pang Ching Court	wong tai sin	https://data.28hse.com/en/kd/wong-tai-sin	https://data.28hse.com/en/kd/wong-tai-sin/5035...	Agreement	20112702410017	\$4.9M	49000
4	2020-11-27	683329	Metro Harbour View	tai kok tsui	https://data.28hse.com/en/kd/tai-kok-tsui	https://data.28hse.com/en/kd/tai-kok-tsui/2564...	Agreement	20112702400032	\$7.15M	71500

5 rows × 29 columns

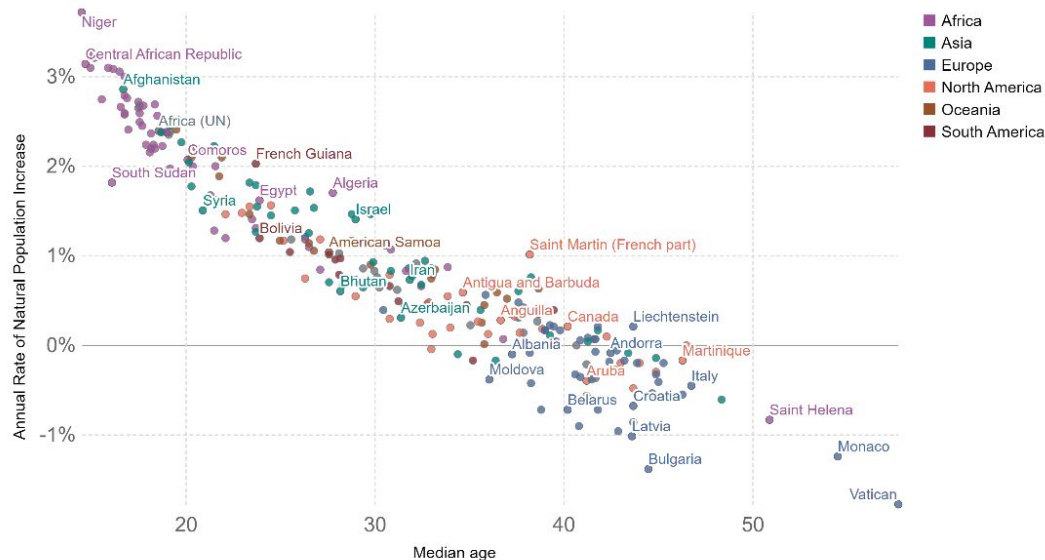
Exploratory Data Analysis(EDA) and Data Visualization

- **EDA is the first step in understanding a dataset.** It uses summary statistics and visualizations to explore patterns, spot anomalies, and understand relationships between variables.
- **Purpose:** find trends, and unexpected behavior before building models or making decisions.
- **Common techniques:** use scatter plots to show the relationship between two variables

Population growth rate vs. median age, 2021

Median age is the age that divides the population into two parts of equal size, that is, there are as many persons with ages above the median as there are with ages below the median. In this metric of population growth, changes due to migration are excluded and only births and deaths are considered.

Our World
in Data



Source: United Nations, World Population Prospects (2022)

OurWorldInData.org/age-structure • CC BY

Exploratory Data Analysis(EDA) and Data Visualization

Exploring your data is a crucial step in data analysis. It involves:

Organising the data set

Plotting aspects of the data set

Maybe producing some numerical summaries; central tendency and spread, etc.

“Exploratory data analysis can never be the whole story, but nothing else can serve as the foundation stone.”

- John Tukey.

Exploratory Data Analysis(EDA) and Data Visualization

Why?

- EDA encompasses the “*explore* data” part of the data science process
- EDA is crucial but often overlooked:
 - If your data is bad, your results will be bad
 - Conversely, understanding your data well can help you create smart, appropriate models

Exploratory Data Analysis(EDA) and Data Visualization

What?

1. Store data in data structure(s) that will be convenient for exploring/processing
(Memory is fast. Storage is slow)
2. Clean/format the data so that:
 - Each row represents a single object/observation/entry
 - Each column represents an attribute/property/feature of that entry
 - Values are numeric whenever possible
 - Columns contain atomic properties that cannot be further decomposed*

* Unlike food waste, which can be composted.
Please consider composting food scraps.

Exploratory Data Analysis(EDA) and Data Visualization

What? (continued)

3. Explore **global** properties: use histograms, scatter plots, and aggregation functions to summarize the data
4. Explore **group** properties: group like-items together to compare subsets of the data (are the comparison results reasonable/expected?)

This process transforms your data into a format which is easier to work with, gives you a basic overview of the data's properties, and likely generates several questions for you to follow-up in subsequent analysis.

Exploratory Data Analysis(EDA) and Data Visualization

Exploratory Data Analysis and Visualization

```
#filter the data with period "2020-1-1" to "2020-12-31"
```

```
# Create a full date column from 'month', 'day', and 'date_y'
```

```
hkbuildingdata['full_date'] = pd.to_datetime(hkbuildingdata['month'] + '-' + hkbuildingdata['day'].astype(str) + '-' +  
                                             hkbuildingdata['date_y'].astype(str))
```

```
# Filter the DataFrame for the date range "2020-01-01" to "2020-12-31"
```

```
start_date = '2020-01-01'
```

```
end_date = '2020-12-31'
```

```
covidfiltered_data = hkbuildingdata[(hkbuildingdata['full_date'] >= start_date) & (hkbuildingdata['full_date'] <= end_date)]
```

```
# Display the first few rows of the filtered DataFrame
```

```
print(covidfiltered_data.head(5))
```

```
   date      id  Building Name  District \  
0  2020-11-27  683333  Bel Air Heights  diamond hill  
1  2020-11-27  683332  Fa Yuen Plaza    mong kok  
2  2020-11-27  683331  Caldecott Hill   yau yat tsuen  
3  2020-11-27  683330  Pang Ching Court wong tai sin  
4  2020-11-27  683329  Metro Harbour View tai kok tsui  
  
   url_father \  
0  https://data.28hse.com/en/kl/diamond-hill  
1  https://data.28hse.com/en/kl/mong-kok  
2  https://data.28hse.com/en/kl/yau-yat-tsuen  
3  https://data.28hse.com/en/kl/wong-tai-sin  
4  https://data.28hse.com/en/kl/tai-kok-tsui  
  
   url_cat  contract \  
0  https://data.28hse.com/en/kl/diamond-hill/2058... Agreement  
1  https://data.28hse.com/en/kl/mong-kok/2712_fa-... Agreement  
2  https://data.28hse.com/en/kl/yau-yat-tsuen/309... Agreement  
3  https://data.28hse.com/en/kl/wong-tai-sin/5035... Agreement  
4  https://data.28hse.com/en/kl/tai-kok-tsui/2564... Agreement
```

```
# Filter the DataFrame for the date range "2019-01-01" to "2019-12-31"
```

```
start_date = '2019-01-01'
```

```
end_date = '2019-12-31'
```

```
filtered_data = hkbuildingdata[(hkbuildingdata['full_date'] >= start_date) & (hkbuildingdata['full_date'] <= end_date)]
```

```
# Display the first few rows of the filtered DataFrame
```

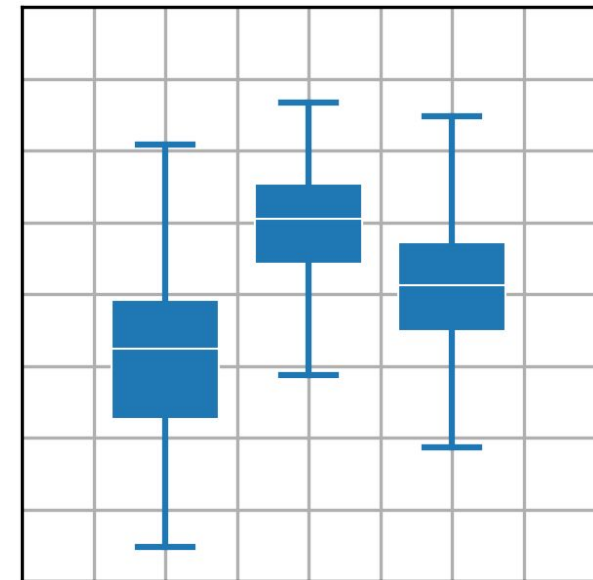
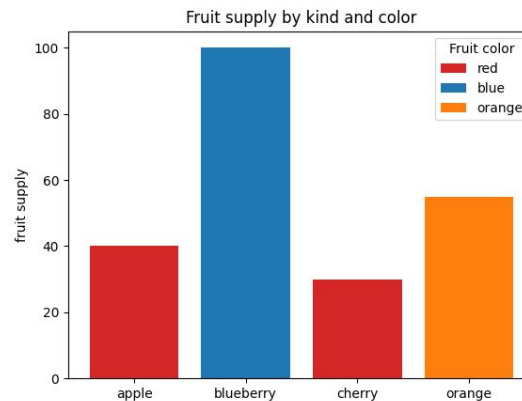
```
print(filtered_data.head(5))
```

Matplotlib: Visualization with Python

Matplotlib is a comprehensive library for creating static, animated, and interactive visualizations in Python. Matplotlib makes easy things easy and hard things possible.

Features:

1. Create publication quality plots.
2. Make interactive figures that can zoom, pan, update.
3. Customize visual style and layout.
4. Export to many file formats.
5. Embed in JupyterLab and Graphical User Interfaces.
6. Use a rich array of third-party packages built on Matplotlib.



<https://matplotlib.org/>

Exploratory Data Analysis(EDA) and Data Visualization

```
import matplotlib.pyplot as plt

# Create a scatter plot using matplotlib
plt.figure(figsize=(10, 6))

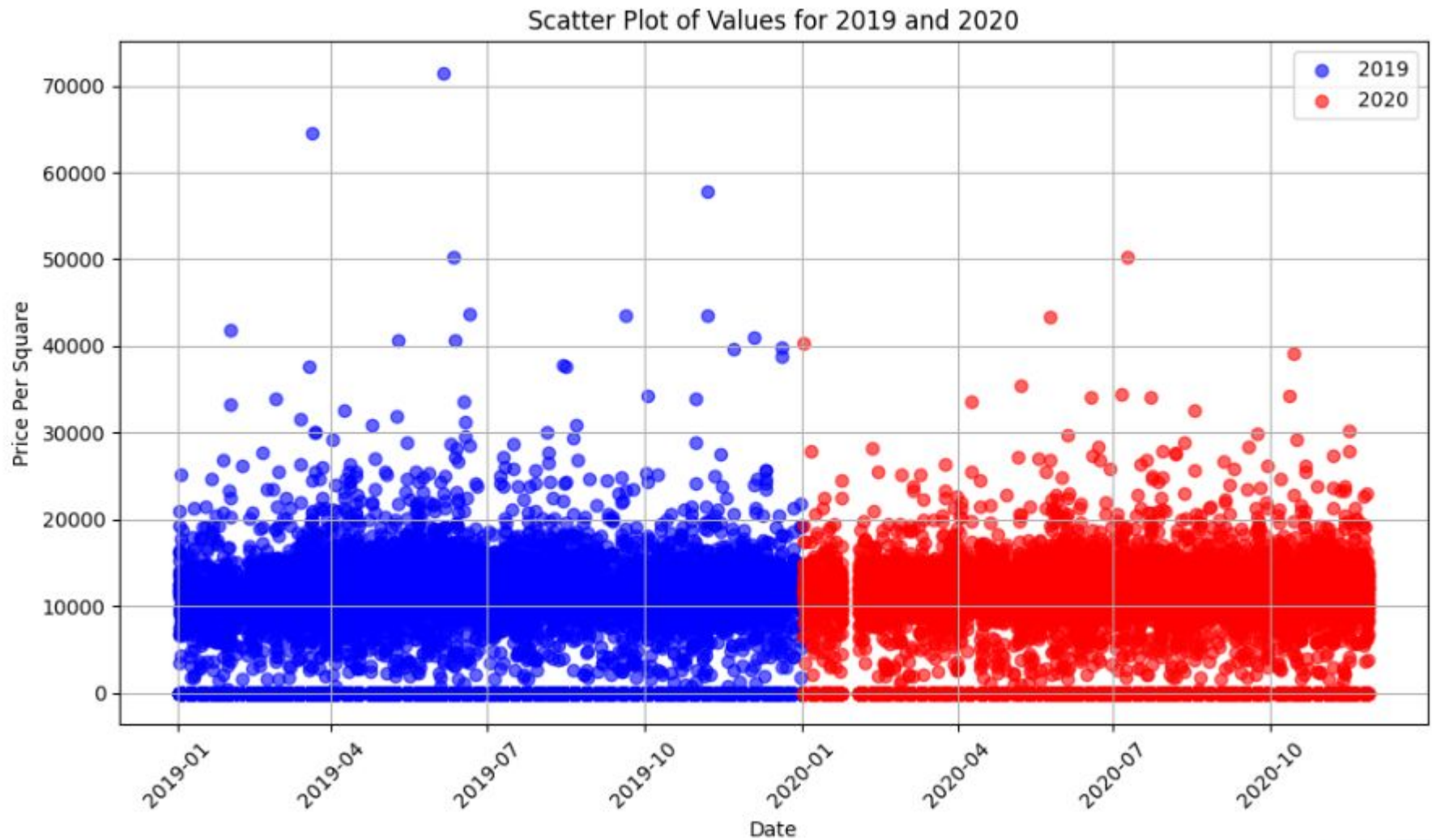
# Plot for 2019
axis_value = filtered_data['full_date']
axis_value = filtered_data['Price Per Square']
plt.scatter(axis_value, axis_value, color='blue', label='2019', alpha=0.6)

# Plot for 2020
axis_value = covidfiltered_data['full_date']
axis_value = covidfiltered_data['Price Per Square']
plt.scatter(axis_value, axis_value, color='red', label='2020', alpha=0.6)

# Adding titles and labels
plt.title('Scatter Plot of Values for 2019 and 2020')
plt.xlabel('Date')
plt.ylabel('Price Per Square')
plt.xticks(rotation=45)
plt.legend()
plt.grid()

# Show the plot
plt.tight_layout()
plt.show()
```

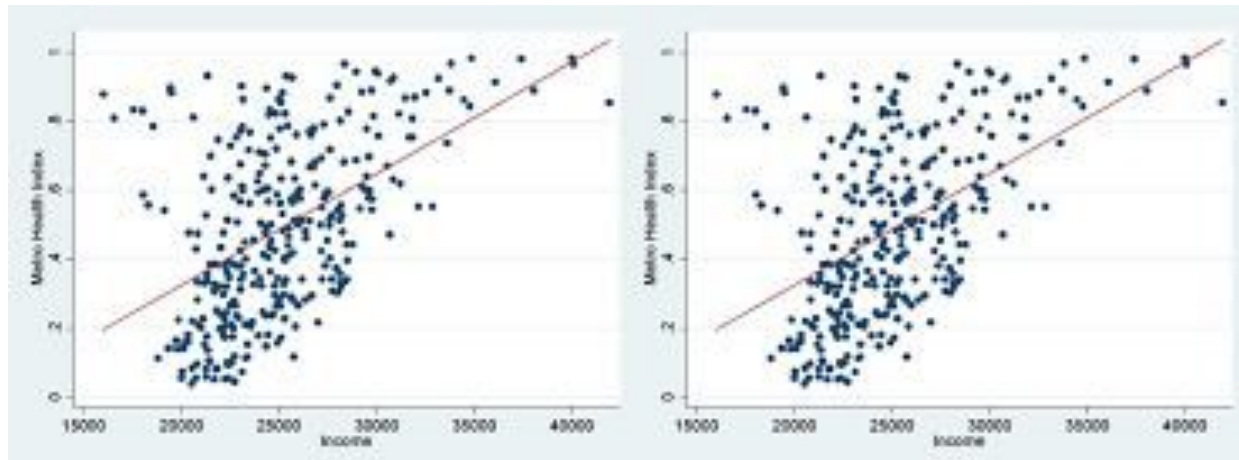
Exploratory Data Analysis(EDA) and Data Visualization



Exploratory Data Analysis(EDA) and Data Visualization

Tasks 1 - Housing Analysis

1. Change to different period from 2019-2020 and 2020-2021 to 2014-2017, 2018-2021
2. Filter the data with school net number “34”, “35”, or others
3. Visualise the result in Scatter Plot
4. Change the chart color, legend.



Exploratory Data Analysis(EDA): with Pandas

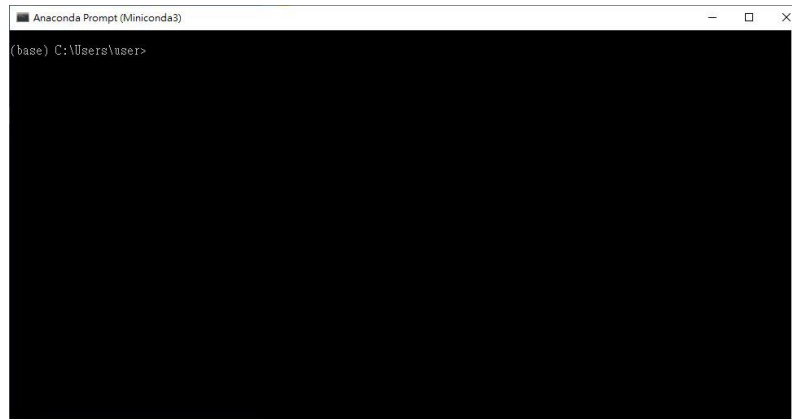
Common Panda functions

Grouping/Splitting/Aggregating:

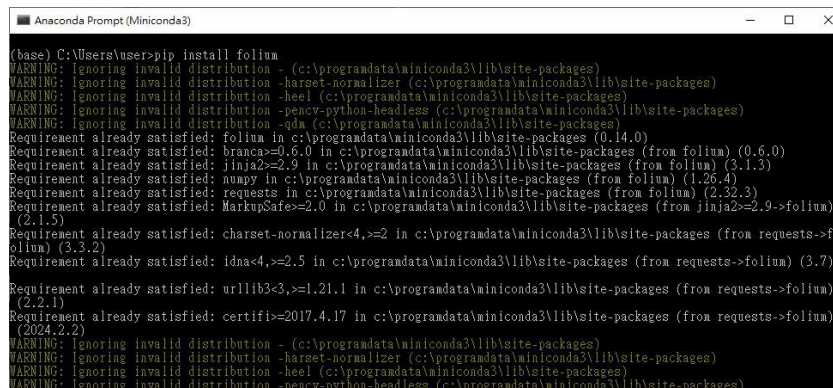
- `groupby()`, `.get_groups()`
- `.merge()`
- `.concat()`
- `.aggagate()`
- `.append()`

Exploratory Data Analysis(ETL), Dynamic Data Visualization

1. Open tools “Anaconda Prompt” which will direct you to the command prompt.



1. Install python package requests by entering “pip install pygwalker”



Exploratory Data Analysis(ETL), Dynamic Data Visualization

We mainly use PyGWalker (Python Library) for Dynamic Data Visualization, similar to use PowerBI or Tableau.

Official Information can be found in <https://kanaries.net/pygwalker>

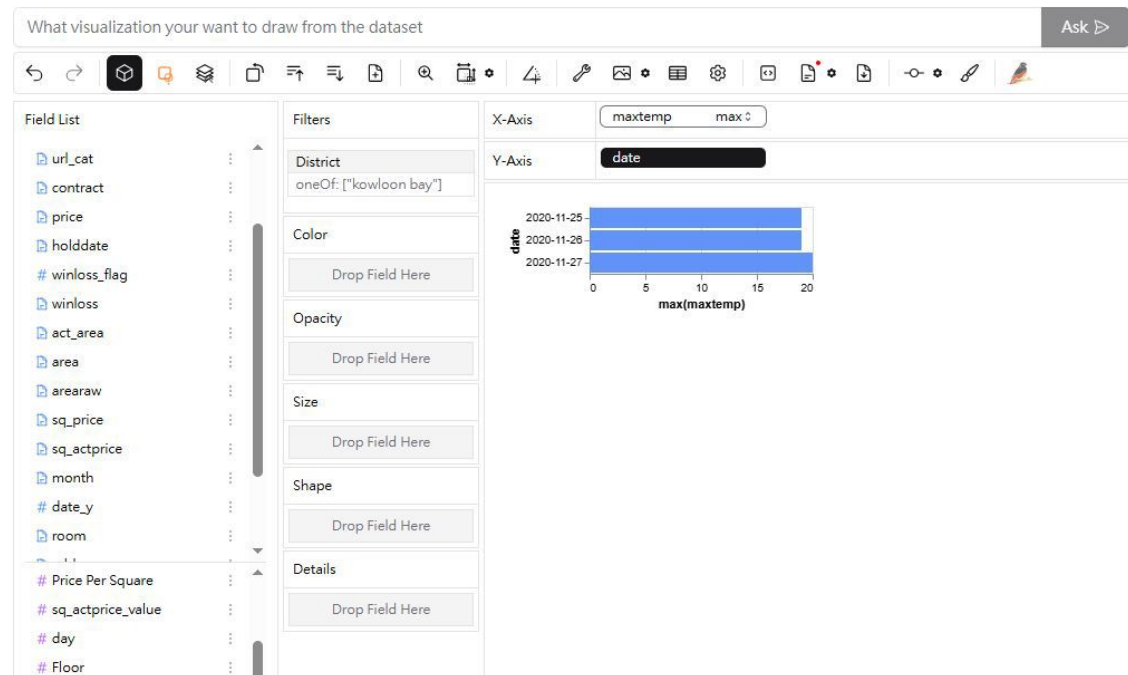
Write python code

```
import pandas as pd
```

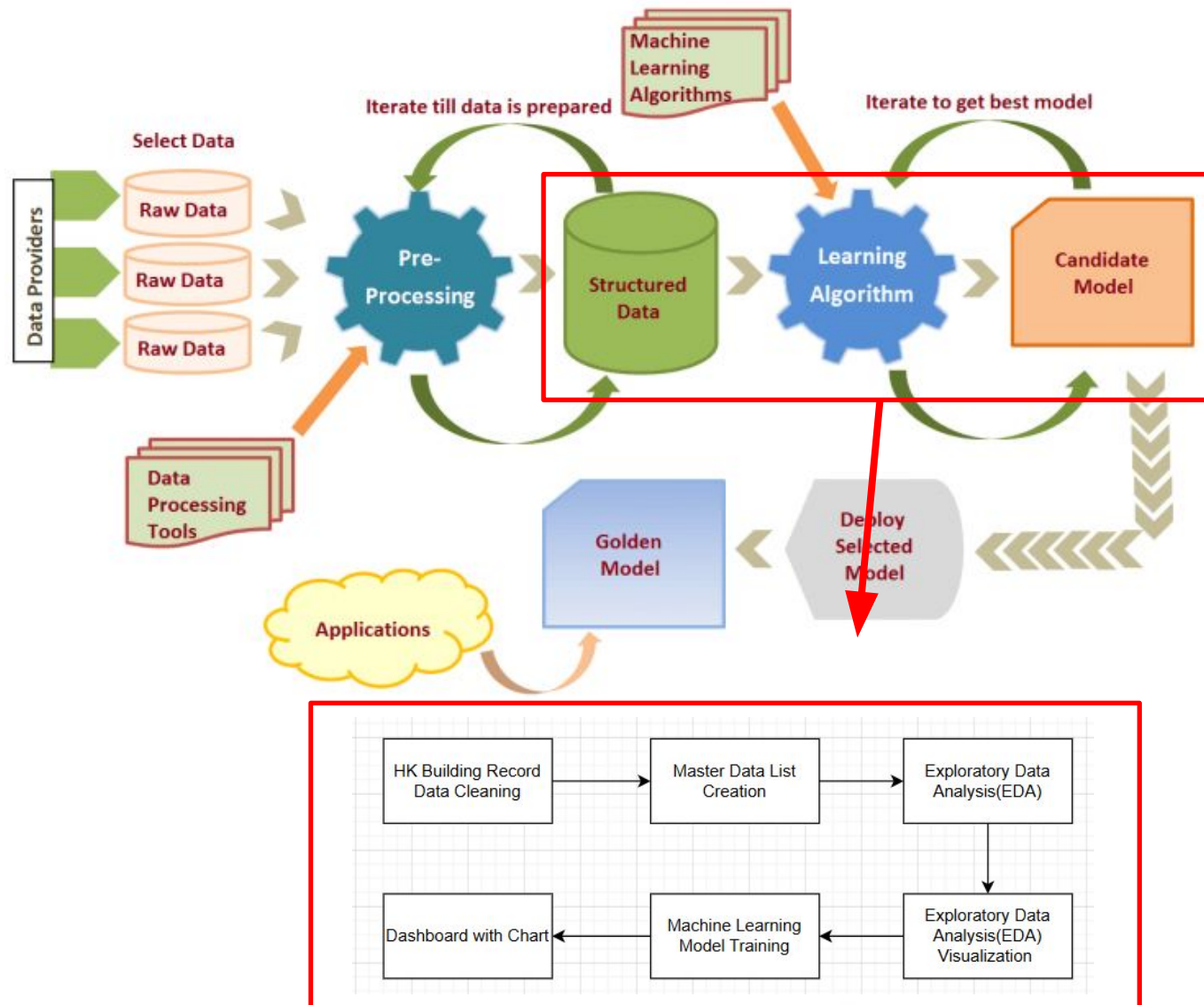
```
import pygwalker as pyg
```

```
df = pd.read_csv("combined_building_withweather_school.csv")
```

```
pyg.walk(df)
```



Applied Machine Learning Pipeline Overview



How To Do Data Mining: Machine Learning Techniques

Features / Attributes

Are used to train
Learning algorithm

Sales Price

Saleable

Area

Number of Bedrooms

Year built

(Refurbished) Schools
in the area

Transportation (MTR, buses)

Address: 270-280 QUEEN'S ROAD
WEST

Price per sqft.: 17,841

Qty transaction: 2

Year: 2017



Using a Home Valuation Prediction Model as an example

How To Do Data Mining: Machine Learning Techniques

Machine Learning

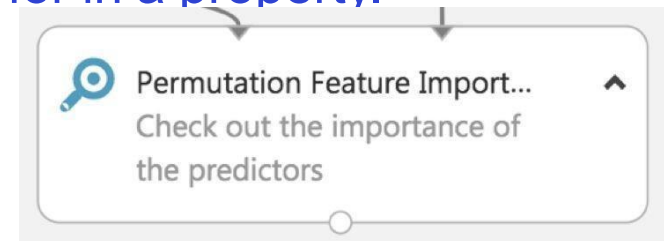
Choosing a 'useful' or **significant** feature (predictors) **makes** or **breaks** a data science project.

Domain experts, property agents will tell you what property buyers value in a home.

Data scientists would **NOT** have a clue what home buyers look for in a property.

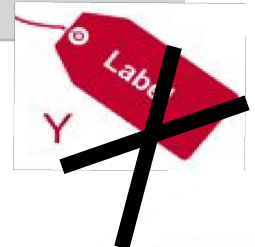
But not all is lost!

Feature	Score
	
SaleableArea	0.858545
Eff_Build_Ages	0.166877
Bus_Route	0.076519
Parks	0.019011



Data Processing & Analysis - What Is Machine Learning?

Supervised Learning	Unsupervised Learning
Used as an advanced form of <i>predictive modeling</i>	Used either as a form of automated data analysis or automated signal extraction
Each observation must be labeled with a "correct answer"	Unlabeled data has no predetermined "correct answer"
Only then can you build a predictive model because you must tell the algorithm what's "correct" while training it (hence, "supervising" it)	You'll allow the algorithm to directly learn patterns from the data (without "supervision")
Regression is the task for modeling continuous target variables	Clustering is the most common unsupervised learning task, and it's for <i>finding groups</i> within your data
Classification is the task for modeling categorical (a.k.a. "class") target variables	



Data Processing & Analysis

Supervised Learning: Classification For Debit Card Fraud

If it Walks/Swims/Quacks Like a Duck ... Then It Must Be a Duck

... with pre-determined features of fraud

Classification takes a set of data

Building a classifier model, you extract the features of interest that most contribute to the classification

And learns how to label new records based on that information

Known ~~labels~~ (Not Fraud)

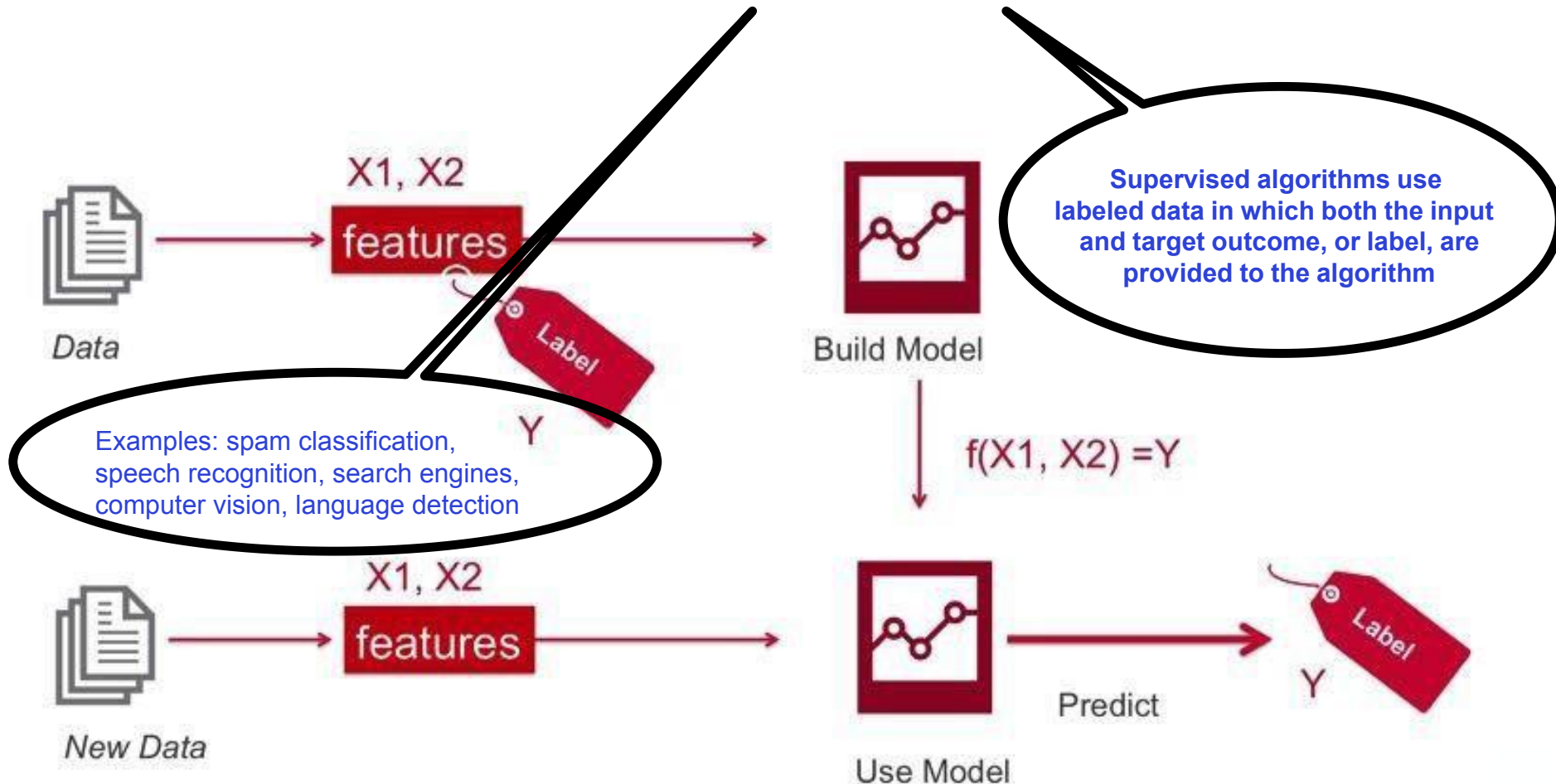
Known ~~labels~~ (Fraud)

Source:
<https://dzone.com/articles/demystifying-ai-machine-learning-and-deep-learning>

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Data Processing & Analysis

Machine Learning: Supervised Learning



Source:
<https://dzone.com/articles/demystifying-ai-machine-learning-and-deep-learning>

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Data Processing & Analysis

Machine Learning: Decision Trees

decision
tree for predicting debit
card fraud

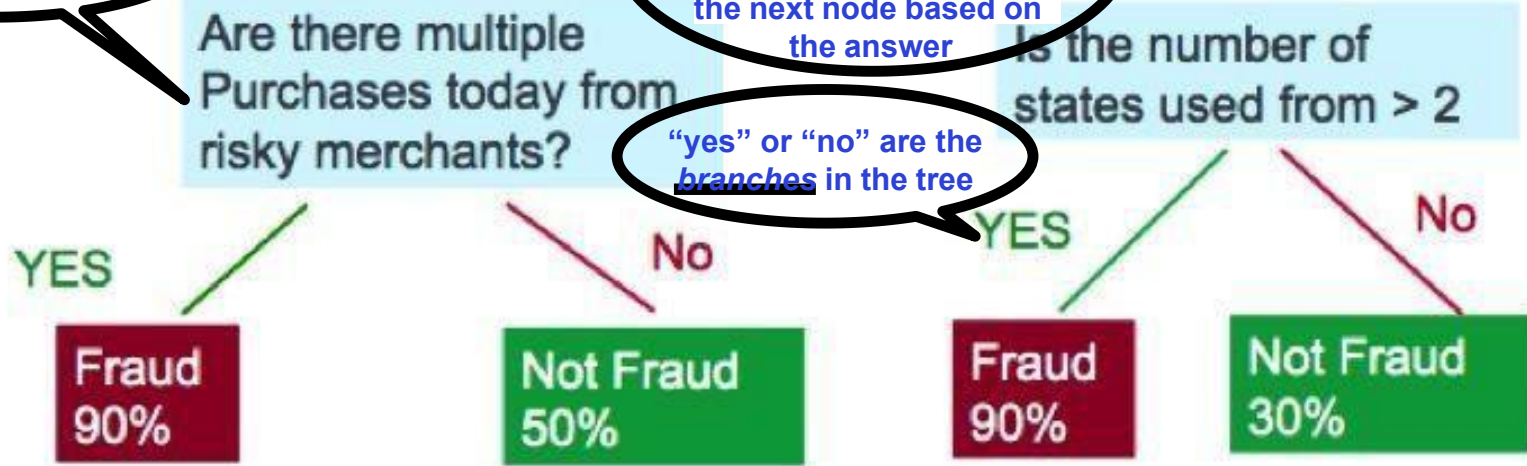
Decision trees create a model that
predicts the class or label based on several
input features

questions are the nodes

Is the amount spent in 24
hours > average

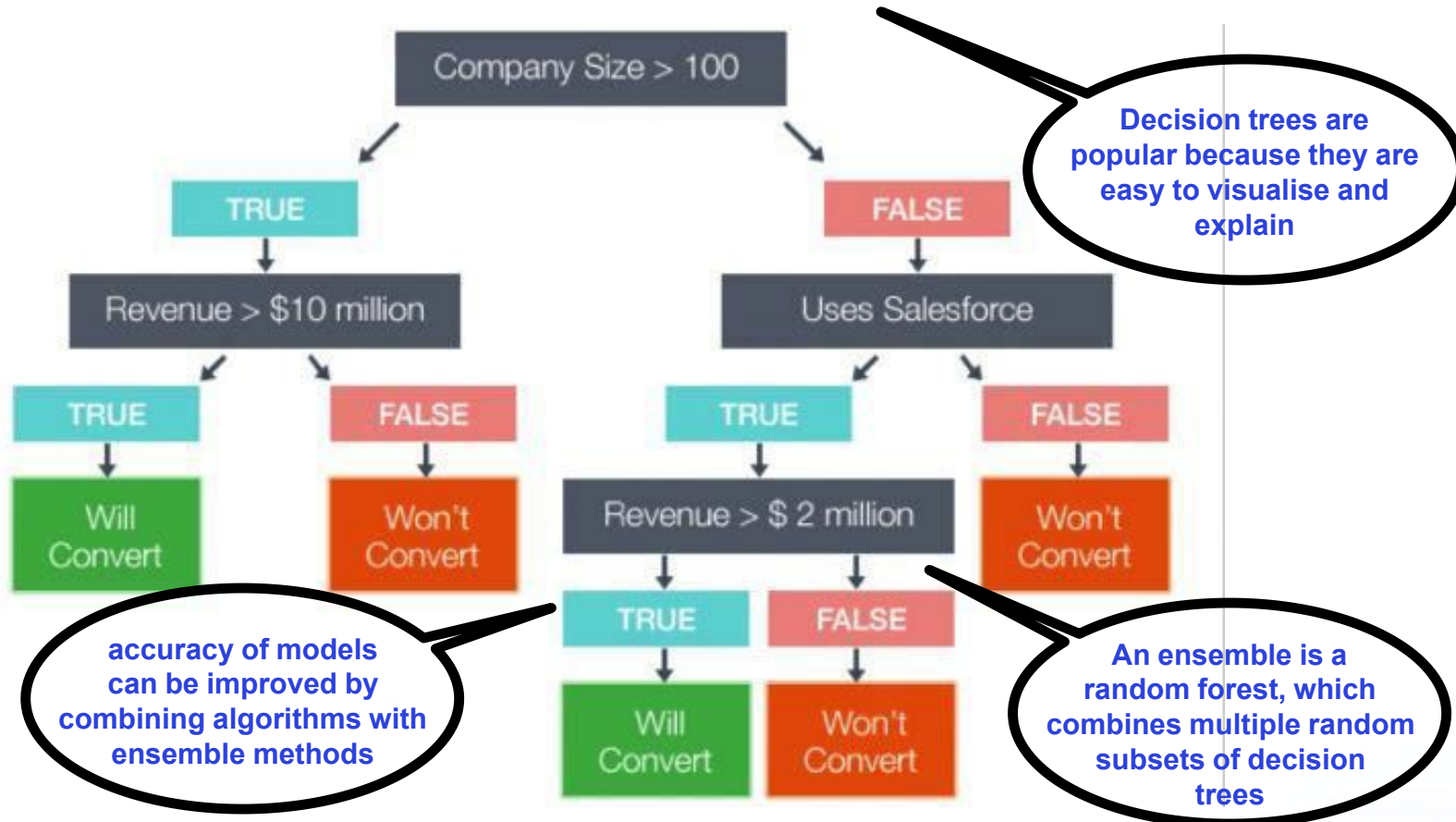
work
by evaluating a question
containing a feature at every
node and selecting a branch to
the next node based on
the answer

“yes” or “no” are the
branches in the tree



Data Processing & Analysis

Decision Trees: Ensemble Random Forests



Source: <http://dataconomy.com/2015/05/predictive-machine-learning-behind-the-scenes-at-fliptop-and-predictions-for-the-future-of-martech/>

How To Do Data Mining: Machine Learning Techniques

Features / Attributes

Set X using RandomForestRegressor
Parameter

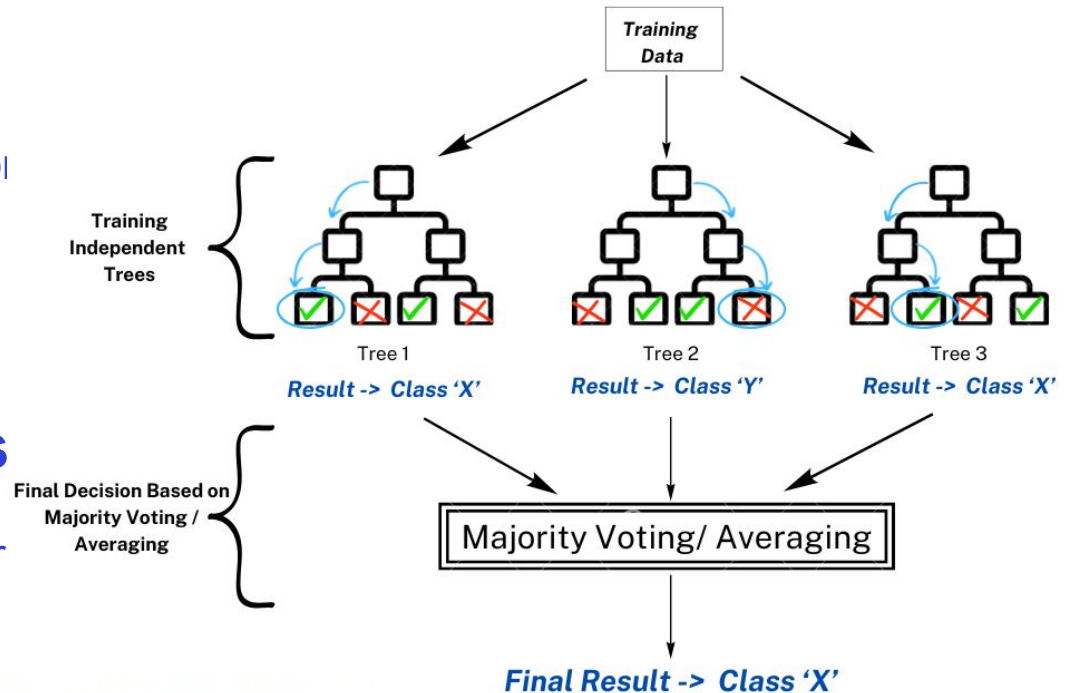
1. District
2. Floor
3. School Net Numbers

Set y using RandomForestRegressor
Outcome - Price Per Square

Top 5 Features and their Importance Scores:

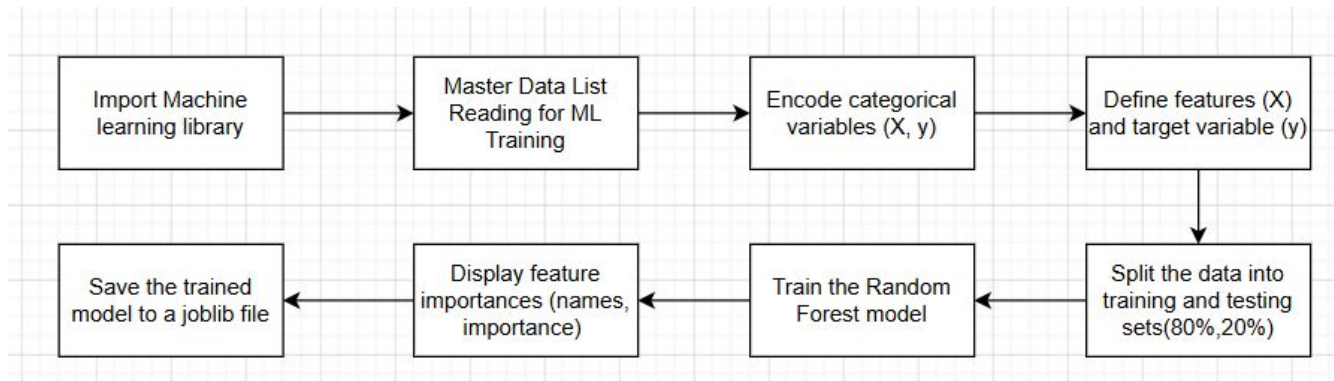
	Feature	Importance
0	District	0.728459
2	School_Netnumber	0.199190
1	Floor	0.072351

Random Forest Algorithm in Machine Learning



How To Do Data Mining: Machine Learning Techniques

Below are the Flow Chart with the Process of Machine Learning Random Forest Model Training



#This code is going to train an random forest model to find top important features

#Step 1 : Import necessary library

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestRegressor
from sklearn.tree import plot_tree
import joblib # Import joblib for saving the model
```

Machine Learning Random Forest Model training

#Step 2 : Master Data List Reading for ML Training

```
data = pd.read_csv("combined_building_withweather_school.csv")
```

#Step 3 : Encode categorical variables (X, y)

```
label_encoder = LabelEncoder()
```

```
data['District'] = label_encoder.fit_transform(data['District'])
```

```
data['School_Netnumber'] = label_encoder.fit_transform(data['School_Netnumber'])
```

#Step 4: Define features and target variable

```
X = data[['District', 'Floor', 'School_Netnumber']] # Features
```

```
y = data['Price Per Square'] # Target variable
```

```
# Check the distribution of the target variable
```

```
print(y.value_counts())
```

```
print(X)
```

```
# Ensure the target variable is continuous
```

```
print(f"Target variable type: {y.dtype}")
```

```
Price Per Square
0.00      30508
10000.00    217
8333.33    109
8000.00     81
12500.00     70
...
8554.05      1
13764.04      1
10443.86      1
15421.30      1
4672.90       1
Name: count, Length: 31331, dtype: int64
   District  Floor  School_Netnumber
0         1    2.0             8
1        16    1.0             2
2        31    7.0             5
3        28    1.0             7
4        24    1.0             2
...      ...    ...             ...
79995      6    2.0             9
79996      9    9.0             6
79997      3    4.0             4
79998      6    1.0             9
79999     26    3.0             1
[80000 rows x 3 columns]
```


Machine Learning Random Forest Model training

#Step 5: Split the dataset into training and testing sets

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

#Step 6: Train the Random Forest model

```
rf_model = RandomForestRegressor(n_estimators=100, random_state=42)
```

```
rf_model.fit(X_train, y_train)
```

#Display a decision tree from the Random Forest with reduced size

```
plt.figure(figsize=(6, 6)) # Reduced by 10%
```

```
plot_tree(rf_model.estimators_[5], feature_names=X.columns.tolist(), filled=True)
```

```
plt.title("Decision Tree from Random Forest")
```

```
plt.show()
```

```
RandomForestRegressor
RandomForestRegressor(random_state=42)
```

Machine Learning Random Forest Model training

#Step 8: Display feature importances

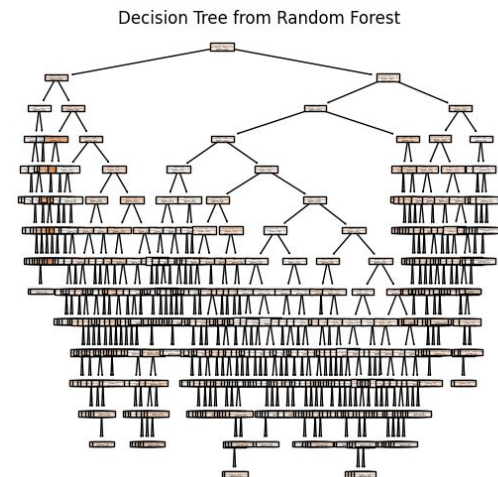
```
importances = rf_model.feature_importances_  
feature_names = X.columns.tolist()  
feature_importances = pd.DataFrame({'Feature': feature_names, 'Importance':  
    importances})
```

Sort by importance and display the top 5 features

```
top_features = feature_importances.sort_values(by='Importance',  
    ascending=False).head(5)  
print("Top 5 Features and their Importance Scores:")  
print(top_features)
```

#Step 9: Save the trained model to a joblib file

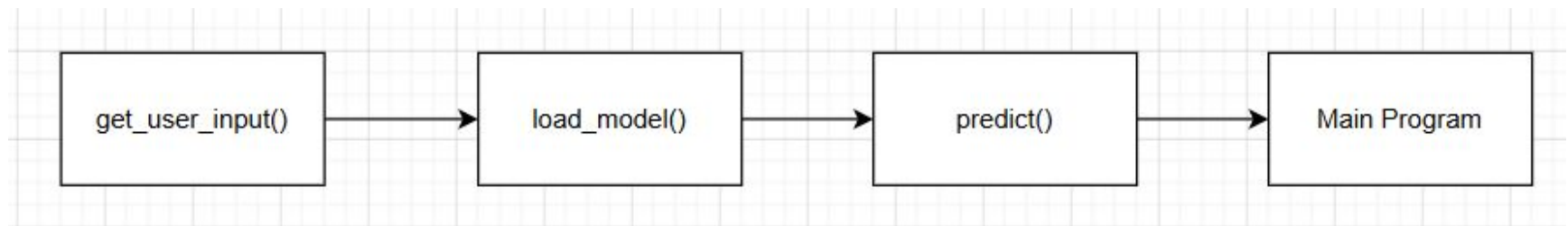
```
model_filename = 'housing_valuation_model.joblib' # Specify the filename  
joblib.dump(rf_model, model_filename) # Save the model  
print(f'Model saved to {model_filename}')
```



Top 5 Features and their Importance Scores:

	Feature	Importance
0	District	0.728459
2	School_Netnumber	0.199190
1	Floor	0.072351

Machine Learning Random Forest Inference Model : Display the predicted housing price per square



```
#add necessary library
import joblib
```

```
#Define function to get user input
```

```
def get_user_input():
    district = float(input("Enter District Number : "))
    schoolnet_no = float(input("Enter School Net Number: "))
    floor = float(input("Enter Floor "))
    input_list = [[district, schoolnet_no, floor]] #assign in data type {list}
    return input_list
```

Machine Learning Random Forest Inference Model : Display the predicted housing price per square

```
#Define function for loading random forest model
def load_model(model_path):
    return joblib.load(model_path)

#Define function to predict housing price per square based on 3 data input
def predict(model, input_data):
    return model.predict(input_data)

#Main Program to display the predicted housing price per square
input_data = get_user_input()
model_path = 'housing_valuation_model.joblib'
model = load_model(model_path)
result = predict(model, input_data)
print(f"Predicted Hosing Price Per Square is : HKD ${round(result[0],2)}")
```

```
Enter District Number : 12
```

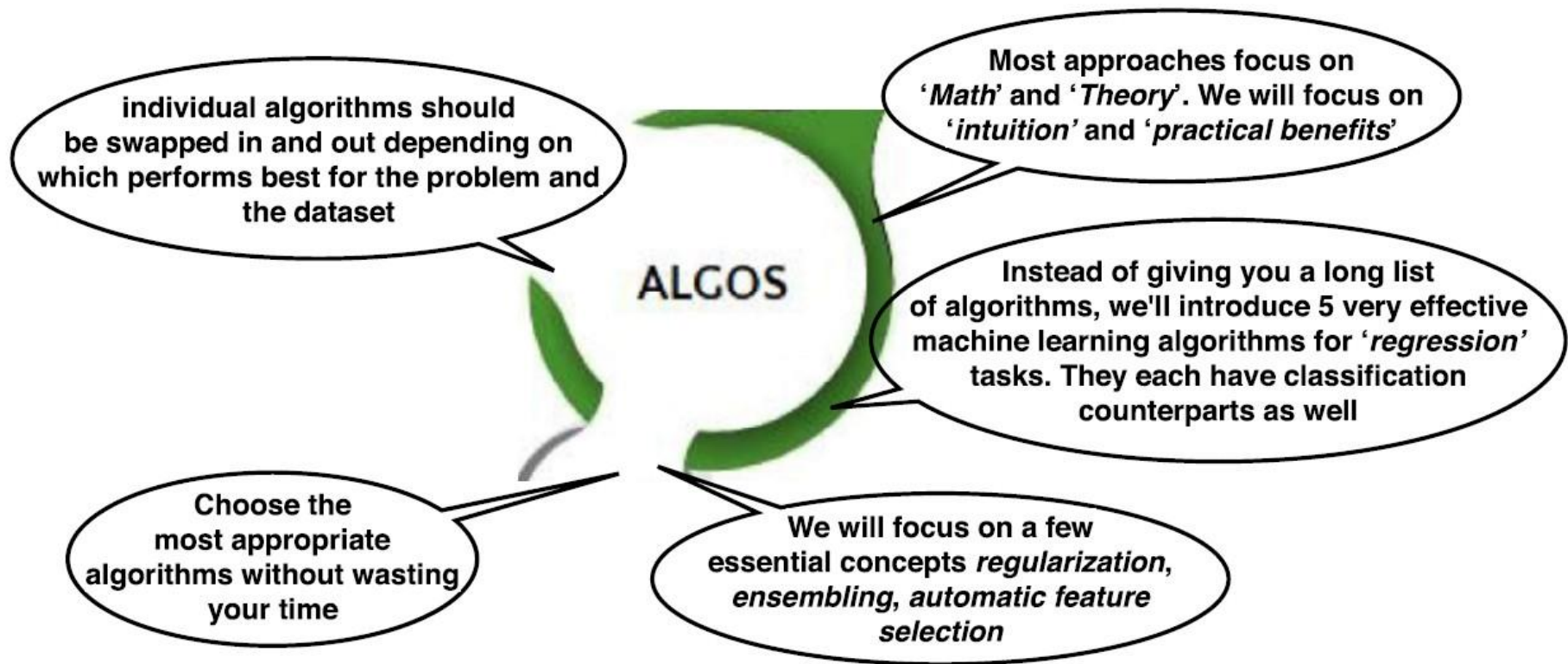
```
Enter School Net Number: 24
```

```
Enter Floor 3
```

```
Predicted Hosing Price Per Square is : HKD $10839.77
```

Big Data Tools and Algorithm Exploration

Applied Machine Learning Algorithm Selection Overview



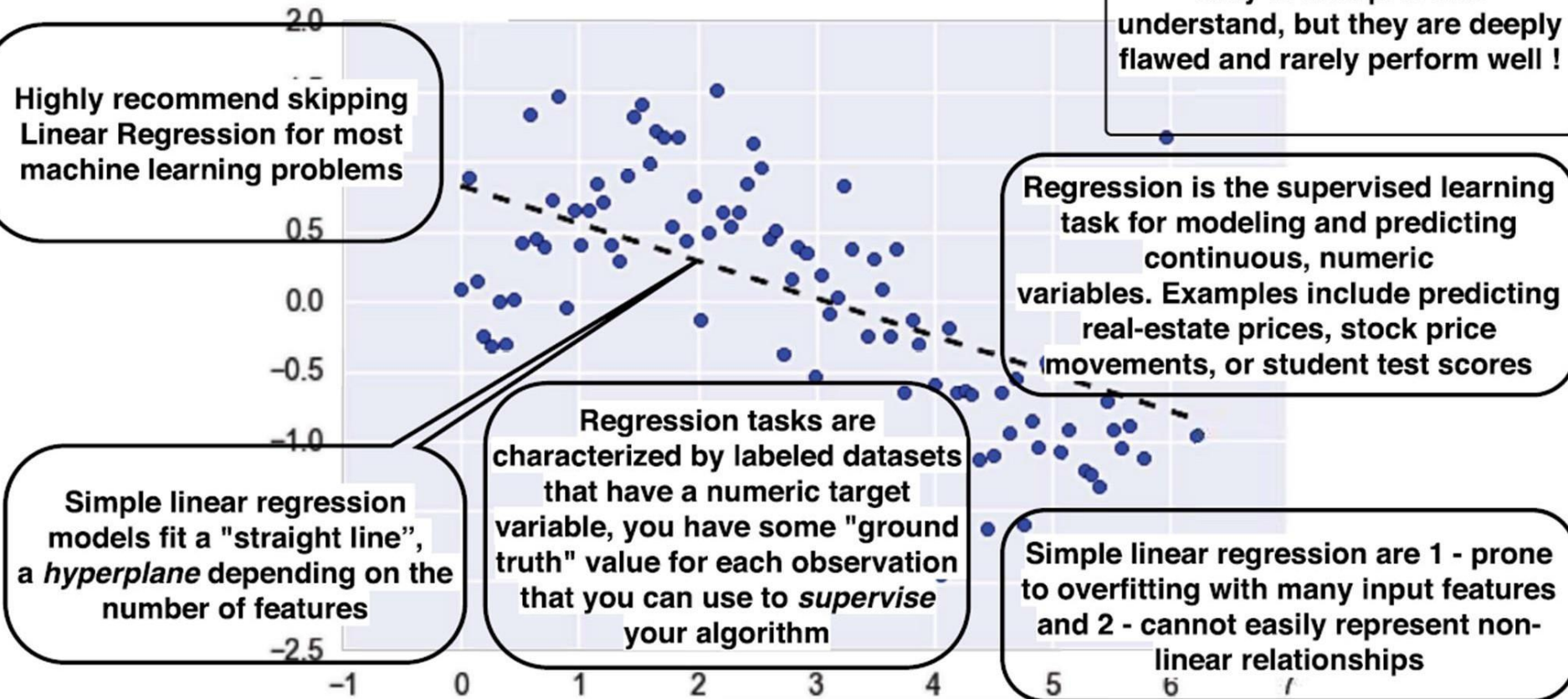
Source: <https://elitedatascience.com/algorithm-selection>

Big Data Tools and Algorithm Exploration

Applied Machine Learning Algorithm Selection

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Applied Machine Learning Algorithm Selection



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Source: <https://elitedatascience.com/algorithm-selection>

Big Data Tools and Algorithm Exploration

Applied Machine Learning Algorithm Selection

Each tree is only trained on a random subset of observations (a process called resampling)

Each tree is only allowed to choose from a random subset of features to split on (leading to feature selection)



Random Forests

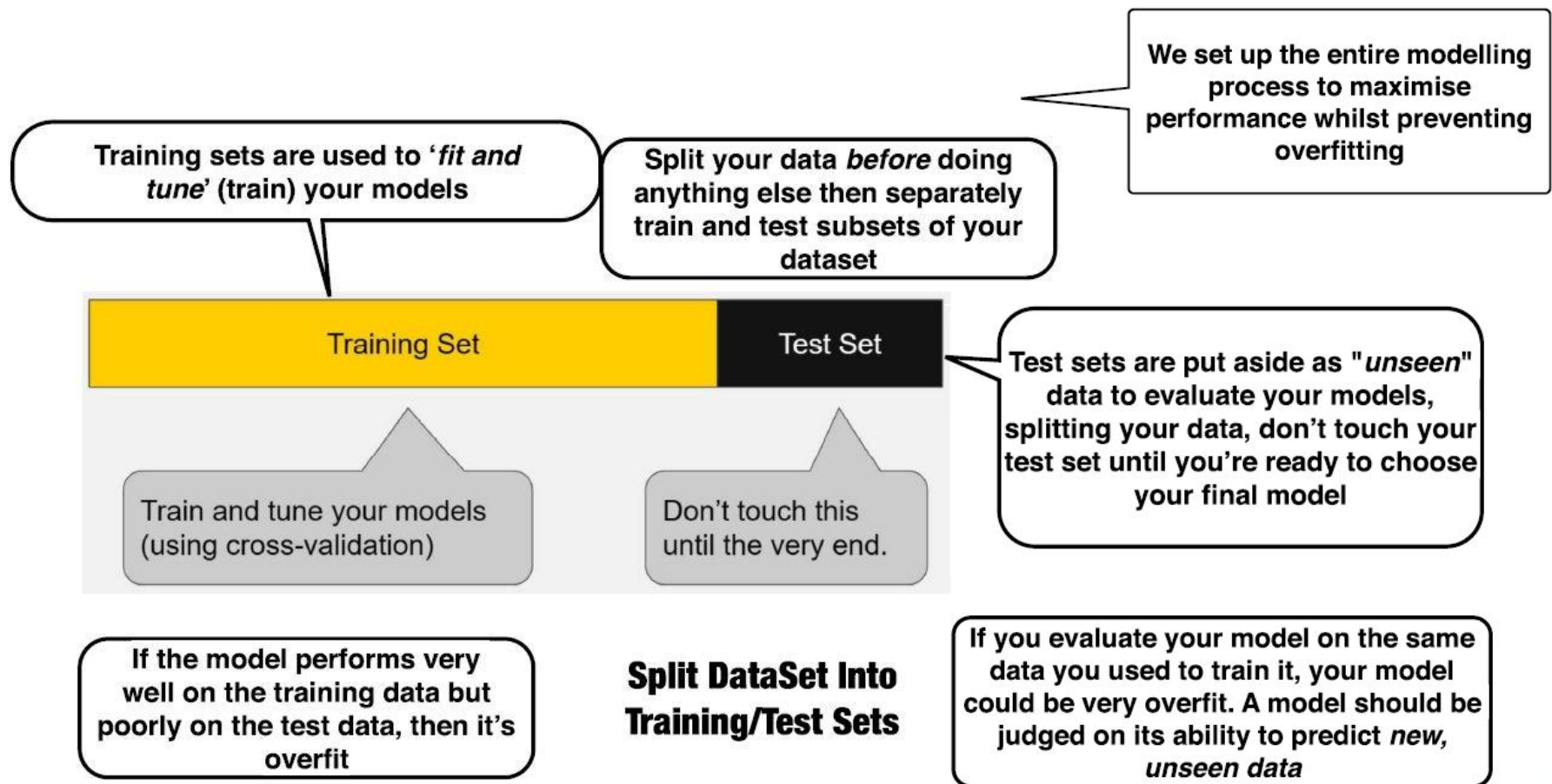
Random forests train a large number of "strong" decision trees and combine their predictions through bagging

random forests tend to perform very well right out of the box often beat many other models that take up to weeks to develop, and are the perfect "*swiss-army-knife*" algorithm that almost always gets good results

Source: <https://elitedatascience.com/algorithm-selection>

Big Data Tools and Algorithm Exploration

Model Training



Source: <https://elitedatascience.com/model-training>

Big Data Tools and Algorithm Exploration

Model Training

Examples of Model parameters include regression coefficients, decision tree split locations

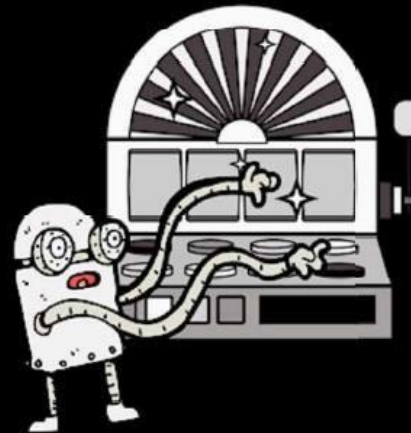
Hyperparameters: When we talk of tuning models, we specifically mean tuning hyperparameters

"Don't mind me. Just randomly pressing buttons carefully tuning parameters."

Hyperparameters express "higher-level" *structural settings* for algorithms. They are decided before fitting the model because they can't be learned from the data

Model parameters are attributes that define *individual models*, and are learned directly from the training data

There are two types of parameters in machine learning algorithms: Model Parameters and hyperparameters



E.g. Hyperparameters "higher-level" *structural settings* are strength of the penalty used in regularized regression and the number of trees to include in a random forest

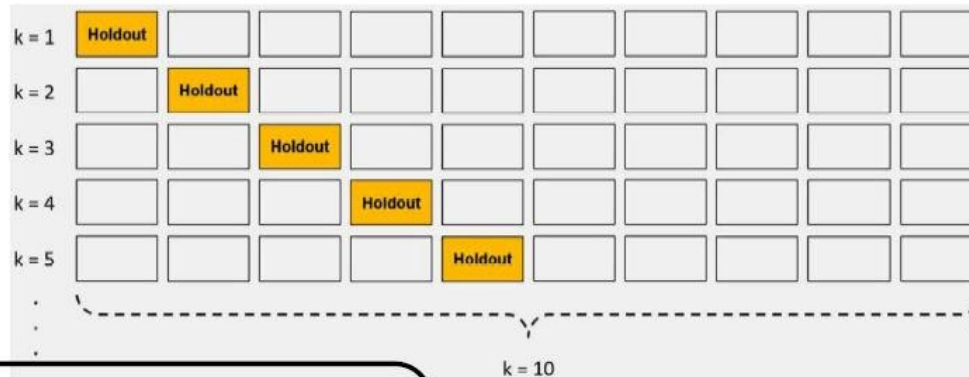
Source: <https://elitedatascience.com/model-training>

Big Data Tools and Algorithm Exploration

Model Training

1- Split your data into 10 equal parts, or “folds”, 2- Train your model on 9 folds (e.g. the first 9 folds)

Cross-validation is a tuning method for getting a reliable estimate of model performance using only training data



3 - Evaluate it on the 1 remaining "hold-out" fold, Perform steps (2) and (3) 10 times, each time holding out a different fold

10-fold cross-validation, breaks your training data into 10 equal parts (or *folds*), essentially creating 10 miniature train/test splits

Average the performance across all 10 hold-out folds, average performance across the 10 hold-out folds is your final performance estimate, also called your *cross-validated score*

Source: <https://elitedatascience.com/model-training>

Big Data Tools and Algorithm Exploration

Model Training

perform the entire cross-validation loop detailed above on each set of hyperparameter values using a high-level pseudo-code

We've split our dataset into training and test sets, and we've learned about hyperparameters and cross-validation, we're ready to fit and tune our models

For each algorithm (i.e. regularized regression, random forest, etc.):

For each set of hyperparameter values to try:

Perform cross-validation using the training set.

Calculate cross-validated score.

At the end of this process, you will have a cross-validated score for each set of hyperparameter values... for each algorithm

Elastic-Net		
Penalty Ratio	Penalty Strength	CV-Score
75/25	0.01	0.63
75/25	0.05	0.64
75/25	0.10	0.67
50/50	0.01	0.62
50/50	0.05	0.63
50/50	0.10	0.66

Source: <https://elitedatascience.com/model-training>

Big Data Tools and Algorithm Exploration

Model Training

Keep the set of hyperparameter values with best cross-validated score. Re-train the algorithm on the entire training set (without cross-validation)

By now, you'll have 1 "best" model *for each algorithm* that has been tuned through cross-validation. Most importantly, you've only used the training data so far

Because you've saved your test set as a truly unseen dataset, you can now use it to get a reliable estimate of each model's performance. There are a variety of performance metrics you could choose from

For classification tasks, we recommend Area Under ROC Curve (AUROC). *Higher values are better*

For regression tasks, we recommend Mean Squared Error (MSE) or Mean Absolute Error (MAE). *(Lower values are better)*



like the Hunger Games... each algorithm sends its own "representatives" (i.e. model trained on the best set of hyperparameter values) to the final selection

Source: <https://elitedatascience.com/model-training>

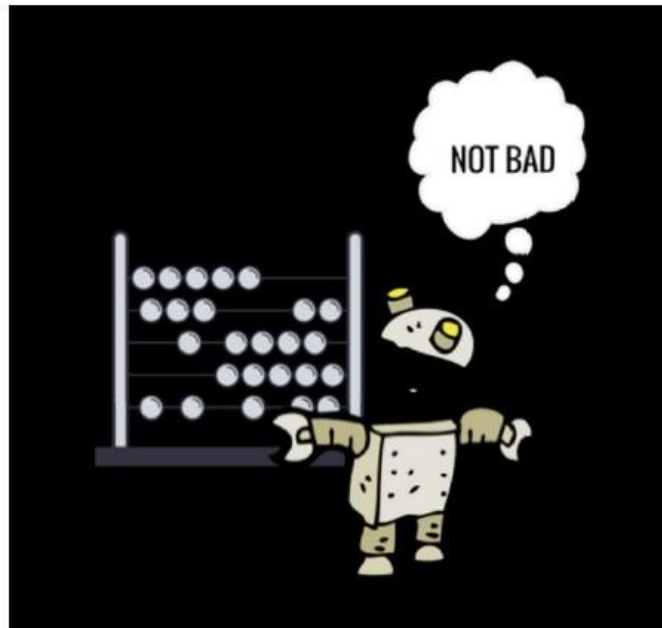
Big Data Tools and Algorithm Exploration

Model Training Conclusion

Pick the winning model by asking: Which model had the best performance on the test set? (performance)

Calculate performance metrics using those predictions and the "ground truth" target variable from the test set

For each of your models, make predictions on your test set



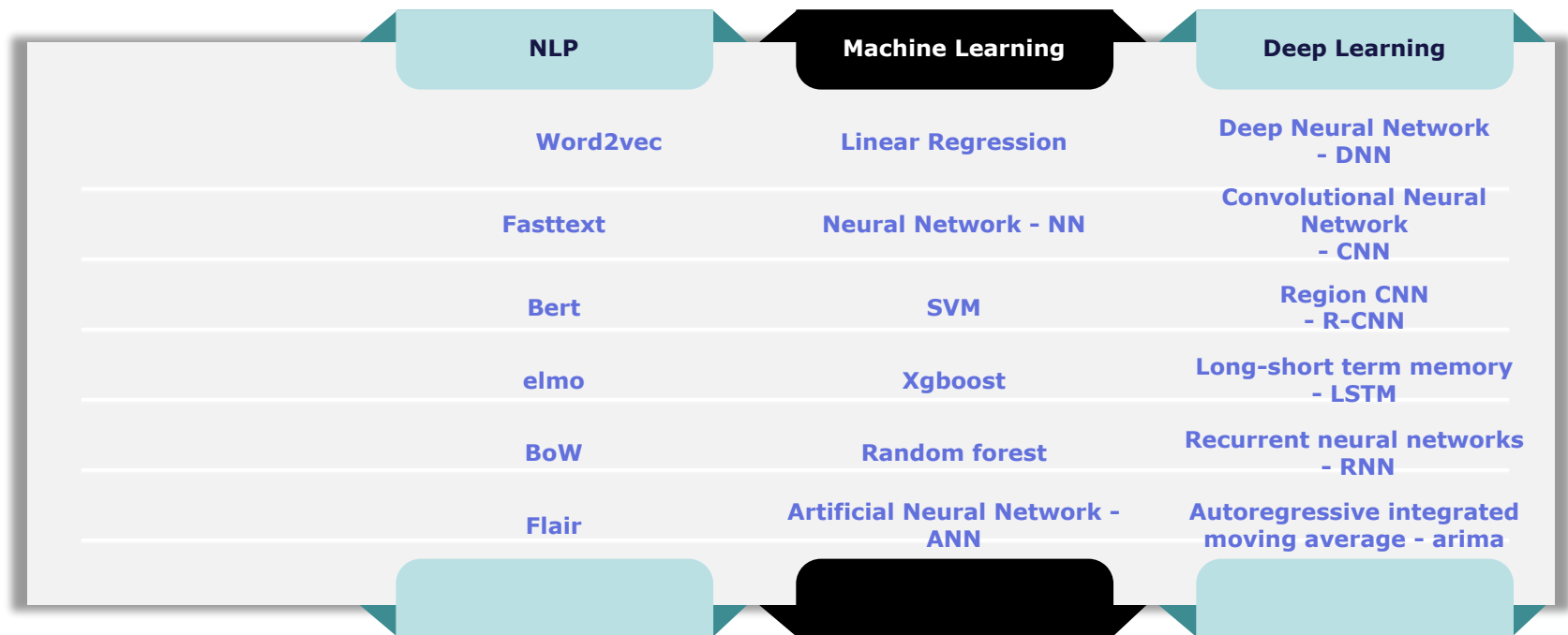
Pick the winning model by asking: Does it perform well across various performance metrics? (robustness)

Pick the winning model by asking: Did it also have (one of) the best cross-validated scores from the training set? (consistency)

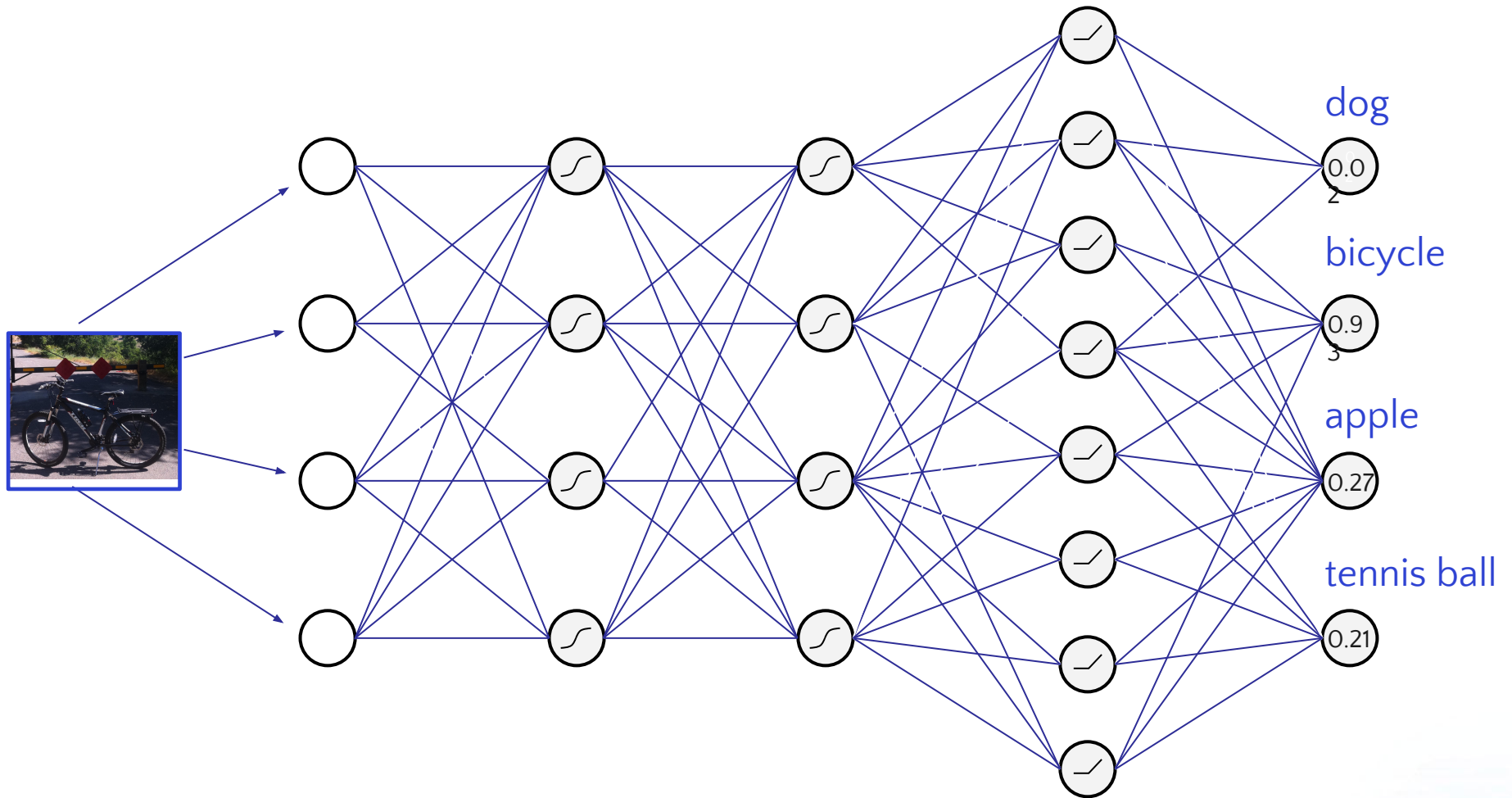
Pick the winning model by asking: Does it solve the original business problem? (win condition)

Source: <https://elitedatascience.com/model-training>

Modelling Techniques



Trained Convolutional NN



Machine Learning Model Training Visualisation

Select a metric to see a visualization or table of the data.

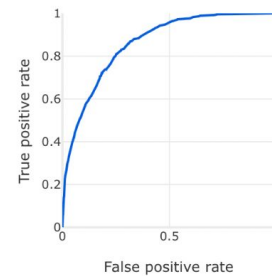
Search

- ☒ Accuracy
- ☒ AUC
- ☒ Confusion matrix
- ☒ F1 Score
- ☒ Lift curve
- ☒ Precision
- ☒ Precision-recall curve
- ☒ Recall
- ☒ ROC curve
- ☒ Scored bins

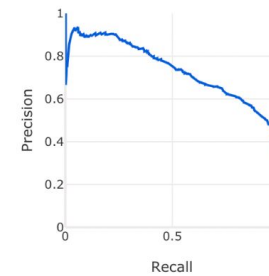
Evaluation_results

● Scored dataset (left port)

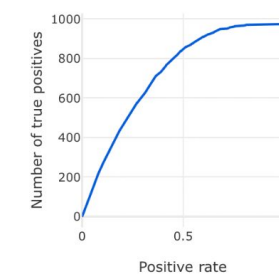
ROC curve



Precision-recall curve

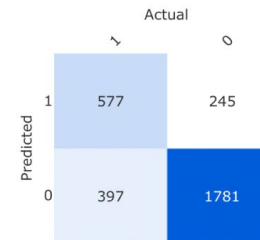


Lift curve



Threshold 0.5

Accuracy 0.786
Precision 0.702
Recall 0.592
F1 Score 0.643
AUC 0.86



Score bin ↓	Positive exam...	Negative exam...	Fraction above thres...	Accura...	F1 Score	Precisi...	Recall	Negative precis...	Negative recall	Cumulative AUC
(0.900,1.000]	95	11	0.035	0.703	0.176	0.896	0.098	0.696	0.995	0.000
(0.800,0.900]	149	21	0.092	0.746	0.390	0.884	0.251	0.732	0.984	0.002

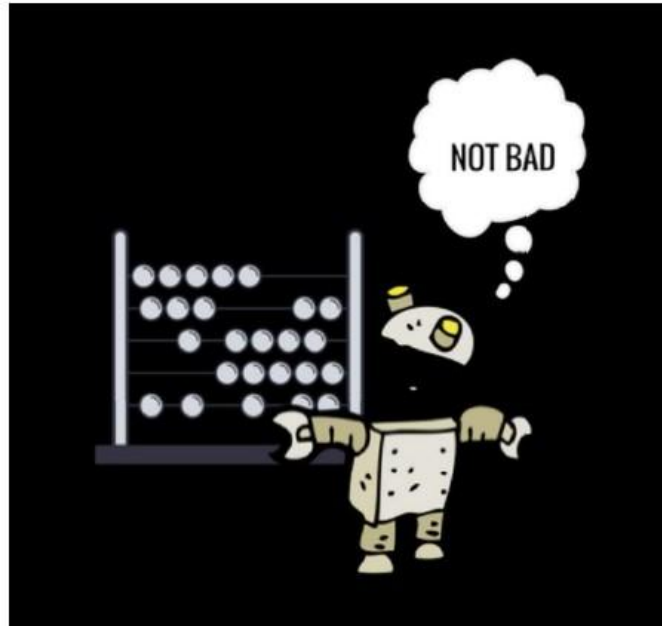
<https://www.analyticsvidhya.com/blog/2021/09/a-comprehensive-guide-on-using-azure-machine-learning/>

Machine Learning Model Training Conclusion

Pick the winning model by asking: Which model had the best performance on the test set? (performance)

Calculate performance metrics using those predictions and the "ground truth" target variable from the test set

For each of your models, make predictions on your test set



Pick the winning model by asking: Does it perform well across various performance metrics? (robustness)

Pick the winning model by asking: Did it also have (one of) the best cross-validated scores from the training set? (consistency)

Pick the winning model by asking: Does it solve the original business problem? (win condition)

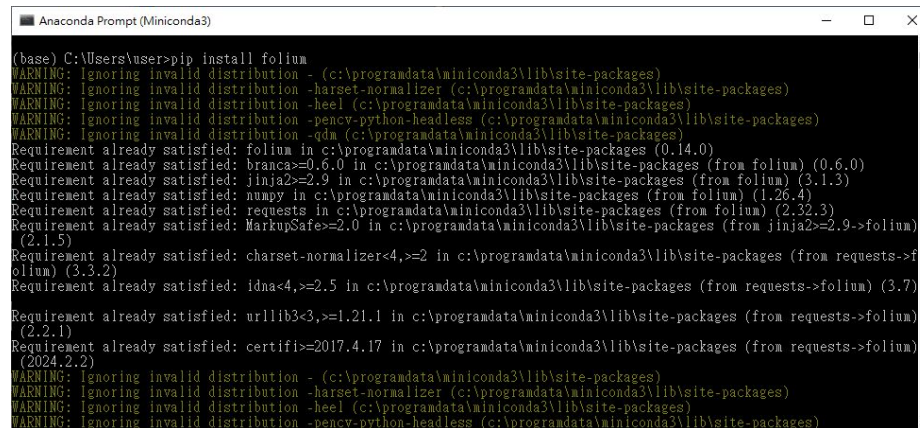
Source: <https://elitedatascience.com/model-training>

Python Google Map Tutorial

1. Open tools “Anaconda Prompt” which will direct you to the command prompt.



2. Install python package requests by entering “pip install folium”



Python Google Map Tutorial

```
import folium
```

```
# List of 23 coordinates (latitude, longitude)
```

```
coordinates = [
```

```
    (37.7749, -122.4194), # San Francisco, CA
```

```
    (34.0522, -118.2437), # Los Angeles, CA
```

```
    (40.7128, -74.0060), # New York, NY
```

```
    (41.8781, -87.6298), # Chicago, IL
```

```
    (29.7604, -95.3698), # Houston, TX
```

```
    (33.4484, -112.0740), # Phoenix, AZ
```

```
    (39.7392, -104.9903), # Denver, CO
```

```
    (47.6062, -122.3321), # Seattle, WA
```

```
    (38.9072, -77.0369), # Washington, D.C.
```

```
    (39.9612, -82.9988), # Columbus, OH
```

```
    (30.2672, -97.7431), # Austin, TX
```

```
    (32.7767, -96.7970), # Dallas, TX
```

```
    (37.3382, -121.8863), # San Jose, CA
```

```
    (39.7392, -104.9903), # Denver, CO
```

```
    (36.1699, -115.1398), # Las Vegas, NV
```

```
    (25.7617, -80.1918), # Miami, FL
```

```
    (38.2527, -85.7585), # Louisville, KY
```

```
    (39.9612, -82.9988), # Columbus, OH
```

```
    (36.8508, -76.2859), # Norfolk, VA
```

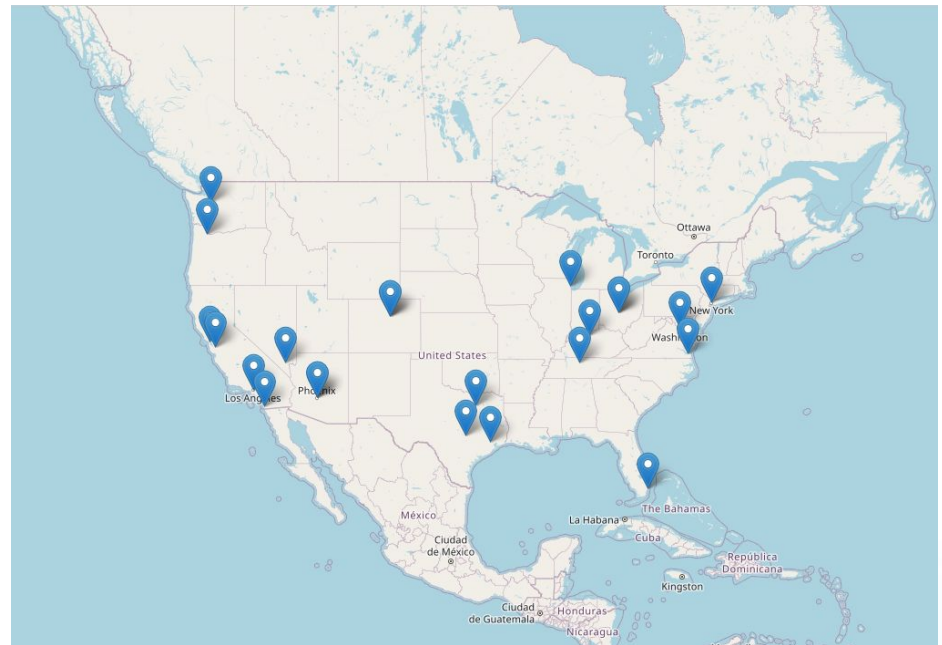
```
    (33.4484, -112.0740), # Phoenix, AZ
```

```
    (45.5155, -122.6793), # Portland, OR
```

```
    (36.1627, -86.7816), # Nashville, TN
```

```
    (32.7157, -117.1611), # San Diego, CA
```

```
]
```



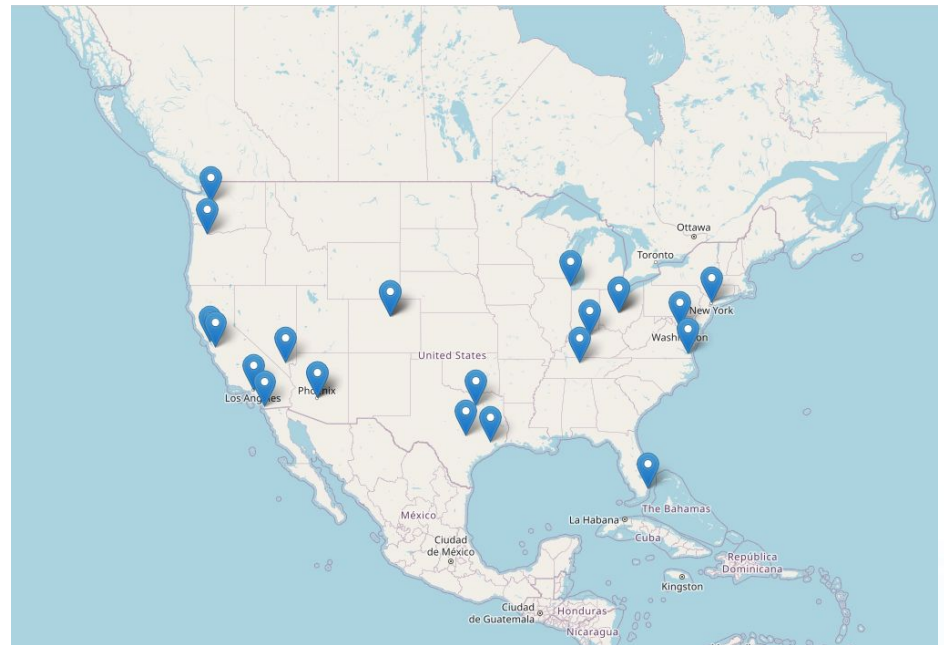
Python Google Map Tutorial

```
# Create a map centered at the first coordinate
my_map = folium.Map(location=coordinates[0],
                    zoom_start=4)
```

```
# Add markers for each coordinate
for lat, lon in coordinates:
    folium.Marker(location=(lat,
                           lon)).add_to(my_map)
```

```
# Save the map to an HTML file
my_map.save("my_map.html")
```

```
# If you're using Jupyter Notebook, you can
    display the map inline
my_map
```



Google Sheet Content Update

1. Open Anaconda Prompt, enter "pip install gspread google-auth google-api-python-client" to install google library
2. Open Jupyter notebook and type following python code

```
import gspread
from google.oauth2.service_account import Credentials
from googleapiclient.discovery import build

def update_values(sheet_id, range_name, value_input_option, values):
    SCOPES = ['https://www.googleapis.com/auth/spreadsheets']
    SERVICES_ACCOUNT_FILE = "graceful-disk-484104-a2-152ebd9c692e.json"

    credentials = Credentials.from_service_account_file(SERVICES_ACCOUNT_FILE,
                                                       scopes=SCOPES)
    service = build('sheets', 'v4', credentials=credentials)

    body = {"values": values}
```

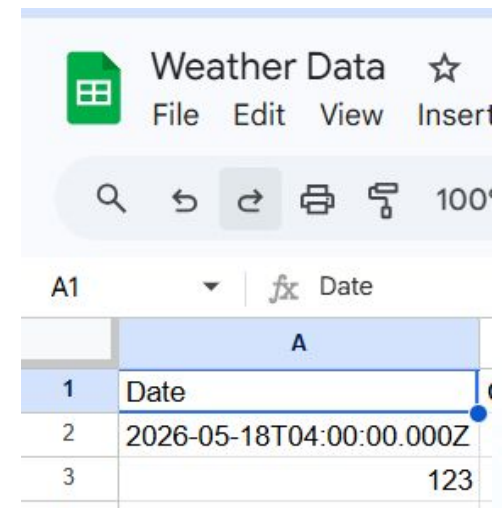
Google Sheet Content Update

```
# Update the range to just the starting cell for appending
result = service.spreadsheets().values().update(
    spreadsheetId=sheet_id,
    range=range_name, # Change range here
    valueInputOption=value_input_option,
    body=body
).execute()
```

```
print(f"{result.get('updatedCells')} cells updated.")
return result
```

```
sheet_id = '1mqrumbU1Si5aGmkg2jpffu7zn9uWxFvXX1Wnuwkas'
range_name = "Sheet1!A3"
value_input_option = 'USER_ENTERED'
values = [['123']]
```

```
update_values(sheet_id, range_name, value_input_option, values)
```



Weather Data	
File	Edit View Insert
A1 fx Date	
1	Date
2	2026-05-18T04:00:00.000Z
3	123



Data Visualizations

1. describe the contemporary trends of process automation and explain the opportunities and challenges of business process automation;
2. outline key steps in process automation project management and illustrate the significance of each step;
3. apply computational tools to implement process automation;
4. discuss the development of process automation and practical cases for business.

Exercise 1

Step 1

```
from matplotlib import pyplot as plt
```

```
# x-axis values
```

```
x = [5, 2, 9, 4, 7]
```

```
# Y-axis values
```

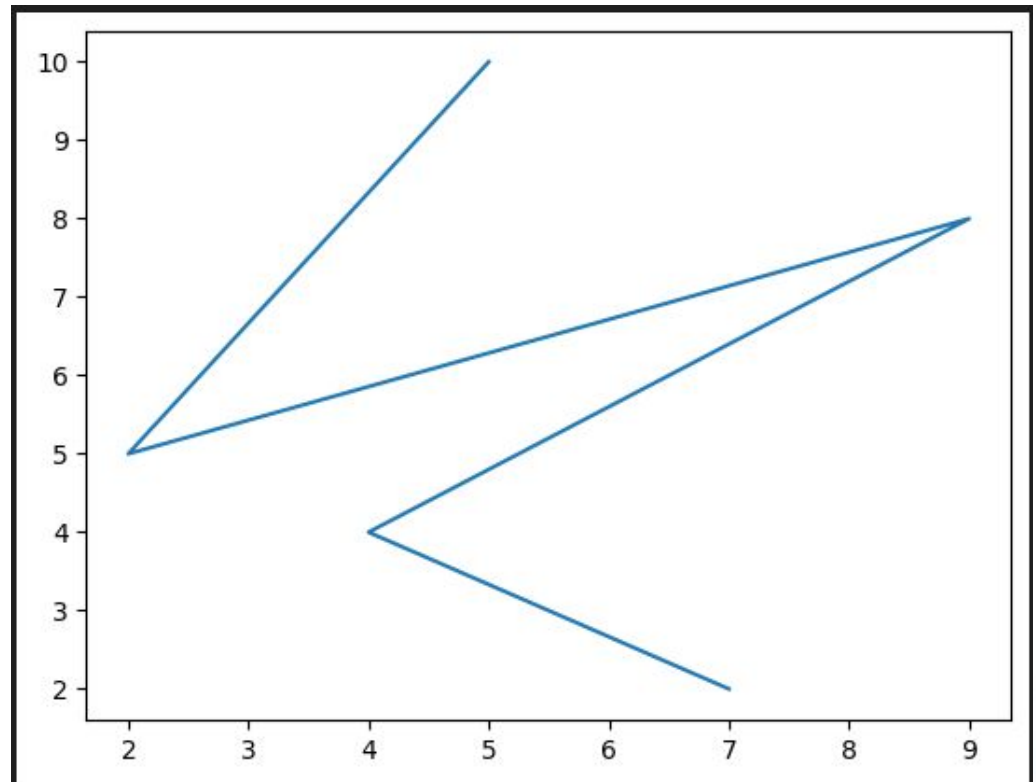
```
y = [10, 5, 8, 4, 2]
```

```
# Function to plot
```

```
plt.plot(x, y)
```

```
# function to show the plot
```

```
plt.show()
```



Exercise 1

Step 2

#scatter

x-axis values

x = [5, 2, 9, 4, 7]

Y-axis values

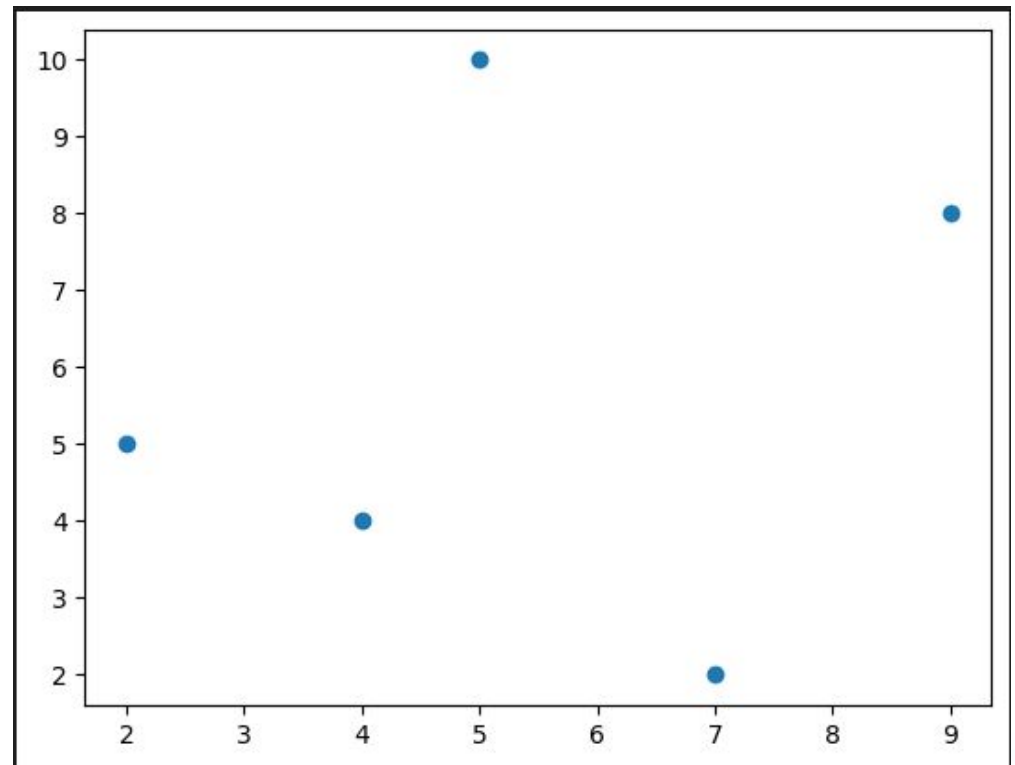
y = [10, 5, 8, 4, 2]

Function to plot scatter

plt.scatter(x, y)

function to show the plot

plt.show()





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THANK YOU

